Competitive Persuasion with a Noisy Judge

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Abstract

Can a firm choosing between two agents gain from judging them with error? I show that she can when the agents design their own evidence. Noiseless judging makes even a narrow lead decisive, so agents conceal bad news not to lose; noise makes a narrow lead worth little, so revealing pays more. For a broad class of noise, if agents are either good or bad, the equilibrium is unique: each agent sometimes admits being bad and otherwise pools, and if agents are likely good, both reveal fully, unlike under noiseless judging. Moderate noise is best: it curbs concealment but never reverses the ranking of the signals sent. The best such judge beats a noiseless one unless agents are very likely bad, and attains the first best when they are at least as likely good as bad. With a continuum of types, the case for noise survives: even a designer who can measure evidence perfectly and commit to any rule prefers to judge with error for many priors.

Keywords: Bayesian persuasion, multiple senders, competition in persuasion, full disclosure, noisy evaluation

JEL Classification: C72, D82, D83

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1. Introduction

A capital committee must fund one of two projects this quarter. Each sponsor decides how rigorous a pilot to run before the meeting, knowing that the committee's verdict will turn partly on things that have nothing to do with the projects: who presents second, how long the meeting has run. The committee would like to see past all of that. This paper asks whether it should want to.

Such imperfect verdicts, biased or merely arbitrary, are well recognized. In Scharfstein and Stein (2000), divisional rent-seeking leads to "socialist" internal capital markets, in which weaker divisions are subsidized by stronger ones. In Prendergast and Topel (1996), evaluators who lack objective performance measures act on personal preferences, favoring some subordinates beyond their true performance. And Pier et al. (2018) find essentially no agreement among experienced reviewers rating the same NIH grant applications, all of which were eventually funded: ratings of the same application were no more similar than ratings of different ones. The usual response is to engineer such imperfections away, through more reviewers, more structured evaluations, and more objective criteria. A few research funders have nonetheless moved the other way: they now fund by lot among proposals that clear a quality bar, citing reviewer bias, reviewer burden, and the hope of bolder proposals (Fang and Casadevall, 2016; Heyard et al., 2022; Röbbecke and Simon, 2023). Only the hope of bolder proposals concerns how applicants respond, and none concerns what the competitors choose to reveal about themselves, the channel I study.

The question is whether a decision-maker should eliminate all noise from her evaluation when the competitors know that it is noisy and adapt their reports to it. It arises whenever three features come together. First, a single prize is at stake, with no price attached: the decision-maker cannot pay more for the better competitor, and a close second earns nothing. Second, the competitors decide what and how much to

reveal about themselves after learning how the decision-maker will choose between them. Third, the decision-maker can make the verdict more or less accurate, so the competitor with the stronger evidence need not always win. I call the decision-maker *the firm*, a noisy judge when her verdict carries error, and the competitors *agents*. Each agent is either a good or a bad type and simultaneously commits, at no cost, to a signal, that is, to a distribution of the posterior probability that he is good; the firm observes the resulting posterior with a random error and selects the agent with the higher score. For many standard noise distributions and equal priors, this is also what she would do on seeing only the scores, so committing to the rule costs her nothing there.

My key observation is that noise of a suitable shape weakens each agent's incentive to conceal. An agent's chance of winning depends only on how far he looks ahead of his rival, in the same way for both agents; I call this dependence the firm's selection rule, and every such rule is generated by some noise. A noiseless firm always selects the agent who looks better, so her rule is a step: a narrow lead is as good as a wide one, and an agent revealed as bad loses unless his rival is too. Each agent therefore spreads his posterior to land just above his rival's, blending his bad type with his good one: Boleslavsky and Cotton (2015) show that both agents reveal only partially, randomizing over a continuum of posteriors, and the firm obtains less than 80% of her first best whenever agents are good with probability below one third.

With noise, the rule instead rises gradually with the lead. A narrow lead is worth little, conceding bad news no longer means certain defeat, and the shape of the noise decides whether an agent gains more from gambling on being revealed as good or from pooling his evidence, and so how much information reaches the firm. The price is accuracy: a rule near one half is close to a coin flip, so the firm passes over the agent who looks better more often, and noise trades selection accuracy for information.

An example. Suppose that each agent is equally likely to be good and bad. Under

the noiseless rule, each agent's posterior is uniformly distributed on $[0, 1]$, so the firm sometimes selects a bad agent over a good one and obtains about 89% of her first best. Now suppose the firm scores each agent by his signal plus an independent draw from $U[0, 1]$. Against a rival who reveals his type, an agent's chance of winning is linear in his own signal, so revealing his type is a best reply, if not the only one; against any other signal it is not, and, since the agents' game is constant-sum, full revelation by both agents is the unique equilibrium. Since a revealed good type always outscores a revealed bad one, the noise only decides between agents of the same type, and the firm attains her first best.

Main results. The example relies on uniform noise and an even prior; the results below show how far this extends. The first set of results, in Section 3, characterizes the equilibrium when the difference between the two evaluation errors has a concave density with a single peak over possible signal gaps, which I call a *well S-shaped* selection rule; normal and logistic noise qualify if its variance is large enough. Under such a rule, whatever his rival reveals, an agent's chance of winning first accelerates, then slows as his own signal rises, so below some signal he gains from gambling and above it from playing safe; concavity preserves this shape when the rival randomizes. Each agent thus uses a *bad news* signal: he reveals the bad type with some probability and otherwise sends one pooled signal, with full revelation and pooling as edge cases.

The equilibrium is unique whenever the priors are equal, and for every pair of priors unless the density is linear next to its peak. In the symmetric case, a single statistic governs it, the elasticity of the normalized selection rule at the pooled signal: an agent concedes bad news with unconditional probability equal to its inverse, and both agents reveal fully once the prior is high enough. With unequal priors (Subsection 3.2.4), each agent reveals more when his rival is better, and when one agent is much better, the favorite reveals nothing while the underdog reveals fully. Does the firm gain from

better agents? Under every well S-shaped rule she does, although better agents may reveal less, since an agent's own strength can outweigh his rival's; when the density of the noise is convex instead, agents reveal good news, and better agents can hurt her (Appendix D.1). With unequal priors, a better agent can hurt the firm only where both agents concede bad news.

The second set of results, in Section 4, concerns the design of the selection rule with equal priors. Within the well S-shaped class, the firm's best rule is generated by i.i.d. uniform shocks and sits exactly at its edge. It beats the noiseless benchmark once agents are good with probability above roughly 21%, and delivers the first best once they are at least as likely good as bad; a continuum of types changes this.

This does not mean that any noise is worth keeping. Scaling the uniform shocks shows the trade-off: more noise leaves the signals unchanged and only blurs selection, while less noise lowers her utility and pushes the equilibrium toward the noiseless one once signals stop being binary. The firm does best at the least noise that keeps signals binary, which lies at or beyond the class's edge. There, in the equilibrium I construct, noise never overturns a difference between signals agents send, so a moderately noisy judge beats both a noiseless and an excessively blurry one.

Section 4 then asks when the firm can attain her first best with any rule of the signal gap. With binary types and equal priors, she needs full revelation and a rule that always picks a revealed good agent over a revealed bad one. For a known prior, the rules that implement full revelation have rather vague structure, so I ask which rules do so robustly. If agents may be promising or unpromising – with priors on either side of one half – only one rule depending on the gap between signals works: the linear one, which leaves agents indifferent among all signals, so robustness to the prior costs uniqueness. If she knows whether agents are promising or unpromising, a rich family of rules makes full revelation the unique equilibrium. For promising agents, i.i.d.

shocks can generate some of these rules; for unpromising agents, they cannot, since under i.i.d. shocks full revelation is never an equilibrium once an agent is more likely bad than good. These limits are specific to rules of the signal gap. As a benchmark, a designer who could measure signals perfectly and commit to any rule avoids them: a pass/fail rule, which treats every signal short of conclusive evidence alike, attains the first best for every pair of priors in the unique equilibrium. It requires a strong commitment, since it tosses a coin between an agent who is almost surely good and one who is almost surely bad, and with binary types the linear rule of the signal gap does as well in its firm-preferred equilibrium, so an imperfect measure loses nothing.

The third set of results, in Section 5, relaxes the binary prior, the most dispersed prior with a given mean, so that no conclusion hinges on it. With a continuum of types, each agent still reveals his type below a threshold and pools all types above it, for every prior, and the best well S-shaped rule induces median censorship. The comparison with the noiseless rule changes: under a convex prior CDF, the noiseless rule attains the first best, which no noisy rule of the signal gap can, and under a concave one, the ranking depends on how dispersed the prior is. The benchmark designer settles the question for concave priors. Her best rule among those that induce full revelation adds noise that makes truthful scores uniform, as in the example, and for every concave prior other than the uniform, her best rule overall, a random pass mark, lies strictly between this rule and the noiseless one: it compares the agents exactly at the top and tosses a coin only among weak signals. So the case for an imperfect evaluation survives even when perfect measurement and full commitment are available.

Appendix D collects extensions: under the mirror-image, reverse S-shaped rules, agents reveal good news instead; beyond the well S-shaped class, the equilibrium stays unique, with at most two signals per agent, under a weaker condition on the noise, while further from the class agents randomize over more signals; and with a single

agent, noise works through the same channel.

1.1 Literature review

This paper contributes to the Bayesian persuasion literature (Kamenica and Gentzkow, 2011; Kamenica, 2019), in particular to its strand on competing senders. Most results of Gentzkow and Kamenica (2017) require a Blackwell-connected information environment, which fails when each sender can only reveal information about himself, as here, and their comparison with collusion is vacuous, since the agents' game is constant-sum. Each agent's best reply solves a single-sender problem with an S-shaped value function, so it is an upper censorship as characterized by Kolotilin et al. (2022); restated in quantiles, their result gives the equilibrium under an arbitrary prior. With a general prior, I verify best replies through a convex function lying above the value function (Dworczak and Martini, 2019), and restrict selection rules to posterior means, as is standard when the receiver's payoff is linear in the state (Gentzkow and Kamenica, 2016; Kolotilin, 2018). To a single agent, the rule acts like a receiver with a privately known threshold (Kolotilin et al., 2017), here the rival's signal plus noise, and for a single sender, making a bell-shaped threshold density more diffuse leads to more information (Curello and Sinander, 2025). Here, the threshold's distribution is an equilibrium object, set by the rival's signal.

The closest paper is Boleslavsky and Cotton (2015), part of the literature on grading policies (Ostrovsky and Schwarz, 2010; Chan et al., 2007; Popov and Bernhardt, 2013; Bar et al., 2012): two schools design grades, and an evaluator hires the graduate with the higher posterior. I add evaluation noise to this game, and its shape decides what agents reveal. Under the noiseless rule, the equilibrium is unique, posteriors are uniform on an interval, possibly with atoms at 0 and 1, and schools never reveal fully. Informativeness increases with the number of senders, both in this setting, where it

becomes full in the limit (Au and Kawai, 2020), and when senders also compete in prices (Hwang et al., 2026). In search markets, whether competition raises informativeness depends on what sellers observe and how buyers search (Board and Lu, 2018; Au and Whitmeyer, 2023; He and Li, 2023).

An imperfect evaluation can also benefit the evaluator by strengthening investment incentives, as strategic grading does in Boleslavsky and Cotton (2015) and a noisy signal of universities' relative quality does in Boleslavsky and Li (2026). Here the gain comes through information instead: a suitably noisy evaluation makes signals more informative. In this, noise plays a role familiar from other settings: noise in prices preserves the incentive to acquire information that fully revealing prices would destroy (Grossman and Stiglitz, 1980), and noise in a cheap-talk channel makes the receiver less responsive to each message, which curbs the sender's exaggeration and improves communication (Blume et al., 2007). Closer in spirit, an organization that can fund only one of two proposals induces more informative evidence than an unconstrained one and is better off unless both are ex-ante strong (Boleslavsky and Cotton, 2018), and moderate attention costs can make a rationally inattentive consumer select the better of two disclosed products more often than when attention is free (Jain and Whitmeyer, 2026). In both, the gain comes from a constraint on the receiver; I instead treat the evaluation error as a primitive of the selection rule, whose shape determines the equilibrium and how much information the firm extracts.

My normative results also relate to the idea that observing an agent more closely can hurt the principal once the agent responds to it (Prat, 2005), and that a receiver can extract more information by committing to use it less: random costs of approval make a single sender indifferent among signals and elicit full revelation, at the price of burned utility (Tsakas et al., 2021), and when agents manipulate their data at a cost, so do a less responsive allocation rule, a test that passes some noncompliant agents, and a score

that underweights some features (Frankel and Kartik, 2022; Perez-Richet and Skreta, 2022; Ball, 2025). Extra noise in an already noisy channel can benefit the sender and sometimes both players (Tsakas and Tsakas, 2021), and coarse grades can inform the public better by inducing participation (Harbaugh and Rasmusen, 2018). Here signals are costless and flexible, and noise works only through competition, by changing the curvature of each agent's value function; it costs the firm nothing when it decides only between agents of the same type. Closest to the firm's commitment, Gao and Luo (2026) let a designer commit to a mechanism mapping each sender's experiment and posterior into a random action: robust mechanisms depend only on posteriors and induce full revelation, and with competing senders the efficient allocation is robustly implementable. Bergemann et al. (2023) show that a receiver uncertain about the sender's preferences optimally commits to a quota. Here the firm commits only to a comparison of the signals, which, for rules of the signal gap, makes robustness costly (Subsection 4.3) and, with a single agent and an absolute standard, caps her payoff (Appendix D.3). A single receiver who could commit to any decision rule would need no noise; with binary types, neither would a firm comparing two agents, as a pass/fail rule shows (Proposition 8), but with a continuum of types even her best rule blurs the signals (Subsection 5.2). Randomized choice also serves other purposes: breaking ties in school choice (Abdulkadiroğlu and Sönmez, 2003; Erdil and Ergin, 2008), saving on costly signaling (Hoppe et al., 2009), economizing on verification (Border and Sobel, 1987), deterring gaming (Ederer et al., 2018), and keeping improper reasons out of selection (Stone, 2011; Fang and Casadevall, 2016).

The agents' game is also a contest in which each agent chooses a distribution with a fixed mean, and noise gives it a difference-form success function (Lazear and Rosen, 1981; Hirshleifer, 1989; Che and Gale, 2000), the counterpart of the ratio form of Tullock (1980), with more noise playing the role of lower decisiveness. When contestants

choose the riskiness of their performance, limiting competition helps a principal select the better one (Hvide and Kristiansen, 2003; Fang and Noe, 2022), and a noisy selection rule is a new instrument with the same effect. In effort contests, a noisy contest can also be optimal (Wang, 2010; Morgan et al., 2022; Letina et al., 2023), and the shape of the noise matters (Drugov and Ryvkin, 2020); see Fu and Wu (2019) for a survey.

2. Model

Agents. There are 2 agents and the firm (from here on she). Each agent's type θ_i is drawn independently from $\{0, 1\}$, and let p_i be the probability of $\theta_i = 1$. From here on, I refer to p_i as a *prior*. For notational simplicity, if $p_1 = p_2$, I refer to them simply as p.

Information and Timing. Agents play a simultaneous-move game in which, before the realization of his type, each agent i commits to a signal structure. Then, after the realization of each agent's type θ_i, the firm receives a signal s_i according to that structure. Since the prior is binary, I can re-identify the reported signal with its posterior mean $s_i := \mathbb{E}[\theta_i|s_i]$, and hence WLOG agent i chooses a distribution of posterior means G_i (from here on signal). Let $\mathcal{G}(p_i)$ be the set of feasible signals for a prior p_i. Since the prior is binary, $\mathcal{G}(p_i) := \{G_i : \text{supp}(G_i) \subseteq [0, 1], \int_0^1 s_i dG_i(s_i) = p_i\}$, see Gentzkow and Kamenica (2016).

Selection. Agents select signal structures knowing the firm's *selection rule*: a function $v : [-1, 1] \mapsto [0, 1]$, such that $v(x) - 0.5$ is odd, increasing and continuous on $[-1, 1]$.[1] If the firm observes signals s_1, s_2, the respective probabilities of agents 1, 2 being selected are $(v(s_1 - s_2), v(s_2 - s_1))$, and since $v(x) - 0.5$ is odd, $v(s_1 - s_2) + v(s_2 - s_1) \equiv 1$ for all s_1, s_2, so that the firm always selects exactly one agent. The oddness of $v(x) - 0.5$

[1] Throughout, increasing and decreasing are meant weakly; strict monotonicity is stated explicitly.

also means that the selection rule treats the agents anonymously. The selection rule is assumed to be exogenously given to the firm for most of the sections; however, I endogenize the choice of $v(x)$ in Section 4.

Noisy evaluation. Such selection rules arise naturally when the firm judges the agents with noise. Suppose that, instead of observing s_i without noise, the firm observes agent i's *score* $s_i + \varepsilon_i$, where ε_i is a random shock, independent of s_i, s_j, reflecting the idiosyncrasies of whoever is judging – a particular committee member's mood, the order in which proposals were read, an unmodeled personal rapport. The noise is additive on the scale of the firm's payoff, the posterior probability that an agent is good; noise on another scale, such as log-odds, would give rules that depend on more than the gap between signals, which I leave aside. Taking the scores at face value, the firm selects the agent with the higher score. Suppose that $\xi := \varepsilon_j - \varepsilon_i$ has a continuous distribution, symmetric around 0; symmetry holds, for instance, whenever the shocks are exchangeable, i.e., $(\varepsilon_i, \varepsilon_j)$ and $(\varepsilon_j, \varepsilon_i)$ have the same distribution. Then she selects agent i with probability

$$\mathbb{P}(s_i + \varepsilon_i > s_j + \varepsilon_j) = \mathbb{P}(\xi < s_i - s_j) = v(s_i - s_j),$$

where $v(x)$ is the CDF of ξ restricted to $[-1, 1]$, and $v(x)$ is a selection rule: it is increasing as a CDF, continuous as ξ has no atoms, and $v(x) - 0.5$ is odd as ξ is symmetric. Conversely, every selection rule arises in this way, for example from $\varepsilon_j = -\varepsilon_i = \xi/2$, where ξ is symmetric with CDF $v(x)$ on $[-1, 1]$. These shocks are exchangeable but not independent, and independent shocks generate only some selection rules (Corollary 4). I usually give $v(x)$ explicitly, and otherwise say that $v(x)$ *is generated by* a distribution of $(\varepsilon_i, \varepsilon_j)$ with these properties.

Example 1. If $v(x)$ is generated by ε_i i.i.d. Gumbel or normal, then $v(x)$ is the CDF of a logistic or normal distribution, respectively.

Payoffs. WLOG, each agent receives 1 if selected and 0 otherwise. If the firm selects agent i, she receives θ_i, so her ex-ante expected utility equals the reported signal s_i. The shocks ε_i that may generate the selection rule do not enter the firm's utility.

Commitment. The firm commits to her selection rule. The commitment need not be deliberate: some noise is unavoidable and undetectable, so the firm cannot filter it out, and everyone understands how it shapes the verdict. Nor, for the leading noise distributions, is it a restriction: if the shocks are i.i.d. with a log-concave density, such as normal, Gumbel or uniform, and both agents use the same signal, as in the equilibrium for equal priors, then by the monotone likelihood ratio property the firm's expected payoff $\mathbb{E}[s_i|s_i + \varepsilon_i]$ is non-decreasing in the score, so selecting the higher score is optimal even without commitment. For unequal priors, and for rules that i.i.d. shocks cannot generate, commitment can bind.

Agents' Equilibrium. Since θ_i is unobservable to agent i, he should simply maximize the ex-ante probability of being selected when choosing a signal structure $G_i(s_i)$. Hence, an equilibrium of the agents' informational game is a tuple (G_1^*, G_2^*), where

$$G_i^* \in \arg\max_{G_i \in \mathcal{G}(p_i)} \int_0^1 \int_0^1 v(s_i - s_j) dG_j^*(s_j) dG_i(s_i).$$

For any priors and selection rule, an equilibrium exists by Sion's minimax theorem, since the game is constant-sum, as $v(x) - 0.5$ is odd, payoffs are linear and continuous in G_i, and $\mathcal{G}(p_i)$ is a compact convex subset of a topological vector space (Proposition 1 in Kleiner et al. (2021)). Moreover, all equilibria are pairs of max-minimizer strategies

and form a product set.

Interpretation. The model is deliberately abstract and says nothing about a price: agent i's type θ_i is simply the value the firm realizes from picking i, with no wage or other transfer attached to that choice. This fits any setting in which a decision-maker allocates one indivisible prize between two competing options and cannot pay more to secure a better one. My running example is **internal capital budgeting**: a firm's capital committee (*the firm*) must fund exactly one of two project proposals this cycle, and each project's sponsor (*agent i*) controls how much hard evidence – the scope of a pilot, the rigor of a feasibility study, which numbers are led with – to place in front of the committee. The same primitives fit a **grant panel** choosing between two proposals, and **grading**, as in Boleslavsky and Cotton (2015): a university chooses a grading policy for a graduate who competes for a single place and is evaluated by an admissions office. In this reading, the grading policy is the signal, the prior p_i is the strength of the school's cohort, and the noise is the office's judgment of what a transcript leaves out, such as interviews and essays; the school sets its policy before it learns how good its graduates are, which is the commitment the model assumes. I use *the firm* and *agent* as generic labels and switch between these readings wherever a concrete example helps. The binary prior simplifies the analysis and sharpens the results; Section 5 relaxes it.

Under any of these readings, the signal $G_i(s_i)$ is the design of the hard information about the project, applicant or student – the scope and pre-registration of a pilot study, the preliminary results in a grant proposal, a school's grading policy – summarized by the distribution of the posterior s_i that the agent is good. Equivalently, as in Boleslavsky and Cotton (2015), each type generates a more primitive random report – a grade, a pilot outcome, a preliminary result – whose distribution satisfies the monotone

likelihood ratio property, so that, by Bayes' rule, the posterior increases in the report. I read θ_i as fit with what the firm needs, which an agent can hardly assess; what he controls, before any result exists, is how much evidence there will be. If he knew more than the firm, this would be a signaling problem, which I do not study.

The selection rule $v(x)$ translates the two signals into the ex-ante probability of being picked when the firm selects the agent with the higher score. The shocks ε_i that generate $v(x)$ are imperfections of the kind documented in the introduction: idiosyncrasies of the evaluator that are uncorrelated with the substance of the signal and with the firm's realized payoff, but still move her decision. An agent may also have outcome-relevant characteristics that are uncorrelated with his signal and influence the firm's decision, but I set these aside for two reasons. First, the agents care only about the selection rule, not about whether the shock also affects the firm's payoff. Second, comparing the firm's utility across selection rules is easier to interpret when the shock is pure evaluation noise.

3. Equilibrium under Well S-Shaped Selection Rules

In this section, I study the agents' game under the additional assumption that the selection rule $v(x)$ has a concave density with a unique mode. Subsection 3.1 shows that this restricts every best reply to a bad news signal. Subsection 3.2 characterizes the equilibrium: for equal priors, it is unique and governed by a single statistic of the rule, and for unequal priors, it takes one of four forms, in which the favorite plays safe and the underdog gambles. Subsections 3.3 and 3.4 study how the equilibrium responds to the priors and to the rule: how much information the agents reveal, and whether the firm gains from better agents.

3.1 Best replies

I begin with the key definition of this and the following sections.

Definition 1. A function $v : [-1, 1] \to [0, 1]$ is called *well S-shaped* if $v(x) - 0.5$ is odd and increasing, and $v'(x)$ is concave on $[-1, 1]$ and uniquely maximized at $x = 0$.

Since $v(x) - 0.5$ is odd, $v'(x)$ is even, so a concave $v'(x)$ is always maximized at $x = 0$. Requiring this maximum to be unique only rules out densities that are flat around 0, including the linear selection rule. Without this requirement, the equilibrium may fail to be unique: if $v(x)$ is linear on some $[-b, b]$, then for priors (p, p) with $p \leq \frac{b}{2}$, both $(\{p\}; \{p\})$ and $(\{0, 2p\}; \{0, 2p\})$ are equilibria. If $v(x)$ is generated by ε_i, ε_j, well S-shapedness means that ξ has a concave density on $[-1, 1]$ with a unique mode, which holds for the distributions of Example 1 if their tails are wide enough.

Example 2. For $v(x)$ generated by i.i.d. Gumbel or normal shocks, there is a threshold $\overline{V}$ s.t. $v(x)$ is well S-shaped if $\mathrm{Var}[\varepsilon_i] \geq \overline{V}$. So is the rule generated by i.i.d. $\varepsilon_i \sim U[0, u]$ with $u \geq 1$, although its triangular density is linear on $[0, 1]$.

Every result for equal priors holds on the whole well S-shaped class, including the firm's best rule, generated by $U[0, 1]$ shocks, whose density is linear next to its mode. For unequal or non-binary priors, uniqueness also requires that $v'(x)$ is *not linear near* 0, i.e., not linear on any $[0, \varepsilon]$, which makes every best reply unique (Lemma 1).

Let $\overline{v}(s_i, G_j) := \int_0^1 v(s_i - s_j) dG_j(s_j)$, often written $\overline{v}(s_i)$, be the *value function* of agent i. By choosing a signal G_i, agent i solves a Bayesian persuasion problem as in Kamenica and Gentzkow (2011):

$$\max_{G_i \in \mathcal{G}(p_i)} \int_0^1 \overline{v}(s_i, G_j) dG_i(s_i) \tag{BP}$$

I call G_i *an optimal signal* for $(\overline{v}(s_i), p_i)$ if it solves (BP), and *the optimal signal* if, in addition, no other G_i solves it.

To find an optimal signal, I use the *concavification* $\overline{v}^{cv}(s_i)$ of $\overline{v}(s_i)$ for a fixed G_j, the pointwise lowest concave function majorizing $\overline{v}(s_i)$. Then G_i is an optimal signal for $\overline{v}(s_i)$ if and only if it is optimal for $\overline{v}^{cv}(s_i)$ and its support is contained in the *concavification support* $\text{supp}(\overline{v}^{cv}) := \{s_i : \overline{v}(s_i) = \overline{v}^{cv}(s_i)\}$. For the concave $\overline{v}^{cv}(s_i)$, this is simple: if it is linear on some interval around p_i, any $G_i \in \mathcal{G}(p_i)$ supported in that interval is optimal; otherwise, the optimal signal has $\text{supp}(G_i) = \{p_i\}$ (pooling), see Kamenica and Gentzkow (2011).

A well S-shaped $v(x)$ helps because if $v'(x)$ is concave on $[-1, 1]$, then $\overline{v}'(s_i)$ is concave on $[0, 1]$ for every G_j, which gives $\overline{v}(s_i)$ the following shape.

Definition 2. $\overline{v}(s)$ is called *S-shaped* on $[a, b]$, if there exists some $c \in [a, b]$ s.t. $\overline{v}(s)$ is convex on $[a, c]$ and concave on $[c, b]$.

This is the standard definition of an S-shaped curve. As $v(x)$ is continuously differentiable, so is $\overline{v}(s_i)$, and then $\overline{v}(s_i)$ is S-shaped if and only if $\overline{v}'(s_i)$ is quasi-concave. Since $\overline{v}'(s_i)$ is even concave, $\overline{v}(s_i)$ is S-shaped for every well S-shaped $v(x)$ and every G_j. As quasi-concave functions are not closed under addition, there is no simple extension of the well S-shaped class for which $\overline{v}(s_i)$ is always S-shaped. Characterizing the equilibrium, however, needs this shape only against the rival's equilibrium signals. Using this, Proposition 17 in Appendix D.2 goes beyond the class, giving a weaker sufficient condition for a unique bad news equilibrium, though one that is less explicit.

So under a well S-shaped $v(x)$, an agent's marginal gain from a higher signal (posterior mean) is concave, and against any signal G_j, $\overline{v}(s_i)$ has at most two regions of different convexity. Hence its concavification consists of at most two parts, which strongly restricts the equilibrium signals, characterized in Lemma 1 (ii).

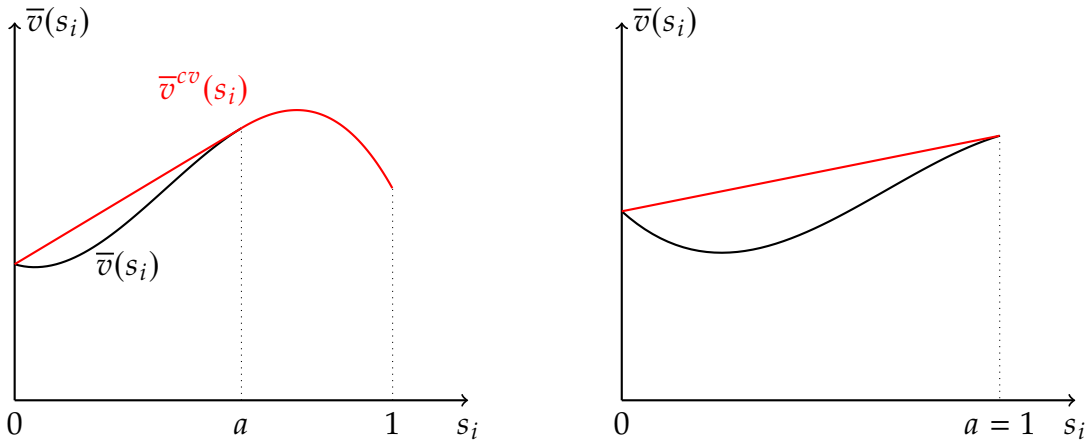
Lemma 1. *If $v(x)$ is well S-shaped, then*

(i) $\overline{v}(s_i, G_j)$ is S-shaped on $[0, 1]$ for all $p_j \in (0, 1)$ and $G_j \in \mathcal{G}(p_j)$.

(ii) An optimal signal for $(\overline{v}(s_i), p_i)$ *is* $\{0, a\}$ *if* $p_i < a$ *and* $\{p_i\}$ *otherwise, where* $a = \max \arg\max_{s_i \in [0,1]} slp(s_i)$, $slp(s_i) := \frac{\overline{v}(s_i) - \overline{v}(0)}{s_i}$ *and* $slp(0) := \overline{v}'(0)$. *If* $v'(x)$ *is not linear near* 0, *it is the optimal signal.*

Remark 1. I identify a signal G_i with its support whenever $|\operatorname{supp}(G_i)| \leq 2$.

Figure 1: Illustration of Lemma 1 (ii)



(a) $\operatorname{supp}(G_i) = \{0, a\}$ or $\{p_i\}$ (b) $\operatorname{supp}(G_i) = \{0, 1\}$

This is not the image of $\overline{v}(s_i)$, *but of* $\overline{v}(s_i) - bs_i$ *for some* $b > 0$ *for clarity. Since* $cav(f(x) + bx) = cav(f(x)) + bx$ *for all* x, b, *this does not influence the optimal signal.*

Lemma 1 thus restricts an agent's optimal signal to a very limited class, the *bad news* signals: the agent reveals the low type with some probability and otherwise sends a pooled signal, with full revelation ($\{0, 1\}$) and pooling ($\{p_i\}$) as the edge cases. Binary support itself is not special: two-point signals are the pure strategies of an agent, every other signal is their mixture, and against any continuous value function some signal with at most two support points is a best reply. What a well S-shaped $v(x)$ adds is that, unless pooling is optimal, such a best reply has 0 in its support, and that it is unique if $v'(x)$ is not linear near 0. Moreover, the point a depends only on the rival's signal, and the agent's own prior only determines whether he pools at p_i or plays $\{0, a\}$. The concavity of the density is what makes agents concede bad news: under the

mirror-image, reverse S-shaped rules of Appendix D.1, they reveal good news instead.

The first claim was shown above. For the second, the concavification of an S-shaped $\overline{v}(s_i)$ coincides with the secant from $\overline{v}(0)$ to $\overline{v}(a)$ on $[0, a]$ and with $\overline{v}(s_i)$ on $[a, 1]$. Hence $\{0, a\}$ is an optimal signal if $a > p_i$, and $\{p_i\}$ if $a \leq p_i$, and if $v'(x)$ is not linear near 0, these are **the** optimal signals. Figure 1 illustrates this claim.

By the Bayesian constraint, positive scaling of $v(x)$ and adding a linear function to it do not change the concavification support, and hence do not change equilibria for any (p_1, p_2). I represent each such equivalence class of selection rules by the normalization

$$\tilde{v}(x) := \frac{v(x) - 0.5 - x \cdot v'(0)}{|v(1) - 0.5 - v'(0)|}$$

It satisfies $\tilde{v}(0) = \tilde{v}'(0) = 0$ and $\tilde{v}(1) = -1$, which simplifies the exposition, and it is well defined for every well S-shaped $v(x)$: as $v'(x)$ is uniquely maximized at $x = 0$, $v(1) - 0.5 = \int_0^1 v'(x)dx < v'(0)$.

The key characteristic of $\tilde{v}(x)$ is its elasticity $\mathcal{E}_{\tilde{v}}(x) = \frac{x\tilde{v}'(x)}{\tilde{v}(x)}$. Since $\tilde{v}(0) = 0$, $\mathcal{E}_{\tilde{v}}(a)$ is the ratio of the slopes of the tangent to $\tilde{v}(x)$ at $x = a$ and of the secant from 0 to a, so it measures how curved $\tilde{v}(x)$ is on $[0, a]$. Hence it governs the curvature of $\overline{v}(s_i)$ and the location of the support point a from Lemma 1, as the results below show.

3.2 Equilibrium

3.2.1 Full revelation

A natural first question is whether the firm can ever get the first best information, full revelation by both agents, in equilibrium. Under a well S-shaped $v(x)$, the answer is simple: both priors must be high enough.

Definition 3. For a given well S-shaped $v(x)$, let $\overline{p} := 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(1)} = 1 - \frac{1}{|\tilde{v}'(1)|}$.

Proposition 1. *Let $v(x)$ be well S-shaped. Then both agents playing full revelation is an equilibrium if and only if $p_1, p_2 \geq \overline{p}$, and then it is the unique equilibrium unless $v'(x)$ is linear near 0 and $p_i > p_j = \overline{p}$ for some $i \neq j$.*

Competing with a sufficiently good rival induces an agent to reveal fully, and the proof of the condition $p_1, p_2 \geq \overline{p}$ is built on exactly that. If agent j plays $\{0, 1\}$, the normalized value function of agent i is

$$\overline{v}(s_i) = (1 - p_j)\tilde{v}(s_i) + p_j\tilde{v}(s_i - 1) = (1 - p_j)\tilde{v}(s_i) - p_j\tilde{v}(1 - s_i).$$

By Lemma 1, it is S-shaped, so $\{0, 1\}$ is optimal if and only if the secant from 0 to 1 is weakly flatter than the tangent at $x = 1$ (see Figure 1b):

$$\overline{v}(1) - \overline{v}(0) \leq \overline{v}'(1) \Leftrightarrow \tilde{v}(1) \leq (1 - p_j)\tilde{v}'(1) \Leftrightarrow p_j \geq 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(1)}.$$

For uniqueness, as the game is constant-sum, all equilibria form a product set, so any signal played in equilibrium is a best reply to full revelation. An S-shaped $\overline{v}(s_i)$ can touch its secant from 0 to 1 at some $s \in (0, 1)$ only by coinciding with it on $[s, 1]$, which non-linearity of $v'(x)$ near 0, or a tangent at 1 strictly steeper than the secant ($p_j > \overline{p}$), rules out, and then full revelation is the only best reply. For $p = \overline{p}$, the best reply need not be unique, but uniqueness follows from Theorem 1.

Remark 2. Uniqueness can fail when both conditions of the exception hold. For $v(x) = 0.5 + x - 0.5x|x|$, $v'(x)$ is linear near 0 and $\overline{p} = 0.5$; for $(p_1, p_2) = (0.75, 0.5)$, agent 1's value function against $\{0, 1\}$ is linear, and $\{0, 1\}$ remains a best reply to any $G_1 \in \mathcal{G}(0.75)$, since $\mathbb{E}_{G_1}[s_1^2] \geq 0.5$. So $(G_1; \{0, 1\})$ is an equilibrium for every such G_1.

Proposition 1 contrasts with the noiseless benchmark[2] of Boleslavsky and Cotton

[2]The noiseless rule $v(x) = 0.5 + 0.5 \cdot \text{sign}(x)$ is not a selection rule, as it is discontinuous at 0; Appendix B.13 gives its equilibria.

(2015), in which schools never use a fully informative grading policy. This is not a feature of well S-shaped rules alone: Section 4 shows that full revelation can be reached by selection rules far beyond this class.

Since $\tilde{v}'(x)$ is concave with $\tilde{v}'(0) = 0$, and $\tilde{v}(x)$ is the (oriented) area under it, $\mathcal{E}_{\tilde{v}}(x) \geq 2$ for every well S-shaped $v(x)$, and $\mathcal{E}_{\tilde{v}}(1) = 2$ only if $\tilde{v}(x) = -x^2$ on $[0,1]$ (Lemma 5). Hence $\overline{p} \geq 0.5$, with equality only in that case: under a well S-shaped rule, $(\{0,1\};\{0,1\})$ is never an equilibrium if $p_1 < 0.5$ or $p_2 < 0.5$, and, unless $\tilde{v}(x) = -x^2$ on $[0,1]$, also if $p_1 = 0.5$ or $p_2 = 0.5$. So full revelation requires both agents to be more likely good than bad.

3.2.2 Equal priors

For equal priors, Lemma 1 and Proposition 1 characterize the equilibrium on the well S-shaped class. Unless it is full revelation, it is a bad news equilibrium where the probability of conceding bad news ($s_i = 0$) is inversely proportional to the elasticity of $\tilde{v}(x)$ at the pooled signal a.[3]

Theorem 1. *Let $v(x)$ be well S-shaped and the priors (p,p). If $p \geq \overline{p}$, the unique equilibrium is $(\{0,1\};\{0,1\})$. If $p < \overline{p}$, the unique equilibrium is $(\{0,a\};\{0,a\})$, where a solves $p = a(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)})$, and $\mathbb{P}(s_i = 0) = \frac{1}{\mathcal{E}_{\tilde{v}}(a)}$.*

The intuition is simple. Against $\{0,a\}$, an agent's normalized value function is

$$\overline{v}(s_i) = \mathbb{P}(s_i = 0)\tilde{v}(s_i) + \mathbb{P}(s_i = a)\tilde{v}(s_i - a) = \frac{a-p}{a}\tilde{v}(s_i) + \frac{p}{a}\tilde{v}(s_i - a),$$

where the second term $\tilde{v}(s_i - a) = -\tilde{v}(a - s_i)$ reflects $\tilde{v}(s_i)$ on $[0,a]$. For $\{0,a\}$ to be a best reply, the secant from $\overline{v}(0)$ to $\overline{v}(a)$ must be tangent at $s_i = a$. Since $\tilde{v}'(0) = 0$, the

[3]The characterization extends to costly signals. If agent i pays $\kappa \operatorname{Var}_{G_i}[s_i]$, the equilibrium is still unique and of the same form, except that it is pooling for a high enough cost; what matters is the cost relative to the curvature of the rule, $\lambda = \frac{\kappa}{|v(1)-0.5-v'(0)|}$ (Proposition 15 in Appendix B.12).

second term has zero derivative at a, so $\overline{v}'(a) = \frac{a-p}{a}\tilde{v}'(a)$, while the secant has the slope $\frac{\tilde{v}(a)}{a}$, the same for both terms. Hence, in equilibrium,

$$\frac{a-p}{a}\tilde{v}'(a) = \frac{\tilde{v}(a)}{a} \Leftrightarrow \frac{a-p}{a} = \frac{1}{\mathcal{E}_{\tilde{v}}(a)} \Leftrightarrow p = a\left(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\right).$$

Such a is unique, continuous and strictly increasing in p: the pooled signal rises with the prior. Indeed, $p = a - \frac{\tilde{v}(a)}{\tilde{v}'(a)}$ is the point where the tangent to $\tilde{v}(x)$ at $x = a$, $y = \tilde{v}(a) + (x-a)\tilde{v}'(a)$, crosses the x-axis. Since $\tilde{v}(x)$ is strictly concave and decreasing on $[0,1]$ with $\tilde{v}(0) = 0$, this point is continuous and strictly increasing in a, and so is its inverse $a(p)$. As $\mathbb{P}(s_i = 0) = \frac{a-p}{a}$, this also gives the inverse elasticity result.

Uniqueness needs one more step, since $\{0, a\}$ need not be the unique best reply to itself: if $\tilde{v}(x)$ is quadratic on $[0, 2p]$, every signal supported there is a best reply to $\{0, 2p\}$. But then $\{0, 2p\}$ is a best reply to none of them except itself, so, as equilibria form a product set, the equilibrium is still unique.

3.2.3 Pooling

Pooling is the other edge case of a bad news signal, and, unlike full revelation, it is never played by both agents.

Lemma 2. *Let $v(x)$ be well S-shaped. Then $(\{p_1\}; \{p_2\})$ is never an equilibrium.*

Against pooling, an agent's value function $v(s_i - p_j)$ is convex on $[0, p_j]$ and concave on $[p_j, 1]$, so pooling is a best reply only for the agent with the strictly higher prior. Still, pooling can be approached in the limit: for $\tilde{v}_n(x) = -x^n$ on $[0,1]$, Theorem 1 gives $a = \frac{np}{n-1} \to p$. More generally, $a \to p$ whenever the elasticity of $\tilde{v}(x)$ diverges uniformly on $(0,1)$; Lemma 7 in Appendix A states this formally.

3.2.4 Unequal priors

For unequal priors, let $p_2 \leq p_1$, so that agent 1 is the favorite and agent 2 the underdog, and let $v'(x)$ be not linear near 0. By Lemma 1, every best reply is unique and takes one of three forms: pooling $\{p_i\}$, full revelation $\{0, 1\}$, or bad news $\{0, c\}$ with $p_i < c < 1$. In general, an agent's best reply depends on his rival's whole signal, but in the following example it depends only on the rival's prior, which makes the equilibrium explicit.

Example 3. Let $\tilde{v}(x) = -x^3$. Agent i's best reply is $\{0, \min(\frac{3p_j}{2}, 1)\}$, or $\{p_i\}$ if $p_i \geq \frac{3p_j}{2}$. So for $p_2 \leq p_1$, the unique equilibrium is $(\{0, 1\}; \{0, 1\})$ if $p_2 \geq \frac{2}{3}$, and otherwise

(i) $(\{p_1\}; \{0, 1\})$ if $p_1 \geq \max(\frac{3p_2}{2}, \frac{2}{3})$;

(ii) $(\{p_1\}; \{0, \frac{3p_1}{2}\})$ if $\frac{3p_2}{2} \leq p_1 < \frac{2}{3}$;

(iii) $(\{0, \frac{3p_2}{2}\}; \{0, 1\})$ if $\frac{2}{3} \leq p_1 < \frac{3p_2}{2}$;

(iv) $(\{0, \frac{3p_2}{2}\}; \{0, \frac{3p_1}{2}\})$ if $p_1 < \min(\frac{3p_2}{2}, \frac{2}{3})$.

The example shows what asymmetry does. Each agent would ideally pool at one and a half times his rival's prior. When the priors are close and low enough, both concede bad news, and each pooled signal is set by the rival: the underdog, facing the better rival, pools higher and so reveals more. As the favorite's lead grows, the signal at which he would pool falls below his own prior, so once $p_1 \geq \frac{3p_2}{2}$ he pools at his prior. The underdog, facing a rival who reveals nothing, gambles harder and reveals fully once $p_1 \geq \frac{2}{3}$. In the example, the $\frac{2}{3}$ threshold is the same whether the favorite pools or concedes bad news, because an agent's best reply depends only on his rival's prior. In general, it depends on the rival's signal, so the underdog's threshold for full revelation differs, and the boundaries are no longer multiples of the priors. The same logic still produces the same four regions.

Proposition 2. *Let $v(x)$ be well S-shaped with $v'(x)$ not linear near 0, and let $p_1, p_2 \in (0,1)$ with $p_2 \leq p_1$. Then the equilibrium is unique and continuous in (p_1, p_2). It is $(\{0,1\};\{0,1\})$ if $p_2 \geq \overline{p}$, and otherwise it is one of*

(i) $(\{p_1\};\{0,1\})$*: the favorite pools, and the underdog reveals fully;*

(ii) $(\{p_1\};\{0,b\})$ *with $p_2 < b < 1$: the favorite pools, and the underdog concedes bad news;*

(iii) $(\{0,a\};\{0,1\})$ *with $p_1 < a$: the favorite concedes bad news, and the underdog reveals fully;*

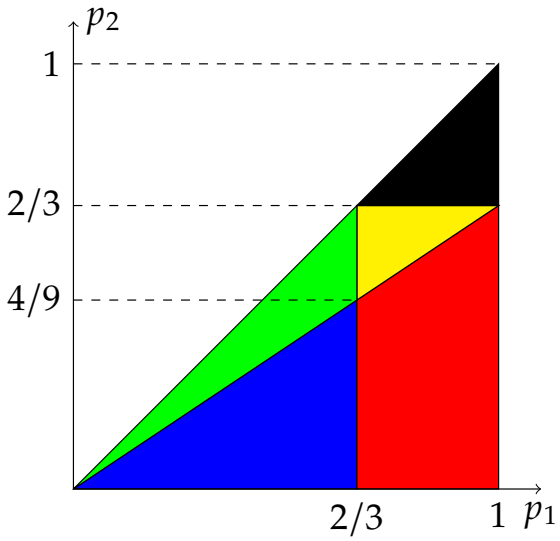
(iv) $(\{0,a\};\{0,b\})$ *with $p_1 < a \leq b < 1$: both concede bad news, and $a = b$ if and only if $p_1 = p_2$.*

In Example 3, regions (i)–(iv) are exactly the cases listed there, and Figure 2 draws them: the favorite pools if and only if his lead is large, $p_1 \geq \frac{3p_2}{2}$, and the underdog reveals fully if and only if the favorite is strong, $p_1 \geq \frac{2}{3}$. So an agent who is nearly sure to win plays safe, while one who is likely to lose gambles; under the reverse S-shaped rules of Appendix D.1, the roles are reversed. For a general rule, these two thresholds depend on the rival's whole signal rather than on his prior, so the regions have curved boundaries, given in Appendix A.7. Every pair of forms except pooling by both agents (Lemma 2) is played for some priors, and the underdog never pools at his prior. Non-linearity of $v'(x)$ near 0 cannot be dropped: without it, the equilibrium need not be unique, and signals need not take these forms (Remark 6 in Appendix A.7).

3.3 Informativeness of the signals

This subsection studies how the prior and the selection rule affect the informativeness of the signals in the symmetric bad news equilibrium, which is unique for every well S-shaped $v(x)$ by Theorem 1, and, for unequal priors, how an agent's signal depends on his rival's prior.

Figure 2: Equilibria for unequal priors in Example 3, $\tilde{v}(x) = -x^3$ on $[0, 1]$



Colors are for the corresponding equilibrium type: black – $(\{0,1\};\{0,1\})$*; red* – $(\{p_1\};\{0,1\})$*; green* – $(\{0,a\};\{0,b\})$*; blue* – $(\{p_1\};\{0,b\})$*; yellow* – $(\{0,a\};\{0,1\})$.

3.3.1 Informativeness and change in prior

For a fixed p, a higher a means a more informative bad news signal. In equilibrium, a rises with p, but this does not imply a more informative signal: if a rises only a little relative to p, the signal may become less informative. To compare signals for different a and p, I represent them as experiments $\pi_i : \{0,1\} \to [0,1]$ and use the Blackwell order (see Blackwell (1953)).

A bad news experiment sends $s_i = 0$ or $s_i = a$ if $\theta_i = 0$, and always $s_i = a$ if $\theta_i = 1$, so it is characterized by the conditional probability of revealing the bad type, $\frac{\mathbb{P}(s_i=0)}{\mathbb{P}(\theta_i=0)}$. A higher probability gives a more informative experiment, since a lower one is obtained by garbling. Hence, I say that agent i *reveals more for* p_{i1} *than for* p_{i2} if $\frac{\mathbb{P}(s_{i1}=0)}{1-p_{i1}} > \frac{\mathbb{P}(s_{i2}=0)}{1-p_{i2}}$ for the corresponding equilibrium strategies G_{i1}, G_{i2}. By Theorem 1, this measure depends on a single variable:

$$\mathbb{P}(s_i = 0) = \frac{1}{\mathcal{E}_{\tilde{v}}(a)} \text{ and } p = a\left(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\right) \text{ hence } \frac{\mathbb{P}(s_i = 0)}{1-p} = \frac{1}{1 + (\mathcal{E}_{\tilde{v}}(a) - 1)(1-a)}.$$

This gives the following lemma.

Lemma 3. *Let $v(x)$ be well S-shaped and $(\{0, a\}; \{0, a\})$ the equilibrium for $p \in (0, \overline{p})$. Then informativeness strictly increases in the prior at p if and only if $(\mathcal{E}_{\tilde{v}}(x) - 1)(1 - x)$ strictly decreases at $x = a$.*

Corollary 1. *Let $v(x)$ be well S-shaped with $\mathcal{E}_{\tilde{v}}(x)$ non-increasing. Then in equilibrium agents reveal more for p_2 than for p_1 whenever $0 < p_1 < p_2 \leq \overline{p}$.*

Some algebra shows that Corollary 1 applies to the natural examples.

Example 4. For $v(x)$ generated by i.i.d. Gumbel or normal shocks with $\text{Var}[\varepsilon_i] \geq \overline{V}$, $\mathcal{E}_{\tilde{v}}(x)$ is non-increasing, so in equilibrium agents reveal more as p increases.

However, the elasticity of a well S-shaped rule can vary flexibly, and ex-ante better agents may then reveal less.

Example 5. There exists a well S-shaped $v(x)$ with $\tilde{v}(x) = -\frac{x^2}{16(1-x)^2}$ on $[0, 0.5]$ s.t. in equilibrium agents reveal less for p_2 than for p_1 if $0 < p_1 < p_2 \leq 0.375$.

3.3.2 Informativeness and change in selection rule

Next, fix the prior and change the selection rule. If $\tilde{v}(x)$ has a constant elasticity, a higher elasticity lowers $\mathbb{P}(s_i = 0)$ for a fixed p, and hence a, making the signal less informative. The next corollary extends this argument to any two selection rules.

Corollary 2. *Let $v_e(x)$ and $v_n(x)$ be well S-shaped, let the priors be (p, p) with $p \in (0, \min(\overline{p}_e, \overline{p}_n))$, and let $(\{0, a_i\}; \{0, a_i\})$ be the equilibrium for v_i, $i \in \{e, n\}$. If $\tilde{v}_e(x)$ is more elastic than $\tilde{v}_n(x)$, i.e. $\mathcal{E}_{\tilde{v}_e}(x) > \mathcal{E}_{\tilde{v}_n}(x)$ for all $x \in (0, 1)$, then $a_e < a_n$.*

So a more elastic selection rule suppresses information, and within the well S-shaped class, the most informative signal is induced by the rules with $\mathcal{E}_{\tilde{v}}(x) = 2$, i.e. $\tilde{v}(x) = -x^2$ on $[0, 1]$. For the natural examples, this reads intuitively in terms of noise.

Example 6. For $v(x)$ generated by i.i.d. Gumbel or normal shocks, $\mathcal{E}_{\tilde{v}}(x)$ increases in $\mathrm{Var}[\varepsilon_i]$ on $[\overline{V}, +\infty)$ for every $x \in (0,1)$, so noisier shocks lead to less information in equilibrium.

3.3.3 Informativeness and the rival's prior

With unequal priors, in regions (ii) and (iii) of Proposition 2, an agent's bad news signal depends only on his rival's prior. More generally, it is the rival who forces an agent to reveal.

Proposition 3. *Let $v(x)$ be well S-shaped with $v'(x)$ not linear near 0. Then the equilibrium signal of agent i is non-decreasing in p_j, and strictly increasing whenever agent i plays $\{0, a\}$ with $p_i < a < 1$.*

An agent's own prior usually has the opposite effect: in Example 3, his high signal does not depend on p_i, so he reveals less as p_i increases. This effect need not be monotone, though (Example 9 in Appendix A.7).

3.3.4 Selection reversals

Informativeness also governs how often noise makes the firm select the agent with the lower posterior. For $p < \overline{p}$, posteriors differ with probability $\frac{2}{\mathcal{E}_{\tilde{v}}(a)}\left(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\right)$, and the firm then selects the agent with $s_i = 0$ with probability $1 - v(a)$. If $\mathcal{E}_{\tilde{v}}(x)$ is non-decreasing, the first factor does not increase in p, as $\frac{1}{\mathcal{E}_{\tilde{v}}(a)} \leq \frac{1}{2}$, and the second decreases, so such reversals become rarer. Under the condition of Corollary 1, which makes informativeness increase in p, the two factors move in opposite directions, and either can dominate. For the two classes from Example 1, which satisfy it by Example 4, a direct computation shows that reversals still become rarer, but for $v(x) = 0.5 + 0.6x - 0.1e^{5-\frac{5}{x}}$ on $[0,1]$, the probability of a reversal increases in p for $p < 0.36$.

3.4 Better agents and the firm

It is natural to expect the firm to benefit from ex-ante better agents. But better agents do not always reveal more, and this effect could outweigh the higher prior.

For a well S-shaped $v(x)$ and priors (p, p), let (G, G) be the unique equilibrium and $u_F(p) := \mathbb{E}_G[s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1)]$ the firm's expected utility.

Proposition 4. *Let $v(x)$ be well S-shaped. Then $u_F(p)$ is strictly increasing in p.*

So, despite possibly decreasing informativeness, a firm that uses a well S-shaped selection rule is never harmed by ex-ante better agents. The result is not immediate: the equilibrium signals for different priors need not be ranked by first-order stochastic dominance or in the Blackwell order, and under a noisy selection rule the firm's payoff need not be monotone or convex in the signals. However, the change from p to $p' > p$ splits into two steps within the bad news class. First, keeping the prior p, the pooled signal rises from $a(p)$ to $a(p')$: this spread reveals the bad type more often and widens the gap between the signals, so the firm selects the worse agent less often. Second, keeping the pooled signal $a(p')$, the prior rises to p', which moves mass from 0 to $a(p')$, and on $\{0, a\}$ the firm's payoff is monotone. Both steps benefit the firm. This logic relies on the pooled signal moving away from the revealed one as p increases. Under the reverse S-shaped selection rules of Appendix D.1, it moves toward it, and there exist $p_1 < p_2$ s.t. $u_F(p_1) > u_F(p_2)$ in the unique equilibrium.

For unequal priors, this monotonicity extends to every region of Proposition 2 except region (iv), where both agents concede bad news: there, once the underdog stops revealing fully, his pooled signal can fall as his prior rises, and the firm's utility with it (Proposition 13 and Example 10 in Appendix A.7). For the agents, a better own prior always helps and a better rival always hurts (Lemma 8 in Appendix A.7).

4. Designing the Selection Rule

This section turns to the normative side: the firm's choice of the selection rule. Subsection 4.1 finds the best rule within the well S-shaped class, and Subsection 4.2 the optimal level of noise beyond it. Both depend on the prior, so Subsection 4.3 asks which rules implement full revelation robustly, and Subsection 4.4 compares rules of the signal gap with a benchmark designer who may use any rule.

4.1 The best well S-shaped rule

I now depart from an exogenous $v(x)$: for a fixed p, the firm chooses the selection rule to maximize $u_F(p)$, for instance, by choosing between committee structures, evaluation protocols or review panels. By changing the biases that shape her selection rule, she changes the equilibrium signals and her utility.

Two questions arise. First, what is the best equilibrium utility within the well S-shaped class for each p, and which rules attain it? Second, how does it compare with the noiseless rule? The answer to the latter is not obvious: in the noiseless equilibrium of Boleslavsky and Cotton (2015), an agent with some probability uses a uniformly distributed posterior, which is less informative than the binary signal under a well S-shaped rule, so noise trades selection accuracy for more information.

The natural examples show an important pattern.

Example 7. For $v(x)$ generated by i.i.d. Gumbel or normal shocks, $u_F(p)$ decreases in $\mathrm{Var}[\varepsilon_i]$ on $[\overline{V}, +\infty)$ for every $p \in (0, 1)$, so noisier shocks lower the firm's equilibrium utility.

So within each of these classes, the best rule lies at the "edge" of the well S-shaped class: the lowest level of noise that keeps $v(x)$ well S-shaped. The next theorem gives a tight upper bound on the firm's utility over the whole class and the rules that attain it.

Theorem 2. *Let $v(x)$ be well S-shaped. Then*

(i) $u_F(p) \le u_F^{ws}(p) = p + \min(2p, 1) \cdot p(1-p)$.

(ii) If $p \ge 0.5$, *then* $u_F(p) = u_F^{ws}(p)$ *if and only if* $p \ge \overline{p}$ *and* $v(1) = 1$.

(iii) If $p < 0.5$, *then* $u_F(p) = u_F^{ws}(p)$ *if and only if* $v(x) = 0.5 + x - 0.5x|x|$.

Since the first best utility is $u_F^*(p) = 2p - p^2 = p + p(1-p)$, the best well S-shaped rule captures the share $\min(2p, 1)$ of the first best gain over random selection.

Corollary 3. *Let* $v(x) = 0.5 + x - 0.5x|x|$. *Then*

(i) $u_F(p) = u_F^{ws}(p)$ *for all* $p \in (0, 1)$;

(ii) $u_F(p)$ *is strictly higher than under the noiseless rule if and only if* $p > \frac{3-\sqrt{3}}{6} \approx 0.21$.

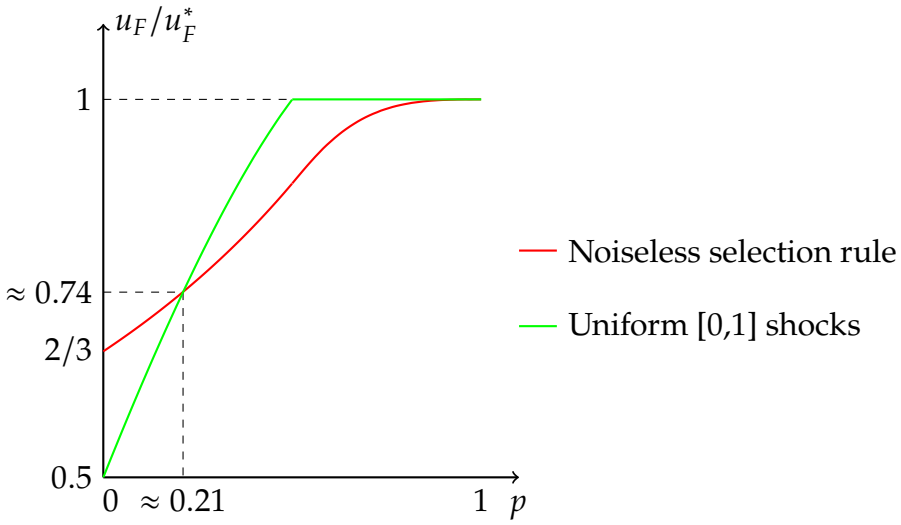


Figure 3: Losses from the best well S-shaped selection rules

So the selection rule $v^*(x) = 0.5 + x - 0.5x|x|$ is optimal within the well S-shaped class for every p, and uniquely so for $p < 0.5$. It has a triangular density and is generated by two independent $\varepsilon_i \sim U[0, 1]$. Its optimality has a simple source. In the equilibrium $(\{0, a\}; \{0, a\})$, the firm's utility is $p + \mathbb{P}(s_1 \ne s_2) \cdot a \cdot \big(v(a) - 0.5\big)$: she obtains

p from a random pick and gains only when the signals differ, by their gap times her excess probability of selecting the higher one. This gain rises with the pooled signal a and, at a given a, with the accuracy $v(a)$, and $v^*(x)$ is best on both counts. It induces the highest possible pooled signal, $a = \min(2p, 1)$, and since a concave density on $[0, 1]$ declines no faster, in proportion, than the triangular density $1 - x$, and its mass there is at most 0.5, $v^*(x)$ is also the most accurate well S-shaped rule at every gap: $v(x) \leq v^*(x)$ on $[0, 1]$. So $v^*(x)$ lies at the far edge of the class, as in the natural examples, and there the firm does not trade sharper selection against more information; this trade-off arises only beyond the edge.

Figure 3 shows the share of the first best utility $u_F^*(p)$ attained by $v^*(x)$ and by the noiseless rule. By Corollary 3, $v^*(x)$ outperforms the noiseless rule if and only if $p > \frac{3-\sqrt{3}}{6} \approx 0.21$, and attains the first best for $p \geq 0.5$: the extra information it induces outweighs the cost of sometimes selecting a bad agent over a good one, most of all at high priors, where it induces full revelation.

4.2 The optimal level of noise

Theorem 2 leaves open what happens beyond the edge of the well S-shaped class, where the noise is too small for the density of the shock difference to be concave. The selection rules generated by i.i.d. $\varepsilon_i \sim U[0, u]$, which have a triangular density on $[-u, u]$, cover both sides of this edge. For $u \geq 1$, $v(x)$ is well S-shaped with $\tilde{v}(x) = -x^2$ on $[0, 1]$, and $u = 1$ gives $v^*(x)$; for $u < 1$, $v'(x)$ is not concave on $[0, 1]$, and as $u \to 0$, $v(x)$ converges pointwise to the noiseless rule. For every u and p, a symmetric equilibrium can be constructed explicitly, and it has binary support if and only if $u \geq 2\min(p, 1 - p)$; for smaller u, each agent randomizes over more than two posteriors.[4] The construction, in Appendix B.4, rests on u-partitions: signals on an interval $[a, b]$ such that, at every

[4]For $u < 1$, I do not prove uniqueness, but the equilibria of the discretized game converge to this one.

point s of $[a, b]$, the rival's signal is as likely to lie within u above s as within u below s. Under the triangular rule, the curvature of an agent's value function at s is proportional to the difference of these two probabilities, so against a u-partition it is linear on $[a, b]$. The next proposition concerns only the informativeness of the signals and the firm's utility.

Proposition 5. *Let $v(x)$ be the CDF of a triangular distribution on $[-u, u]$, $u \in (0, 1]$, and let $u_F(p, u)$ be the firm's utility in the equilibrium (G_u, G_u) constructed in Appendix B.4.*

(i) If $p \leq 0.5$, G_u is strictly increasing in u in the Blackwell order on $(0, 2p]$, and $G_u = \{0, 2p\}$ for $u \in [2p, 1]$. If $p > 0.5$, $G_u = \{0, 1\}$ for $u \in [2 - 2p, 1]$.

(ii) If $p \leq 0.5$, $u_F(p, u)$ is uniquely maximized at $u = 2p$ and strictly decreasing on $[2p, 1]$. If $p > 0.5$, $u_F(p, u)$ equals the first best $u_F^*(p) = 2p - p^2$ if and only if $u \in [2 - 2p, 1]$.

(iii) As $u \to 0$, G_u converges weakly to the equilibrium signal for the noiseless rule.

The proposition considers only $u \leq 1$: for $u \geq 1$, $v(x)$ is well S-shaped with $\tilde{v}(x) = -x^2$ on $[0, 1]$, so the equilibrium signal does not change with u, while $v(x) = 1 - \frac{(u-x)^2}{2u^2}$ decreases in u on $(0, 1]$, and so does the firm's utility.

Part (i) shows that for $p \leq 0.5$, less noise only suppresses information. For $p > 0.5$ and $u < 2 - 2p$, this ranking breaks down. For instance, for $p = \frac{2}{3}$, G_u puts $(\frac{1}{5}, \frac{1}{5}, \frac{3}{5})$ on $\{0, \frac{1}{3}, 1\}$ for $u = \frac{1}{3}$ and $(\frac{4}{21}, \frac{4}{21}, \frac{13}{21})$ on $\{0, \frac{1}{4}, 1\}$ for $u = \frac{1}{2}$. The latter reveals the good type more often but the bad type less often, and since a mean-preserving spread on $[0, 1]$ cannot lower the probabilities of 0 and 1, neither is more informative. The same holds for any two $u \in [1 - p, 2 - 2p)$ if $p \geq \frac{5}{9}$ (see the proof of Proposition 5). Still, on this range, the probability of revealing the good type and the variance increase in u.

Part (ii) shows that the firm's utility is maximized at the least noise at which agents play a binary signal, $\{0, 2p\}$ or $\{0, 1\}$, and less noise, which breaks the binary structure, strictly lowers it. So the optimal level of noise is intermediate: it trades off information

from the agents against accuracy of selection by the firm. Moreover, at least within this family, the firm never benefits from non-binary signals, so going far beyond the well S-shaped class may be unnecessary.

Part (ii) also compares the best rule in this family with the noiseless rule at every prior. For $p \leq 0.5$, the noise $u = 2p$ gives $v(2p) = 1$, so the firm selects the higher of two different signals for sure and noise changes only what agents reveal. Her utility is then $\frac{3p}{2}$, against $\frac{4p}{3}$ under the noiseless rule and $u_F^{ws}(p)$ under the best well S-shaped rule; as $p \to 0$, this is three quarters of the first best, against two thirds. For $p > 0.5$, every $u \in [2 - 2p, 1]$ attains the first best. So the right amount of noise beats the noiseless rule at every prior, whereas $v^*(x)$ does so only for $p > 0.21$.

Part (iii) shows that there is no discontinuity at zero noise: as $u \to 0$, $v(x)$ converges pointwise to $0.5 + 0.5 \cdot \text{sign}(x)$, and the equilibrium to the noiseless one of Boleslavsky and Cotton (2015). Simulations show the same for logistic and normal noise.

Two caveats, however, point to robust design. First, the optimal noise is tied to the prior: for $p \leq 0.5$, more noise than $u = 2p$ blurs selection and less breaks the binary signals, so a firm that misjudges p loses either way. Second, for $u < 1$ I do not prove that the equilibrium is unique, so this comparison rests on the equilibrium of Proposition 14. A firm that does not know the priors exactly thus needs a design that is robust to them, which is the subject of Subsection 4.3.

4.3 Robust design

Subsection 4.1 found the best well S-shaped rule, which outperforms the noiseless rule for sufficiently high priors and attains the first best for all priors $p \geq 0.5$. The next question is which $v(x)$ maximizes the firm's utility if she can choose any rule of the signal gap, and which properties of $v(x)$ are necessary or sufficient for optimality.

I say that $v(x)$ is *revelation-accurate* if $v(1) = 1$, i.e. whenever agent i is revealed to be

good and agent j bad ($s_i = 1$, $s_j = 0$), the firm selects agent i with probability 1. This is a simple necessary condition for optimality: if $v(1) < 1$, an affine modification of $v(x)$ leads to the same equilibria but gives the firm a strictly higher utility.

I also say that $v(x)$ *implements a fully revealing equilibrium (uniquely) for* (p_1, p_2) if, for these priors, full revelation by both agents is an (the unique) equilibrium.

The firm's maximal equilibrium utility[5] under a given selection rule and her first best utility, both for symmetric priors, are defined as follows.

Definition 4. For a selection rule $v(x)$ and priors (p, p), let E be the set of all equilibria. Then the maximal expected utility of the firm among all equilibria is $\overline{u}_F(p) := \max_{(G_1^*, G_2^*) \in E} \mathbb{E}_{G_1^*, G_2^*}[s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1)]$.

Definition 5. Let $\mathcal{V}$ be the set of all selection rules. Then the first best expected utility of the firm is $u_F^*(p) := \sup_{G_1, G_2 \in \mathcal{G}(p), v \in \mathcal{V}} \mathbb{E}_{G_1, G_2}[s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1)]$.

The first result shows that, for symmetric agents, a fully revealing equilibrium and revelation accuracy are not only sufficient, but also necessary for the first best.

Lemma 4. *For every selection rule $v(x)$, $\overline{u}_F(p) \leq u_F^*(p) = 2p - p^2$, with equality if and only if $v(x)$ implements a fully revealing equilibrium for (p, p) and is revelation-accurate.*

The proof is simple. For priors (p, p),

$$u_F^*(p) = \max_{G_1, G_2 \in \mathcal{G}(p)} \mathbb{E}_{G_1, G_2}[\max(s_1, s_2)].$$

As $\max(s_1, s_2)$ is convex and piecewise linear, the maximum is attained if and only if at least one agent plays full revelation. As all equilibria form a symmetric product set, if full revelation and some other signal form an equilibrium, so do two full revelations. Since $s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1) \leq \max(s_1, s_2)$, $\overline{u}_F(p) < u_F^*(p)$ if $v(x)$ does not implement

[5]The maximum is well defined: by the continuity of $v(x)$, each agent's set of max-minimizers is compact, and so is the set of all equilibria.

a fully revealing equilibrium, and in a fully revealing equilibrium, the firm gets the first best if and only if $v(x)$ is revelation-accurate. The result generally fails for $p_1 \neq p_2$: the set of equilibria is then not symmetric and may contain only equilibria in which exactly one agent reveals fully.

For every $p \in (0,1)$, the selection rules that implement a fully revealing equilibrium form a non-empty class, described by an infinite system of linear inequalities:

$$(1-p)\hat{v}(x) - p\hat{v}(1-x) \leq (x-p)\hat{v}(1) \quad \forall x \in (0,1), \quad \text{where } \hat{v}(x) := v(x) - 0.5.$$

Except for $p = 0.5$, this class has no simple structure, and the assumption that the firm knows the priors exactly is restrictive. Instead, suppose that the firm only knows that the priors lie in some exogenous set S. This robust approach leads to sharper results.

Although a fully revealing equilibrium is not necessary for the first best under unequal priors, this approach covers that case too: a rule implements a fully revealing equilibrium for all pairs of priors in S if and only if it does so for all equal priors in S (Lemma 10 in Appendix B). Robust implementation strongly restricts $v(x)$.

Proposition 6. *Let a selection rule $v(x)$ implement a fully revealing equilibrium for (p_l, p_l) and (p_h, p_h), where $p_l < 0.5 < p_h$. Then*

(i) $v(x)$ implements a fully revealing equilibrium for all $(p_1, p_2) \in (0,1)^2$;

(ii) $v(x) = 0.5 + ax$, where $a \in [0, 0.5]$.

To see why, note that the system of inequalities above holds for p_l and p_h. Adding it at x and at $1-x$ with weights $1-p$ and p gives $(1-2p)\big(\hat{v}(x) - x\hat{v}(1)\big) \leq 0$ for all $x \in (0,1)$. So p_l rules out any upward deviation of $\hat{v}(x)$ from the linear function $x\hat{v}(1)$, and p_h rules out any downward one. Hence, $v(x) = 0.5 + x(v(1) - 0.5)$ is linear, so it implements full revelation for all equal priors, and by Lemma 10 for all pairs of priors, and any $a = v(1) - 0.5 \in [0, 0.5]$ is valid.

In particular, as the first best requires revelation accuracy (Lemma 4), a firm that attains her first best at one prior below 0.5 and at one above must use $v(x) = 0.5 + 0.5x$. But this rule has a crucial issue. Since $v(x)$, and hence $\overline{v}(s_i)$, is linear, every pair of signals is an equilibrium, and the firm need not get the first best: robustness comes at the cost of complete indifference. This indifference is also fragile: if signals are costly, any positive cost makes pooling the unique equilibrium (Proposition 16 in Appendix B.12).

The issue disappears under slightly less robustness. Suppose that the firm knows whether both agents are more or less likely to be good than bad. The rules that implement a fully revealing equilibrium in either case are described by the deviation of $\hat{v}(x)$ from its secant from 0 to 1.

Definition 6. Let $v(x)$ be a selection rule, $a := v(1) - 0.5$, and $g(x) := v(x) - 0.5 - ax$, the deviation of $v(x)$ from its secant on $[0, 1]$. Then $v(x)$ is *balanced* if $g(x) = g(1 - x)$ on $(0, 1)$; *positively* [*negatively*] *balanced* if also $g(x) \geq 0$ [≤ 0] there; and *strictly* so if $g(x) > 0$ [< 0].

Remark 3. Like $\tilde{v}(x)$, $g(x)$ removes a linear part of $v(x)$, but it subtracts the secant from 0 to 1 rather than the tangent at 0. So, for every selection rule, equilibria depend on $v(x)$ only through $g(x)$, and revelation accuracy only through a. A well S-shaped $v(x)$ is positively balanced if and only if $\tilde{v}(x) = -x^2$ on $[0, 1]$.

Balance alone is exactly the condition for a fully revealing equilibrium at $p = 0.5$, as shown after Proposition 7; the sign of $g(x)$ then determines the half-interval of priors on which full revelation extends. For a selection rule, $a \in [0, 0.5]$ and $g(0) = g(1) = 0$ hold automatically, and for a balanced $v(x)$, the monotonicity of $v(x)$ is equivalent to $|g(x) - g(y)| \leq a|x - y|$.

Proposition 7. *Let $v(x)$ be a selection rule.*

(i) $v(x)$ *implements a fully revealing equilibrium for all* $(p_1, p_2) \in [0.5, 1)^2$ *[for all* $(p_1, p_2) \in (0, 0.5]^2$*] if and only if it is positively [negatively] balanced.*

(ii) If $v(x)$ *is strictly positively [negatively] balanced, it implements a fully revealing equilibrium uniquely* $\forall (p_1, p_2) \in (0.5, 1)^2$ $\left[\forall (p_1, p_2) \in (0, 0.5)^2\right]$.

Part (i) gives necessary and sufficient conditions for implementing full revelation for all priors on a half-interval. They impose a specific structure: $v(x)$ can deviate from the universal implementation $v(x) = 0.5 + ax$ ($v(1) = 1$ again implies $a = 0.5$) only by a function that is symmetric around 0.5 on $[0, 1]$, non-negative for high priors and non-positive for low ones.

Indeed, the conditions hold for all $p \in (0, 0.5]$ if and only if they hold for $p = 0$ and $p = 0.5$, and the latter gives the symmetry. If agent j with $p = 0.5$ reveals fully, agent i's normalized value function is odd around $s_i = 0.5$:

$$\overline{v}(s_i) = 0.5\hat{v}(s_i) - 0.5\hat{v}(1 - s_i) = -\overline{v}(1 - s_i) \; \forall s_i \in [0, 1].$$

Hence $\{0, 1\}$ is a best reply only if $\overline{v}(s_i)$ is linear:

$$\overline{v}(s_i) = 0.5\hat{v}(s_i) - 0.5\hat{v}(1 - s_i) = (s_i - 0.5)\hat{v}(1) \Leftrightarrow \hat{v}(s_i) - s_i\hat{v}(1) = \hat{v}(1 - s_i) - (1 - s_i)\hat{v}(1).$$

That is, the deviation $g(s_i)$ of $\hat{v}(s_i)$ from the secant from 0 to 1 must be symmetric around 0.5: $g(s_i) = g(1 - s_i)$ for all $s_i \in [0, 1]$, i.e., $v(x)$ must be balanced.

Part (ii) gives a sufficient condition for full revelation to be the unique equilibrium. As all equilibria form a product set, it suffices that $\{0, 1\}$ is each agent's unique best reply to itself. This is also why $p = 0.5$ is excluded: there, any $v(x)$ that implements full revelation makes $\overline{v}(s_i)$ linear, so the best reply is not unique. The condition is not necessary, but since it is only a slight strengthening of (i), the room for other rules

with a unique fully revealing equilibrium is small, and checking uniqueness for them requires showing that no signal supported on $\{s_i : g(s_i) = 0\}$ is a max-minimizer. Unlike the linear rule, a strictly positively or negatively balanced rule also withstands costly signals if $g(x)$ has a non-zero slope at 0: full revelation then remains the unique equilibrium for all priors in its half-interval that are farther from 0.5 than a margin proportional to the cost (Proposition 16).

A firm facing a high prior needs a positively balanced rule, and some subclasses of these rules can be generated by i.i.d. shocks. A firm facing a low prior, however, needs a negatively balanced rule, and additive evaluation noise cannot deliver one: by the next result, which extends this consequence of Proposition 1 beyond the well S-shaped class, i.i.d. shocks rule out full revelation as soon as one prior is below 0.5.

Corollary 4. *Let $v(x)$ be a selection rule with $v'(0) > v'(1)$. Then $(\{0,1\};\{0,1\})$ is never an equilibrium if $p_1 < 0.5$ or $p_2 < 0.5$. In particular, this holds for every $v(x)$ generated by i.i.d. shocks with a square-integrable density.*

4.4 A benchmark: general selection rules

Both limitations, indifference under full robustness and the failure of i.i.d. noise for low priors (Corollary 4), are specific to rules that depend only on the difference between the signals. To see how much they cost, consider a designer who may use any rule with the following properties.

Definition 7. A *general selection rule* is a measurable function $H_1 : [0,1]^2 \to [0,1]$, where $H_1(s_1, s_2)$ is the probability that agent 1 is selected when the signals are (s_1, s_2), such that

(i) (anonymity) agent 2 is selected with probability $H_1(s_2, s_1)$;

(ii) (single prize) $H_1(x, y) + H_1(y, x) = 1$ for all $x, y \in [0,1]$, so that $H_1(x, x) = \frac{1}{2}$.

Rules of the signal gap are the case $H_1(x, y) = v(x - y)$.

Neither continuity nor monotonicity in the own signal is required, and neither restriction would cost the firm anything. Every rule that attains a bound in this section or the next is nondecreasing in the own signal, and the bounds hold among all rules: if a value function decreases somewhere on the support, pooling the lowest signals makes it nondecreasing and weakly raises the firm's payoff. The optimal rules of Section 5 are discontinuous at ties, but smoothing them near ties gives continuous rules whose equilibrium payoffs converge to the same bounds (see Appendix C.6).

Many such rules have no natural interpretation, so I treat them as a benchmark. It is demanding in two ways. The designer commits to any rule, however unnatural ex post, and she measures each agent's signal, the posterior mean of his type, perfectly: the rule is applied to s_i itself rather than to a noisy reading of it. The preceding subsections showed that eliminating all noise from the measurement can hurt the firm. The benchmark asks the converse question: how much could a firm gain if it could measure perfectly and commit to any use of the measurement?

Proposition 8. *Let types be binary, and let H be a general selection rule nondecreasing in the own signal.*

(i) For priors (p, p), H attains the first best $2p - p^2$ in some equilibrium if and only if $H_1(1, 0) = 1$ and

$$(1 - p)H_1(x, 0) + p\, H_1(x, 1) \leq \tfrac{1-p+x}{2} \qquad \text{for all } x \in (0, 1).$$

(ii) H attains the first best in some equilibrium for all priors (p, p) with $p \in (0, 1)$ if and only if $H_1(1, 0) = 1$, $H_1(x, 0) \leq \frac{1+x}{2}$ and $H_1(x, 1) \leq \frac{x}{2}$ for all $x \in (0, 1)$. If the last two inequalities are strict, then for all priors $(p_1, p_2) \in (0, 1)^2$ the firm attains the first best in the unique equilibrium, which is fully revealing.

(iii) In particular, the pass/fail rule*, which selects an agent whose signal is* 1 *over one whose signal is below* 1 *and tosses a coin otherwise, attains the first best in the unique equilibrium for all* $(p_1, p_2) \in (0,1)^2$.

The right-hand side of (i) is the chord through the agent's winning probabilities at 0 and 1 against a fully revealing rival, so (i) says that no intermediate signal may win more often than the chord. For a rule of the signal gap, the two inequalities in (ii) read $v(x) \leq \frac{1+x}{2}$ and $v(1-x) \geq 1 - \frac{x}{2}$, which together force the linear rule, as in Proposition 6. A general rule can satisfy both strictly: the rule $0.5 + 0.5(s_i^2 - s_j^2)$ does, and so does the pass/fail rule.

The pass/fail rule requires a strong commitment. Ex post, it ignores what the firm has learned: between an agent with signal 0.95 and one with signal 0.05, it tosses a coin, which is as noisy as selection can be. Ex ante, this is exactly what makes revelation pay: a partial signal, however close to 1, buys nothing over revealing a bad type. So, with binary types, the firm has two ways to curb concealment: she can commit to disregard every signal short of conclusive evidence, or she can prefer, ex ante, some deliberate noise in her observations, which is what a noisy rule of the signal gap amounts to. So an imperfect measure can be just as good as a perfect one: the linear rule of the signal gap attains the benchmark's first best too. With binary types, the benchmark is thus trivial; Section 5 relaxes the binary prior and shows that with a concave prior the benchmark's own optimum deliberately blurs the measured signals.

5. Non-Binary Priors

Every result up to this point has assumed a binary prior. That assumption buys tractability, but it is not innocuous: the two-point distribution on $\{0,1\}$ is the *largest* element in the convex order among all distributions on $[0,1]$ with mean p, so it gives agents the richest possible set of feasible signals. This section relaxes it. The question

is whether the paper's conclusions are robust to the prior, as the value of single-sender persuasion is: it varies continuously with the prior. Subsection 5.1 keeps rules of the signal gap; Subsection 5.2 returns to the benchmark of general rules.

With a binary prior the only restriction on a signal G is that its mean equal p; with a general prior, G is feasible if and only if it is a mean-preserving contraction of F (Gentzkow and Kamenica, 2016), i.e. $\mathbb{E}_G[\phi] \leq \mathbb{E}_F[\phi]$ for every convex ϕ. I write $\underline{\theta} = \min \operatorname{supp}(F)$ and $\overline{\theta} = \max \operatorname{supp}(F)$, and count the noiseless rule $v(x) = 0.5 + 0.5 \cdot \operatorname{sign}(x)$ among selection rules.

With a binary prior, a posterior is pinned down by its mean, so identifying an agent's signal with his posterior mean is without loss. With a general prior it is not: a signal induces a distribution over posteriors, and two posteriors with the same mean can differ, for instance in how conclusive they are. Throughout, selection rules depend on the signals only through the posterior means. This is the standard restriction in persuasion problems where the receiver's payoff is linear in the state, so that only the posterior mean matters for her decision (Gentzkow and Kamenica, 2016; Kolotilin, 2018; Dworczak and Martini, 2019; Kolotilin et al., 2022), and it fits the firm here, whose payoff is the expected type of the agent she selects. For the benchmark designer of Subsection 4.4 the restriction has bite. A rule that could condition on the whole posterior would attain the first best for every prior: it could select an agent whose posterior is degenerate over one whose posterior is not, the higher of two degenerate posteriors, and toss a coin between two non-degenerate ones; full revelation is then weakly dominant, and the higher type is always selected.

5.1 Rules of the signal gap

Under a well S-shaped rule, an agent's value function is S-shaped against every signal of his rival, so, for a prior with a positive density, an upper censorship is a best reply

(Kolotilin et al., 2022): it reveals the types below a cutoff and pools all types above it into one signal. For an arbitrary prior, censorship is defined on quantiles: the *quantile censorship* G_q reveals the lowest share q of types and pools the rest at their mean, splitting an atom at the cutoff if needed. Against every rival signal, some quantile censorship is a best reply (Lemma 13), and the equilibrium is again a symmetric censorship.

Proposition 9. *Let $v(x)$ be well S-shaped and F any prior. Then (G_{q^*}, G_{q^*}) is an equilibrium for some $q^* \in (0,1]$, unique if $v'(x)$ is not linear near 0. If F has a positive density f on $[\underline{\theta}, \overline{\theta}]$, then $q^* < 1$, and the cutoff $\theta^* = F^{-1}(q^*)$ solves*

$$\int_0^{\theta^*} [\tilde{v}(m^* - s) - \tilde{v}(\theta^* - s) - (m^* - \theta^*)\tilde{v}'(m^* - s)] f(s) ds + \tilde{v}(m^* - \theta^*)(1 - F(\theta^*)) = 0, \quad (1)$$

where $m^* = \mathbb{E}[\theta | \theta \geq \theta^*]$.

With atoms or gaps, (1) holds in quantile form, and each of its solutions gives an equilibrium (Lemma 12). Full revelation, $q^* = 1$, can occur only if F has an atom at $\overline{\theta}$, as the binary prior does. For the best well S-shaped rule $v(x) = 0.5 + x - 0.5x|x|$ from Corollary 3, the equation collapses entirely: $q^* = \frac{1}{2}$ for every prior, so each agent reveals the types below the median and pools those above it, splitting an atom at the median so that exactly half of the mass is revealed (see Appendix C.3).

So, under a well S-shaped rule, the equilibrium keeps the structure of Section 3 with a general prior: low types are revealed, and high types are pooled into one signal. What changes is the comparison with the noiseless rule, which Subsection 5.2 takes up together with the benchmark of general rules.

5.2 General selection rules with a general prior

From here on, the firm chooses any general selection rule H of Definition 7, as in Proposition 8; rules of the signal gap are a special case, and *rule* means a general selection rule throughout. Proposition 8 showed that with binary types the benchmark designer attains the first best robustly. With a general prior she does not, and three questions arise: when the first best is attainable, what is the best she can do while insisting on full revelation, and what is the best she can do at all. Against a rival who plays G, the agent's value function is $V_G(x) := \int H_1(x, y)\, dG(y)$. Without continuity an equilibrium need not exist; when equilibria exist, they are interchangeable and give each agent $\frac{1}{2}$, and I evaluate a rule at its firm-preferred equilibrium.

In a symmetric profile (G, G), the firm's payoff is

$$u_F = p + 2\,\mathrm{Cov}_G\big(s, V_G(s)\big), \tag{2}$$

so she cares only about how steeply an agent's winning probability rises with his signal. A winning-probability profile V is feasible under full revelation, i.e., $V = V_F$ for some rule, if and only if $V(\theta) \leq_{cx} U[0,1]$ for $\theta \sim F$ (see Appendix C.6): noise can only make winning probabilities less dispersed than exact ranking does, and the noiseless rule is the boundary case, with $V_F = F$ and $F(\theta) \sim U[0,1]$.

5.2.1 The first best

Proposition 10. *Let* $\mathrm{supp}(F) = [\underline{\theta}, \overline{\theta}]$. *A general selection rule nondecreasing in the own signal attains the first best in some equilibrium if and only if F is convex on $[\underline{\theta}, \overline{\theta})$ and has no atom except possibly at $\overline{\theta}$. Then the noiseless rule attains it, with full revelation as an equilibrium, and every equilibrium that attains the first best is fully revealing. If F is moreover atomless, full revelation is the unique equilibrium of the noiseless rule.*

A noisy rule of the signal gap selects the higher of two close signals with probability near one half, so among such rules only the noiseless one attains the first best. With a general prior, the comparison in Subsection 4.1 thus reverses sharply for convex priors: no noisy rule of the signal gap attains the first best, and the noiseless rule does.

Remark 4. Proposition 10 extends to priors whose support is not an interval. With $\bar{F}(x) := \frac{1}{2}\big(F(x^-) + F(x)\big)$, the first best is attained if some function, convex on $[\underline{\theta}, \overline{\theta}]$, coincides with $\bar{F}$ on $\operatorname{supp}(F)$. The rule that does so rounds each signal down to the support and selects the higher rounded signal. Every binary prior satisfies the condition, and the rounding rule is then the pass/fail rule of Proposition 8 (see Appendix C.6 for the converse).

5.2.2 Full revelation and robustness

For any prior, full revelation is an equilibrium if there is a convex $\phi \geq V_F$ on $[\underline{\theta}, \overline{\theta}]$ with $\phi = V_F$ on $\operatorname{supp}(F)$, by Jensen's inequality, and, when V_F is upper semicontinuous, only if (Dworczak and Martini, 2019); it is the unique equilibrium if V_F is strictly convex. Define the affine value function

$$L(x) := \frac{1}{2} + \frac{x - p}{2(\overline{\theta} - p)}, \qquad \text{so } L(p) = \tfrac{1}{2},\ L(\overline{\theta}) = 1,$$

and let $\bar{U} := U[2p - \overline{\theta}, \overline{\theta}]$, the uniform distribution with the prior's mean and top.

Theorem 3. *Let F be any prior. For every general selection rule with upper semicontinuous V_F under which full revelation is an equilibrium,*

$$u_F \leq p + \frac{\operatorname{Var}_F(\theta)}{\overline{\theta} - p},$$

with equality if and only if $V_F = L$. The bound is attained if and only if $F \leq_{cx} \bar{U}$, i.e., if and

only if L is feasible; it is then attained by unbiased noise that spreads the truthful signals into $\bar{U}$, and approached by rules with full revelation as the unique equilibrium.

So the best fully revealing rule is *the narrowest uniformizing noise*: it makes truthful scores uniform on the narrowest interval that still contains the prior's top and spreads the prior. This is the logic of the example in the introduction, where uniform shocks make truthful scores uniform. Every concave prior satisfies $F \leq_{cx} \bar{U}$; the condition fails for convex priors other than the uniform, for which the noiseless rule attains the first best anyway.

The rule attaining the bound can be chosen monotone. Every feasible nondecreasing value function, L included, is produced by a lottery over coarse gradings: the firm draws a partition of the signals into intervals, selects the agent in the higher interval, and tosses a coin within an interval. For concave priors, a single random pass mark suffices: draw a bar from a fixed distribution, select the agent with the higher signal if it clears the bar, and toss a coin otherwise (see Appendix C.6).

Full revelation can also be implemented robustly, without the indifference of Proposition 6.

Proposition 11. *If a general selection rule H has $H_1(\cdot, y)$ convex for every y, full revelation is an equilibrium for every prior on $[0,1]$; conversely, if H is continuous and full revelation is an equilibrium for every prior, each $H_1(\cdot, y)$ is convex. Among rules with $H_1(\cdot, y)$ convex for every y,*

$$H_1^*(x,y) = \frac{1}{2} + \frac{x-y}{2\big(1-\min(x,y)\big)}$$

selects the higher signal with the highest probability at every pair of signals, so it maximizes the firm's utility under full revelation for every prior simultaneously. For every prior on $[0,1]$, full revelation is the unique equilibrium under H^*.

H^* has a simple description: draw a bar uniformly between the lower signal and

1; if the higher signal clears it, that agent is selected, and otherwise the firm tosses a coin. Among rules of the signal gap, only the linear one is robust (Proposition 6); H^* weakly beats it at every pair of signals and, unlike it, makes full revelation the unique equilibrium for every prior.

5.2.3 The second best

A concave prior has a nonincreasing density, possibly unbounded at $\underline{\theta}$, and possibly an atom at $\underline{\theta}$.

The rule behind Theorem 3 has a width. Replacing $\overline{\theta}$ by $\bar{s} > p$ in the definitions of L and $\bar{U}$ gives $L_{\bar{s}}$ and $\bar{U}_{\bar{s}} := U[2p - \bar{s}, \bar{s}]$; let $G^*_{\bar{s}}$ be the greatest common mean-preserving contraction of F and $\bar{U}_{\bar{s}}$ (see Appendix C.6). For $\bar{s} \geq 2p - \underline{\theta}$, the *uniformizing rule of width* $\bar{s}$ blurs each signal so that its winning probability against $G^*_{\bar{s}}$ is $L_{\bar{s}}$.

Proposition 12. *For every $\bar{s} \geq 2p - \underline{\theta}$, $(G^*_{\bar{s}}, G^*_{\bar{s}})$ is an equilibrium of the uniformizing rule of width $\bar{s}$, and*

$$u_F(\bar{s}) = p + \frac{\mathrm{Var}(G^*_{\bar{s}})}{\bar{s} - p}.$$

This is the highest payoff among symmetric equilibria of all general selection rules in which the agent's value function equals $L_{\bar{s}}$ on the support of the equilibrium signal. If F is concave, then for every $\bar{s} \in (2p - \underline{\theta}, \overline{\theta})$, $G^*_{\bar{s}}$ reveals the types below some t_1, puts no mass on some (t_1, t_2), and has the density of $\bar{U}_{\bar{s}}$ on $[t_2, \bar{s}]$.

Corollary 5. *Let F be concave on its support $[\underline{\theta}, \overline{\theta}]$. Then both agents playing $G = U[\underline{\theta}, 2p - \underline{\theta}]$ is the unique equilibrium of the noiseless rule, and the firm's equilibrium utility is $u_F = \frac{4p - \underline{\theta} - \overline{\theta}}{3}$.*

For concave priors, the family runs between two familiar ends: the uniformizing rule of width $2p - \underline{\theta}$ adds no noise and is the noiseless rule, whose equilibrium Corollary 5 gives, since a suitable uniform distribution is always feasible when the density is decreasing, and at width $\overline{\theta}$ the equilibrium is fully revealing and gives the

best fully revealing rule of Theorem 3. In between, agents reveal their low types and spread their high types uniformly, as all types do under noiseless competition.

Remark 5. This equilibrium has the shape of the censorship equilibrium of Proposition 9: low types are revealed and the rest are pooled. The difference lies in the agent's value function on the pooled range. Under a well S-shaped rule, it is strictly concave there, so it touches its supporting line at a single point, and all pooled types are sent to one signal. Under a uniformizing rule, it is linear there, equal to $L_{\bar{s}}$, so it coincides with its supporting line on the whole range, and the pooled types can be spread uniformly over it. In this sense, a uniformizing rule replaces the concave part of the S by a straight line.

Under the noiseless rule, the firm's payoff depends on the prior only through its mean and its lower support point, so, for a given lower support point, it does not grow as the prior becomes more dispersed. The first best does, since $\max(\theta_1, \theta_2)$ is convex, and so may the payoff under a noisy selection rule. So the comparison turns on dispersion. For the two ends of the uniformizing family, it is a variance threshold:

Corollary 6. *Let F be concave on $[\underline{\theta}, \overline{\theta}]$. The best fully revealing rule beats the noiseless rule if and only if*

$$\mathrm{Var}_F(\theta) > \frac{(\overline{\theta} - p)(p - \underline{\theta})}{3}.$$

The threshold is a third of the variance of the binary prior on $\{\underline{\theta}, \overline{\theta}\}$ with mean p, the most dispersed prior with this support and mean. The mean of a concave prior is at most the midpoint of its support, with equality only for the uniform distribution; the uniform meets the threshold with equality, and both rules attain the first best. For every mean below the midpoint, concave priors fall on both sides. The most dispersed concave prior, an atom at $\underline{\theta}$ plus a uniform density on $[\underline{\theta}, \overline{\theta}]$, always exceeds the threshold, while priors close to $U[\underline{\theta}, 2p - \underline{\theta}]$ fall below it.

Example 8. Let $p = \frac{1}{3}$ and $\mathrm{supp}(F) = [0, 1]$, and compare $F(x) = 2x - x^2$ with its

mean-preserving spread $F(x) = \frac{1+2x}{3}$, an atom of $\frac{1}{3}$ at 0 plus a uniform density. Their variances, $\frac{1}{18}$ and $\frac{1}{9}$, lie on opposite sides of the threshold $\frac{2}{27}$. The noiseless rule gives $u_F = \frac{4}{9} \approx 0.444$ under both; the best fully revealing rule gives 0.417 and 0.500; and the rule of the signal gap $v(x) = 0.5 + x - 0.5x|x|$, in its median-censorship equilibrium, gives 0.399 and 0.457.

So, under both the best fully revealing rule and the S-shaped rule of the signal gap, the noiseless rule is better under the first prior and worse under the second.

Neither end of the family is optimal, and for concave priors the family beats every general selection rule.

Theorem 4. *Let F be concave on $[\underline{\theta}, \overline{\theta}]$. In every symmetric equilibrium of every general selection rule,*

$$u_F \leq \max_{\bar{s} \in [2p - \underline{\theta}, \overline{\theta}]} u_F(\bar{s}).$$

If, moreover, F is not uniform, the maximum is attained only at widths in $(2p - \underline{\theta}, \overline{\theta})$: the best uniformizing rule strictly beats both the noiseless rule and the best fully revealing rule.

For the two priors of Example 8, the second best is 0.456 and 0.509, attained at widths $\bar{s} \approx 0.73$ and 0.94. At both ends the logic is first order against second order. Near full revelation, narrowing the width steepens winning probabilities, a first-order gain, while agents pool only a sliver of their best types; near the noiseless rule, widening it lets agents spread their posteriors further, while only a sliver of the lowest types must be revealed.

In a symmetric equilibrium (G, G) with V_G nondecreasing on the support and $\mathrm{Cov}_G(s, V_G(s)) > 0$, let

$$\beta := \frac{\mathrm{Cov}_G(s, V_G(s))}{\mathrm{Var}_G(V_G(s))},$$

the slope of the regression of the signal on the winning probability, which measures how much noise the rule effectively adds. Then u_F is at most the payoff of the

uniformizing rule of width $p + \frac{\beta}{2}$, or of the nearest end of the family if this width lies outside $[2p - \underline{\theta}, \overline{\theta}]$. Concavity of F matters: for priors that are neither concave nor convex, such as Beta$(2, 5)$, other rules beat the family numerically.

The best uniformizing rule is again a random pass mark, now with a bar that never exceeds t_2: agents who pool into the top range always clear it and are compared exactly, and only revealed low types face the coin. For non-uniform concave priors its equilibrium is unique, so the firm gets the second best in every equilibrium (see Appendix C.6).

So the case for not eliminating all randomness survives the move away from binary types, and it does not rest on the gap restriction: for every non-uniform concave prior, even a designer who measures signals perfectly chooses to blur them, and the noiseless rule is strictly suboptimal. What the move does change is the form of the best rule. With binary types, a deterministic coarse grade attains the first best; with a continuum of types, coarse grading discards real distinctions, and the best rule randomizes exactly among the agents it cannot separate cheaply, at the bottom.

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Overview of the Appendices

Appendix A proves the results of Section 3 and collects the details on unequal priors: the boundaries of the regions in Proposition 2, further examples, and how the priors affect the payoffs of the firm and of the agents. Appendix B proves the results of Section 4, constructs the equilibrium for the triangular selection rules of Subsection 4.2, extends the robust design results to costly signals, and gives the equilibria of the noiseless rule. Appendix C proves the results of Subsection 4.4 and Section 5, starting with censorship under an arbitrary prior. Appendix D collects extensions: reverse S-shaped selection rules, equilibria around the boundary of the well S-shaped class, and a single agent.

A. Proofs, Technical Details and Additions for Section 3

A.1 Properties of well S-shaped selection rules

Throughout the proofs, I use the following properties of a well S-shaped $v(x)$.

1) *Regularity.* Since $v'(x)$ is concave, it is continuous on $(-1,1)$ and has one-sided derivatives there; I set $v'(\pm 1)$ equal to its one-sided limits. I write $v''(x)$ and $\tilde{v}''(x)$ for either one-sided derivative, and every statement involving them holds for both.

2) *Shape.* Since $v(x)$ is increasing, $v'(1) \geq 0$, and since $v'(x)$ is even, concave and uniquely maximized at 0, $v'(x) \geq (1-|x|)v'(0) + |x|v'(1) > v'(1) \geq 0$ on $(-1,1)$, so $v(x)$ is strictly increasing. For $0 \leq x < y \leq 1$, the same argument gives $v'(x) \geq \frac{y-x}{y}v'(0) + \frac{x}{y}v'(y) > v'(y)$, so $v'(x)$ is strictly decreasing on $[0,1]$. Since $\tilde{v}'(x) = \frac{v'(x)-v'(0)}{|v(1)-0.5-v'(0)|}$, it is even, concave, and strictly decreasing on $[0,1]$ with $\tilde{v}'(0) = 0$. So $\tilde{v}(x)$ is negative, strictly decreasing and strictly concave on $(0,1]$. By the concavity of $\tilde{v}'(x)$, $\tilde{v}''(x) \leq \frac{\tilde{v}'(x)}{x} < 0$ on $(0,1)$, and $-\tilde{v}''(x)$ is non-decreasing on $(0,1)$.

3) *Elasticity.* The following lemma bounds the elasticity of $\tilde{v}(x)$ from below.

Lemma 5. *Let $v(x)$ be well S-shaped. Then $\mathcal{E}_{\tilde{v}}(x) \geq 2$ for all $x \in (0,1]$, with equality at x only if $\tilde{v}'(t)$ is linear on $[0,x]$. Hence $\frac{\tilde{v}(x)}{x^2}$ is non-increasing on $(0,1]$, so $\tilde{v}(x) \geq -x^2$ on $[0,1]$; if $v'(x)$ is not linear near 0, the ratio is strictly decreasing.*

Proof. Let $x \in (0,1]$ and $\phi := -\tilde{v}'$, which is convex on $[0,x]$ with $\phi(0) = 0$ by 2), so $\phi(t) \leq \frac{t}{x}\phi(x)$ for $t \in [0,x]$. Hence $-\tilde{v}(x) = \int_0^x \phi(t)dt \leq \frac{x\phi(x)}{2}$, i.e., $\mathcal{E}_{\tilde{v}}(x) \geq 2$, with equality only if ϕ is linear on $[0,x]$. Finally, $\left(\frac{\tilde{v}(x)}{x^2}\right)' = \frac{\tilde{v}(x)(\mathcal{E}_{\tilde{v}}(x)-2)}{x^3} \leq 0$, with strict inequality if $v'(x)$ is not linear near 0, and $\tilde{v}(1) = -1$. □

4) *Non-linearity near 0.* If $v'(x)$ is not linear near 0, then it is not affine on any interval whose closure contains 0: on an interval with 0 in its interior, an affine $v'(x)$ would be constant, contradicting the unique maximum at 0, and an affine piece ending at 0 gives,

by evenness, one starting at 0. Hence, for every $G \in \mathcal{G}(p)$, $\overline{v}'_G(s) := \int_0^1 v'(s-t)dG(t)$ is not affine on any interval $[\alpha, \beta]$ with $G([\alpha, \beta]) > 0$. Indeed, $\overline{v}'_G$ is a mixture of the concave functions $v'(s-t)$, so it is affine on $[\alpha, \beta]$ only if $v'(x)$ is affine on $[\alpha - t, \beta - t]$ for G-almost every t, and the closure of this interval contains 0 whenever $t \in [\alpha, \beta]$.

A.2 Proof of Lemma 1

(i) As shown in the text, $\overline{v}'(s_i) = \int_0^1 v'(s_i - s_j)dG_j(s_j)$ is concave on $[0,1]$. For a maximizer c of $\overline{v}'$, $\overline{v}$ is convex on $[0,c]$ and concave on $[c,1]$.

(ii) With $slp(0) = \overline{v}'(0)$, slp is continuous on $[0,1]$, so a exists. For $s \in (0,1]$, differentiating gives $slp'(s) = \frac{N(s)}{s^2}$, where

$$N(s) := s\overline{v}'(s) - \overline{v}(s) + \overline{v}(0) = \int_0^s \left(\overline{v}'(s) - \overline{v}'(t)\right) dt.$$

By (i), $N(s) \geq 0$ on $[0,c]$, and $N(s)$ is non-increasing on $[c,1]$, since for $c \leq s_1 < s_2 \leq 1$,

$$N(s_2) - N(s_1) = s_1 \left(\overline{v}'(s_2) - \overline{v}'(s_1)\right) + \int_{s_1}^{s_2} \left(\overline{v}'(s_2) - \overline{v}'(t)\right) dt \leq 0.$$

Hence slp is non-decreasing on $[0,c]$ and changes its monotonicity on $[c,1]$ at most once, from non-decreasing to non-increasing. So $a \geq c$, and $\overline{v}$ is concave on $[a,1]$. Let $l(s) := \overline{v}(0) + s \cdot slp(a)$. Then $l \geq \overline{v}$ on $[0,a]$ with equality at 0 and a, and if $a < 1$, $N(a) = 0$, i.e., l is tangent to $\overline{v}$ at a. Therefore $\overline{v}^{cv}$ equals l on $[0,a]$ and $\overline{v}$ on $[a,1]$. By Kamenica and Gentzkow (2011), a signal is optimal if and only if it is supported where $\overline{v}$ touches a line supporting $\overline{v}^{cv}$ at p_i. So $\{0,a\}$ is optimal if $p_i < a$, and $\{p_i\}$ is optimal if $p_i \geq a$, where $\overline{v}^{cv}(p_i) = \overline{v}(p_i)$.

If $v'(x)$ is not linear near 0, the concave $\overline{v}'$ is strictly increasing on $[0, z_1]$, constant on $[z_1, z_2]$ and strictly decreasing on $[z_2, 1]$ for some $z_1 \leq z_2$, and $z_1 > 0$ unless $z_2 = 0$: if $\overline{v}'$ were constant on $[0, z_2]$ with $z_2 > 0$, then $G_j([0, z_2]) = 0$ by property 4) in Appendix A.1,

so $\overline{v}''(s) = \int_{z_2}^{1} v''(s - s_j) dG_j(s_j) > 0$ on $(0, z_2)$, as $v'' > 0$ on $[-1, 0)$ by property 2), a contradiction. Hence $N > 0$ on $(0, z_2]$ and N is strictly decreasing on $[z_2, 1]$, so $\overline{v} < l$ on $(0, a)$, and $\overline{v}$ is strictly concave on $[a, 1]$. Hence the supporting line at p_i touches $\overline{v}$ only on $\{0, a\}$ if $p_i \leq a$ and only at p_i if $p_i > a$, so the optimum is unique.

A.3 Proof of Proposition 1

Let $\overline{v}(s_i)$ denote the value function of agent i for $\tilde{v}(x)$ rather than $v(x)$, which does not change best replies. If agent j plays $\{0, 1\}$, then $\overline{v}(s_i) = (1 - p_j)\tilde{v}(s_i) - p_j\tilde{v}(1 - s_i)$, which is S-shaped by Lemma 1, with $\overline{v}(1) - \overline{v}(0) = \tilde{v}(1) = -1$ and $\overline{v}'(1) = (1 - p_j)\tilde{v}'(1)$. By Lemma 1, $\{0, 1\}$ is a best reply if and only if $a = 1$, i.e., if and only if $\overline{v}$ lies below its secant $L(s) := \overline{v}(0) - s$ on $[0, 1]$. This implies $\overline{v}'(1) \geq -1$. Conversely, if $a < 1$, then $\overline{v}$ is concave on $[a, 1]$ by the proof of Lemma 1, so $\overline{v}(a) \leq \overline{v}(1) - (1 - a)\overline{v}'(1)$, while $slp(a) > slp(1)$ gives $\overline{v}(a) > \overline{v}(0) - a$; together, $\overline{v}'(1) < -1$. Hence

$$\{0, 1\} \text{ is a best reply } \Leftrightarrow (1 - p_j)\tilde{v}'(1) \geq -1 \Leftrightarrow p_j \geq 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(1)} = \overline{p},$$

and $(\{0, 1\}; \{0, 1\})$ is an equilibrium if and only if $p_1, p_2 \geq \overline{p}$.

For uniqueness, since all equilibria form a product set, it suffices that $\{0, 1\}$ is the unique best reply to $\{0, 1\}$ for both agents. Another best reply exists only if $D := \overline{v} - L \leq 0$ vanishes at some $s_0 \in (0, 1)$. Since D is convex on $[0, c]$ and concave on $[c, 1]$ for some c, it then vanishes on $[s_0, 1]$: a convex D with an interior maximum at $s_0 < c$ vanishes on $[0, c]$, and a concave D that vanishes at both ends of $[\max(s_0, c), 1]$ is non-negative there. Hence $\overline{v}'(1) = -1$, i.e., $p_j = \overline{p}$, and $\overline{v}$ is affine on $[s_0, 1]$, which property 4) in Appendix A.1 rules out if $v'(x)$ is not linear near 0, as $\{0, 1\}$ puts mass at 1. This proves uniqueness if $v'(x)$ is not linear near 0 or $p_1, p_2 > \overline{p}$. For $p = \overline{p}$, the uniqueness follows from the uniqueness part of the proof of Theorem 1, which covers

$p = \overline{p}$.

A.4 Proof of Theorem 1

If $p \geq \overline{p}$, full revelation is an equilibrium by Proposition 1, and it is unique for $p > \overline{p}$ by Proposition 1 and for $p = \overline{p}$ by the uniqueness part of this proof below. So let $p < \overline{p}$, and suppose agent j plays a signal $\{0, a\}$. Then for i we have

$$\overline{v}(s_i) = \frac{a-p}{a}\tilde{v}(s_i) + \frac{p}{a}\tilde{v}(s_i - a).$$

Then, for $\{0, a\}$ to be a best reply, it should be true that the slope of the secant is equal to the slope of the tangent:

$$\overline{v}'(a) = \frac{a-p}{a}\tilde{v}'(a) = \frac{a-p}{a}\frac{\tilde{v}(a)}{a} + \frac{p}{a}\frac{\tilde{v}(a)}{a},$$

hence $(a-p)\tilde{v}'(a) = \tilde{v}(a)$.

Then, since $\tilde{v}'(a) < 0 \; \forall a \in (0,1]$, a should be a solution of the equation $p = a - \frac{\tilde{v}(a)}{\tilde{v}'(a)}$. The RHS is the zero of the tangent $T_a(x) := \tilde{v}(a) + (x-a)\tilde{v}'(a)$, which is strictly decreasing in x, and it lies in $[0, a)$. For $a_1 < a_2$ and all $x \leq a_1$,

$$T_{a_2}(x) - T_{a_1}(x) = \int_{a_1}^{a_2} \left(\tilde{v}'(t) - \tilde{v}'(a_2)\right) dt + (a_1 - x)\left(\tilde{v}'(a_1) - \tilde{v}'(a_2)\right) > 0,$$

since $\tilde{v}'(x)$ is strictly decreasing on $[0,1]$. So T_{a_2} is positive at the zero of T_{a_1}, and its own zero lies strictly to the right. Hence the RHS is strictly increasing in a, and the solution is unique and monotone in p, if it exists.

To prove existence, notice that the RHS is continuous in a and its value at $a = 1$ is $\overline{p}$.

By the definition of $\mathcal{E}_{\tilde{v}}(x)$ and Lemma 5,

$$a - \frac{\tilde{v}(a)}{\tilde{v}'(a)} = a\left(1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\right) \in \left[\frac{a}{2}, a\right),$$

hence the RHS converges to 0 as $a \to 0_+$. So $\forall p \in (0, \overline{p})$ there always exists a unique monotone solution to this equation, which is exactly the equation from the statement. Moreover, as the inverse of a continuous and strictly increasing function, $a(p)$ is continuous and strictly increasing.

Best reply. Let $p \in (0, \overline{p}]$, where for $p = \overline{p}$, $a = 1$ satisfies $(a - p)\tilde{v}'(a) = \tilde{v}(a)$ by the definition of $\overline{p}$. Let $l(s)$ be the secant from $(0, \overline{v}(0))$ to $(a, \overline{v}(a))$ and $d(s) := l(s) - \overline{v}(s)$, so that $d(0) = d(a) = 0$, and $d'(a) = 0$ by this equation. As $\tilde{v}'(x)$ is even,

$$d'(s) = \frac{\tilde{v}(a)}{a} - \frac{a - p}{a}\tilde{v}'(s) - \frac{p}{a}\tilde{v}'(a - s),$$

which is convex on $[0, a]$, since $\tilde{v}'(x)$ is concave. By the condition above, $d'(0) = \frac{\tilde{v}(a)}{a} - \frac{p}{a}\tilde{v}'(a) = \frac{a-2p}{a}\tilde{v}'(a) \geq 0$, since $\tilde{v}'(a) < 0$ and $\frac{p}{a} = 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)} \in [\frac{1}{2}, 1)$ by Lemma 5, i.e., $p < a \leq 2p$. So $d' \geq 0$ on some $[0, z]$ and $d' \leq 0$ on $[z, a]$, and hence $d \geq 0$ on $[0, a]$. For $s \in (a, 1]$, both s and $s - a$ lie in $(0, 1]$, where $\tilde{v}'(x)$ is strictly decreasing, so

$$\overline{v}'(s) = \frac{a - p}{a}\tilde{v}'(s) + \frac{p}{a}\tilde{v}'(s - a) < \frac{a - p}{a}\tilde{v}'(a) + \frac{p}{a}\tilde{v}'(0) = \overline{v}'(a),$$

and $\overline{v}(s)$ lies strictly below its tangent $l(s)$ on $(a, 1]$. Hence $l(s)$ supports $\overline{v}(s)$ on $[0, 1]$ and touches it at 0 and $a > p$, so $\{0, a\}$ is a best reply to itself, and every best reply to $\{0, a\}$ is supported on $\{s \in [0, a] : d(s) = 0\}$.

Uniqueness. By the constant-sum property, all equilibria form a symmetric product set $E \times E$ with $\{0, a\} \in E$. Hence, if $G \in E$, then both $(G, \{0, a\})$ and $(\{0, a\}, G)$ are equilibria, i.e., G is a best reply to $\{0, a\}$, and $\{0, a\}$ is a best reply to G.

Step 1: best replies to $\{0, a\}$. Suppose that $d(c) = 0$ for some $c \in (0, a)$. Then c minimizes d on $[0, a]$, so $d'(c) = 0$. By convexity, $d'(s) \leq 0$ on $[c, a]$, and since $\int_c^a d'(s)ds = d(a) - d(c) = 0$, $d' \equiv 0$ on $[c, a]$. Again by convexity, $d'(s) \geq 0$ on $[0, c]$, and since $d(0) = d(c) = 0$, $d' \equiv 0$ on $[0, c]$. Hence $\frac{a-p}{a}\tilde{v}'(s) + \frac{p}{a}\tilde{v}'(a - s)$ is constant on $[0, a]$. A sum of two concave functions is affine only if both are affine, so $\tilde{v}'(s)$ is affine on $[0, a]$, and as $\tilde{v}'(0) = 0 > \tilde{v}'(a)$, $\tilde{v}'(s) = -Ks$ on $[0, a]$ for some $K > 0$. Then

$$\frac{a-p}{a}\tilde{v}'(s) + \frac{p}{a}\tilde{v}'(a-s) = -pK - \frac{K(a-2p)}{a}s$$

is constant only if $a = 2p$.

So, either $d(s) > 0$ on $(0, a)$, in which case $\{0, a\}$ is the only best reply to $\{0, a\}$, and the equilibrium is unique, or $\tilde{v}(x) = -\frac{K}{2}x^2$ on $[0, a]$ and $a = 2p$.

Step 2: the quadratic case. Let $\tilde{v}(x) = -\frac{K}{2}x|x|$ on $[-2p, 2p]$ and $a = 2p$, so that $d \equiv 0$ on $[0, 2p]$ and every G with support in $[0, 2p]$ and mean p is a best reply to $\{0, 2p\}$. Take such a $G \neq \{0, 2p\}$ and let $w(s) := \int_0^{2p} \tilde{v}(s - t)dG(t)$ be the normalized value function against it. For $s \in [0, 2p]$, $w'(s) = -K\int_0^{2p} |s - t|dG(t)$, hence $w'(2p) = -Kp$, while $w(0) = \frac{K}{2}\mathbb{E}_G[t^2]$ and $w(2p) = -\frac{K}{2}\mathbb{E}_G[(2p - t)^2] = -\frac{K}{2}\mathbb{E}_G[t^2]$. So the slope of the secant from $(0, w(0))$ to $(2p, w(2p))$ exceeds $w'(2p)$ by

$$-\frac{K\mathbb{E}_G[t^2]}{2p} + Kp = \frac{K}{2p}\left(2p^2 - \mathbb{E}_G[t^2]\right) > 0,$$

since $t^2 \leq 2pt$ on $[0, 2p]$, with equality only at $t \in \{0, 2p\}$. Hence $w(s)$ lies strictly above the secant for $s < 2p$ close enough to $2p$, so, for such $s > p$, the signal $\{0, s\}$ yields a strictly higher payoff against G than $\{0, 2p\}$. Thus $\{0, 2p\}$ is not a best reply to G, $G \notin E$, and the equilibrium is unique.

Let me also introduce a bound on a that is used in later proofs.

Lemma 6. *Let $v(x)$ be well S-shaped and $p \in (0, \overline{p})$. In the symmetric equilibrium*

$(\{0,a\};\{0,a\})$, *$p < a \leq 2p$, and $a < 2p$ if $v'(x)$ is not linear near 0.*

Proof. Since $\frac{p}{a} = 1 - \frac{1}{\mathcal{E}_{\tilde{v}}(a)}$, and $\mathcal{E}_{\tilde{v}}(a) \geq 2$ by Lemma 5, with strict inequality if $v'(x)$ is not linear near 0, $p < a \leq 2p$, and $a < 2p$ in the latter case. □

A.5 Proof of Lemma 2

Against $\{p_j\}$, $\overline{v}(s) = v(s - p_j)$, so $\overline{v}'(s) = v'(s - p_j)$ is strictly increasing on $[0, p_j]$, since $v'(x)$ is even and strictly decreasing on $[0,1]$ by property 2) in Appendix A.1. As in the proof of Lemma 1, let $slp(s) := \frac{\overline{v}(s)-\overline{v}(0)}{s}$, so that $slp'(s) = \frac{N(s)}{s^2}$ with $N(s) = \int_0^s (\overline{v}'(s) - \overline{v}'(t))dt$. Then $N > 0$ on $(0, p_j]$, and, by continuity, slp is strictly increasing on $(0, s]$ for some $s \in (p_j, 1]$. If $p_i \leq p_j$, the signal $\{0, s\}$ with mean p_i yields $\overline{v}(0) + p_i\, slp(s) > \overline{v}(0) + p_i\, slp(p_i) = \overline{v}(p_i)$, so $\{p_i\}$ is not a best reply. Hence $\{p_i\}$ is a best reply only if $p_i > p_j$, which cannot hold for both agents.

A.6 Proof of Lemma 7

Lemma 7. *Let $p \in (0,1)$, let $v_n(x)$ be well S-shaped for all $n \in \mathbb{N}$, and let $(\{0, a_n\};\{0, a_n\})$ be the equilibrium for v_n and priors (p, p). If $\mathcal{E}_n(x) \to \infty$ uniformly on $(0,1)$, then $a_n \to p$.*

Since $\mathcal{E}_n(x) \to \infty$ uniformly on $(0,1)$ and $\mathcal{E}_n(x)$ is continuous on $(0,1]$, $\varepsilon_n := \sup_{x\in(0,1]} \frac{1}{\mathcal{E}_n(x)} \to 0$. Hence $p < \overline{p}_n = 1 - \frac{1}{\mathcal{E}_n(1)}$ for large n, and then, by Theorem 1, $a_n = \frac{p}{1-1/\mathcal{E}_n(a_n)} \in \left[p, \frac{p}{1-\varepsilon_n}\right]$, so $a_n \to p$.

A.7 Unequal priors: regions and further results

The regions of Proposition 2 are bounded by best replies to the three forms of signals, listed below with their values in Example 3.

	definition	*Example 3*
$\overline{p}$	$1 - \frac{1}{\mathcal{E}_{\tilde{v}}(1)}$: full revelation for $p_1, p_2 \geq \overline{p}$	$\frac{2}{3}$
$b(p_1)$	agent 2's best reply to $\{p_1\}$ is $\{0, b(p_1)\}$ if $p_2 < b(p_1)$, else $\{p_2\}$	$\min(\frac{3p_1}{2}, 1)$
p^*	smallest p_1 with $b(p_1) = 1$	$\frac{2}{3}$
$a(p_2)$	agent 1's best reply to $\{0, 1\}$ is $\{0, a(p_2)\}$ if $p_1 < a(p_2)$, else $\{p_1\}$	$\frac{3p_2}{2}$
$\check{p}_2(p_1)$	largest p_2 for which $\{p_1\}$ is agent 1's best reply to $\{0, b(p_1)\}$	$\frac{2p_1}{3}$
$\hat{p}_1(p_2)$	smallest p_1 for which $\{0, 1\}$ is agent 2's best reply to $\{0, a(p_2)\}$	$\frac{2}{3}$

With $p_2 \leq p_1$ and $p_2 < \overline{p}$, the equilibrium is

(i) $(\{p_1\}; \{0, 1\})$ if $p_1 \geq \max\{p^*, a(p_2)\}$;

(ii) $(\{p_1\}; \{0, b(p_1)\})$ if $p_1 < p^*$ and $p_2 \leq \check{p}_2(p_1)$;

(iii) $(\{0, a(p_2)\}; \{0, 1\})$ if $\hat{p}_1(p_2) \leq p_1 < a(p_2)$;

(iv) $(\{0, a\}; \{0, b\})$ with $p_1 < a \leq b < 1$ otherwise.

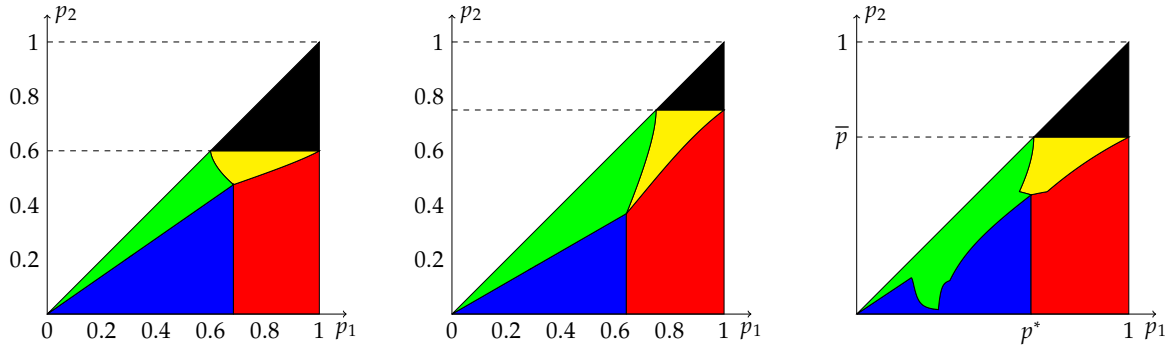
Figure 4 depicts these regions for three other rules. In Example 3, $\overline{p} = p^*$ (Figure 2): it is the knife-edge case between $\overline{p} < p^*$ and $\overline{p} > p^*$ in panels (a) and (b). The boundary $a(\cdot)$ is monotone, but $\hat{p}_1(\cdot)$ and $\check{p}_2(\cdot)$ need not be: in panel (c), neither is monotone.

Region (i) has a simple explanation. Against full revelation, agent 1's value function $(1 - p_2)v(s_1) + p_2 v(s_1 - 1)$ is close to $v(s_1)$ for low p_2, so it is concave except near 0, and pooling at p_1 is optimal. Against pooling, agent 2's value function $v(s_2 - p_1)$ is convex on $[0, p_1]$ and concave on $[p_1, 1]$, and for high p_1, its concavification is supported on $\{0, 1\}$ despite the concave part. For a fixed $v(x)$, not every pair of forms is played: full revelation is a best reply to $\{0, c\}$ only if $c \geq p^*$, so $(\{0, c\}; \{0, 1\})$ is an equilibrium only if $c \geq p^*$.

Remark 6. Let $v(x) = v^*(x)$, for which $v'(x)$ is linear on $[0, 1]$, and $(p_1, p_2) = (0.4, 0.3)$. Then agent 2 plays $\{0, 0.6\}$ in every equilibrium, and agent 1's value function against it is linear on $[0, 0.6]$. So agent 1's equilibrium signals are exactly those with support in

$[0, 0.6]$ and $\mathbb{E}[s_1^2] = 0.18$, which keep $\{0, 0.6\}$ a best reply of agent 2, such as $\{0, 0.45\}$, $\{0.3, 0.6\}$ and their mixtures.

Figure 4: Equilibria for unequal priors, with $\tilde{v}(x)$ given on $[0, 1]$



(a) $\tilde{v}(x) = -x^{2.5}$: $\overline{p} = 0.6 < p^* \approx 0.68$

(b) $\tilde{v}(x) = -x^4$: $\overline{p} = 0.75 > p^* \approx 0.64$

(c) $\tilde{v}(x) = -\frac{x^3+60\max(x-0.3,0)^2}{30.4}$: $p^* \approx 0.641 < \overline{p} \approx 0.651$

Colors are for the corresponding equilibrium type: black – $(\{0,1\};\{0,1\})$; *red* – $(\{p_1\};\{0,1\})$; *green* – $(\{0,a\};\{0,b\})$; *blue* – $(\{p_1\};\{0,b\})$; *yellow* – $(\{0,a\};\{0,1\})$.

Example 9. Let $v(x)$ be well S-shaped with $\tilde{v}(x) = -x^{2.5}$ on $[0, 1]$, and let $p_j = 0.6$. Then agent i's equilibrium signal is non-monotone in p_i.

For the firm, the monotonicity of Proposition 4 extends to unequal priors everywhere except in region (iv) of Proposition 2.

Proposition 13. *Let $v(x)$ be well S-shaped with $v'(x)$ not linear near 0, and let $p_2 \leq p_1$. In the interior of the full revelation region and of each of the regions (i)–(iii) of Proposition 2, both partial derivatives of u_F are positive.*

Example 10. Let $v(x) = 0.5+(20x-x^{20})/38$ on $[0, 1]$, so that $\tilde{v}(x) = -x^{20}$ and $v(1) = 1$, and let $p_1 = 0.924$. Then u_F strictly decreases in p_2 on $[0.839, 0.873]$, where the equilibrium is in region (iv): once agent 2 stops revealing fully, his high signal b decreases in p_2.

For the agents, monotonicity in the priors is trivial under equal priors, as each is selected with probability 0.5. Under unequal priors, since the game is constant-sum,

each agent's equilibrium payoff $u_i(p_1, p_2) = \mathbb{E}_{G_1,G_2}[v(s_i - s_j)]$ is well defined for any selection rule, whether or not the equilibrium is unique.

Lemma 8. *For any selection rule, $u_i(p_1, p_2)$ is non-decreasing in p_i and non-increasing in p_j. If $v(x)$ is strictly increasing, in particular if it is well S-shaped, $u_i(p_1, p_2)$ is strictly increasing in p_i and strictly decreasing in p_j.*

A.8 Proof of Proposition 2

If $p_2 \geq \overline{p}$, the claim follows from Proposition 1, so let $p_2 < \overline{p}$.

Preliminaries. For a signal of agent j let $N(s) := s\overline{v}'(s) - \overline{v}(s) + \overline{v}(0)$, so that $slp'(s) = N(s)/s^2$. By the proof of Lemma 1 (ii), $N > 0$ on $(0, a)$ and $N < 0$ on $(a, 1]$, with a from Lemma 1. Hence agent i plays $\{p_i\}$ if and only if $N(p_i) \leq 0$, and $\{0, 1\}$ if and only if $N(1) \geq 0$. For $y, c \in [0, 1]$, let

$$A(y) := \tilde{v}(y) - y\tilde{v}'(y) = -\int_0^y x\tilde{v}''(x)dx,$$
$$B(y; c) := y\tilde{v}'(c - y) + \tilde{v}(c - y) - \tilde{v}(c) = -\int_0^y x\tilde{v}''(c - x)dx.$$

Since $-\tilde{v}'' > 0$ on $(0, 1]$, $A(y) > 0$, and $B(y; c) > 0$ for $0 < y \leq c$. If agent j plays $\{0, c\}$, so that $\mathbb{P}(s_j = c) = q := \frac{p_j}{c}$, then $\overline{v}(s) = (1 - q)\tilde{v}(s) + q\tilde{v}(s - c)$ and

$$N(s) = -(1 - q)A(s) + qB(s; c); \tag{3}$$

pooling is the case $c = p_j$, $q = 1$.

Step 1: $b(\cdot)$ and p^*. Against $\{p_1\}$, $N(s) = B(s; p_1)$. Hence agent 2 never plays $\{p_2\}$, as $p_2 \leq p_1$, and plays $\{0, b(p_1)\}$, where $b(p_1)$ solves $B(b; p_1) = 0$, i.e. $b\tilde{v}'(b - p_1) = \tilde{v}(b - p_1) + \tilde{v}(p_1)$, or $b(p_1) = 1$ if $B(1; p_1) \geq 0$. With $b = p_1 + \tilde{b}$ this equation reads

$$\varphi(\tilde{b}) := \tilde{v}(\tilde{b}) - (p_1 + \tilde{b})\tilde{v}'(\tilde{b}) + \tilde{v}(p_1) = 0,$$

where $\varphi'(\tilde{b}) = -(p_1 + \tilde{b})\tilde{v}''(\tilde{b}) > 0$, $\varphi(0) = \tilde{v}(p_1) < 0 < 2A(p_1) = \varphi(p_1)$ and $\partial_{p_1}\varphi = \tilde{v}'(p_1) - \tilde{v}'(\tilde{b}) < 0$ for $\tilde{b} < p_1$. So the root is unique, $p_1 < b(p_1) < 2p_1$, and b is continuous, strictly increasing while $b < 1$. Let p^* be the smallest p_1 with $b(p_1) = 1$, solving $\tilde{v}(1 - p^*) + \tilde{v}(p^*) = \tilde{v}'(1 - p^*)$; it exists and $p^* \in (\frac{1}{2}, 1)$, since $b(\frac{1}{2}) < 1$ and $B(1;1) = 1 > 0$. Thus against $\{x\}$ the other agent plays $\{0,1\}$ if and only if $B(1;x) \geq 0$, i.e. $x \geq p^*$, as b is non-decreasing.

Step 2: $a(\cdot)$. Against $\{0,1\}$, by (3),

$$N(s) = \big(A(s) + B(s;1)\big)\big(p_2 - \Phi(s)\big), \qquad \Phi := \frac{A}{A + B(\cdot\,;1)}.$$

Φ is strictly increasing: writing $w = -\tilde{v}''$,

$$\partial_y B(y;1)\,A(y) - B(y;1)\,A'(y) = y\int_0^y x\big[w(1-y)w(x) - w(1-x)w(y)\big]dx < 0,$$

since $w(x) \leq w(y)$ and $w(1-y) \leq w(1-x)$ for $x < y$, and $w(x) < w(y)$ for x close to 0, as $v'(x)$ is not linear near 0. Moreover, $\Phi(1) = \frac{|\tilde{v}'(1)|-1}{|\tilde{v}'(1)|} = \overline{p}$. Hence agent 1 plays $\{p_1\}$ if $p_1 \geq a(p_2)$ and $\{0, a(p_2)\}$ otherwise, where $a(p_2) = 0$ if $p_2 \leq \Phi(0^+)$, $a(p_2) = \Phi^{-1}(p_2)$ if $\Phi(0^+) < p_2 < \overline{p}$, and $a(p_2) = 1$ if $p_2 \geq \overline{p}$; a is continuous and non-decreasing, since Φ is continuous and strictly increasing.

Step 3: $\check{p}_2(\cdot)$ and $\hat{p}_1(\cdot)$. By (3), against $\{0, b(p_1)\}$, agent 1 plays $\{p_1\}$ if and only if

$$p_2 \leq \check{p}_2(p_1) := \frac{b(p_1)A(p_1)}{A(p_1) + B\big(p_1; b(p_1)\big)},$$

and against $\{0, a\}$ with $a > p_1$, agent 2 plays $\{0,1\}$ if and only if $-(a - p_1)A(1) + p_1 B(1;a) \geq 0$. By Step 1, $B(1;a) \geq 0$ if and only if $a \geq p^*$, so for $a < p^*$ this fails, and for $a \geq p^*$ it reads $p_1 \geq \frac{aA(1)}{A(1)+B(1;a)}$. For $a = a(p_2) \geq p^*$ denote this bound by $\hat{p}_1(p_2)$.

Step 4: cases (i)–(iii). A pair of signals is an equilibrium if and only if each is a best

reply to the other. By Steps 1 and 2, $(\{p_1\};\{0,1\})$ is an equilibrium if and only if $p_1 \geq p^*$ and $p_1 \geq a(p_2)$; by Steps 1 and 3, $(\{p_1\};\{0,b\})$ with $b<1$ is an equilibrium if and only if $b=b(p_1)$, $p_1<p^*$ and $p_2 \leq \check{p}_2(p_1)$; by Steps 2 and 3, $(\{0,a\};\{0,1\})$ is an equilibrium if and only if $a=a(p_2)\geq p^*$ and $\hat{p}_1(p_2)\leq p_1<a(p_2)$. An equilibrium exists by the minimax theorem, and as the game is constant-sum, all equilibria form a product set. By Lemma 1 every best reply is unique, hence so is the equilibrium.

Step 5: case (iv). For equal priors the game is symmetric, so the unique equilibrium is symmetric and none of the forms in (i)–(iii) is an equilibrium. By Step 4 this gives: $a(p)>p$ if $p\geq p^*$; $\check{p}_2(p)<p$ if $p<p^*$; and $\hat{p}_1(p)>p$ if $a(p)\geq p^*$ and $p<\overline{p}$, since then $p<a(p)$. Now let $p_2\leq p_1$ and suppose that none of (i)–(iii) holds. By Lemma 1 each agent plays $\{p_i\}$, $\{0,1\}$ or $\{0,c\}$ with $p_i<c<1$, and by Lemma 2 not both pool at their priors. Apart from (i)–(iv), this leaves the following forms, and each requires, by Steps 1–3 with the roles of the agents exchanged (the functions b, a, $\check{p}_2$ and $\hat{p}_1$ stay the same, only their arguments change), a condition incompatible with $p_2\leq p_1$:

- $(\{0,b(p_2)\};\{p_2\})$ requires $p_2<p^*$ and $p_1\leq\check{p}_2(p_2)<p_2$;
- $(\{0,1\};\{p_2\})$ requires $p_2\geq p^*$ and $p_2\geq a(p_1)\geq a(p_2)>p_2$;
- $(\{0,1\};\{0,a(p_1)\})$ with $a(p_1)<1$ requires $a(p_1)\geq p^*$, $p_1<\overline{p}$ and $p_2\geq\hat{p}_1(p_1)>p_1$;
- $(\{0,1\};\{0,1\})$ requires $p_2\geq\overline{p}$ by Proposition 1.

Hence the equilibrium is $(\{0,a\};\{0,b\})$ with $p_1<a<1$ and $p_2<b<1$.

Step 6: $a\leq b$ and continuity. The correspondence $p\mapsto\mathcal{G}(p)$ is continuous: upper hemicontinuous since the mean is weakly continuous, and lower hemicontinuous since any $G\in\mathcal{G}(p)$ is approximated by mixtures of G with δ_0 or δ_1. As payoffs are continuous, the equilibrium correspondence has a closed graph, and being single-valued with values in a compact set, it is continuous. If $a=b=c$ in (iv), then, since $B(c;c)=-\tilde{v}(c)$, the indifference conditions $N(c)=0$ of the two agents, with $q=\frac{p_2}{c}$ and

$q = \frac{p_1}{c}$, read $\frac{c}{p_2} - 1 = -\frac{\tilde{v}(c)}{A(c)} = \frac{c}{p_1} - 1$, so $p_1 = p_2$; conversely $a = b$ on the diagonal by symmetry. On the convex set $U = \{0 < p_2 < p_1 < 1,\ p_2 < \overline{p}\}$, let R be the set where (iv) holds. R is open in U, as the strategies in (iv) are not weak limits of $\{0, 1\}$ or of point masses, and $a - b \neq 0$ is continuous on R. Since U is connected and (i) holds for p_1 close to 1, each connected component of R has a boundary point in $U \setminus R$. There one of (i)–(iii) holds, and continuity gives $(a, b) \to (p_1, 1)$, $(p_1, b(p_1))$ or $(a(p_2), 1)$ respectively, with $a(p_2) < 1$ as $p_2 < \overline{p}$. Each limit has $a < b$, hence $a < b$ on R.

A.9 Proof of Proposition 3

Describe a signal of agent i by its high signal $c_i \in [p_i, 1]$, where $c_i = p_i$ stands for $\{p_i\}$. For a fixed p_i, $\frac{\mathbb{P}(s_i=0)}{1-p_i} = \frac{c_i - p_i}{c_i(1-p_i)}$ is increasing in c_i, so it suffices to show that c_i is non-decreasing in p_j. By Lemma 1, the best reply to c_j is $c_i = \max\{r(c_j, \frac{p_j}{c_j}), p_i\}$, where $r(c, q)$ is the point at which N from (3) changes sign; it is continuous in (c, q). With $w = -\tilde{v}''$, which is odd and non-decreasing,

$$\partial_q N(s) = A(s) + B(s; c) = \int_0^s x\big[w(x) + w(c - x)\big]dx > 0,$$

since $w(x) + w(c - x)$ is at least $w(x) > 0$ if $x \leq c$ and equals $w(x) - w(x - c) \geq 0$ if $x > c$. Hence $r(c, q)$ is non-decreasing in q, and strictly increasing where $r(c, q) \in (0, 1)$, and $r(c, 1) = b(c)$ from Step 1 of the proof of Proposition 2.

Let $\rho(a) := r(a, \frac{p_1}{a})$, so that agent 2's best reply to a is $\max\{\rho(a), p_2\}$, and let $\Gamma(a) := \max\{r(c, \frac{p_2}{c}), p_1\}$ with $c = \max\{\rho(a), p_2\}$, for $a \in [p_1, 1]$. Γ is continuous and its fixed points are exactly the equilibria, so by Step 4 of the proof of Proposition 2 it has a unique fixed point a^*; since $\Gamma(p_1) \geq p_1$, $\Gamma(a) > a$ on $[p_1, a^*)$. As p_2 increases to p_2', $\Gamma(a)$ does not decrease for any a: if $\rho(a) \geq p_2'$, only the weight $\frac{p_2}{\rho(a)}$ increases; if $\rho(a) \leq p_2$, agent 2 pools and $b(p_2) \leq b(p_2')$; and if $p_2 < \rho(a) < p_2'$, then $r(\rho, \frac{p_2}{\rho}) \leq r(\rho, 1) = b(\rho) \leq b(p_2')$. Hence the fixed point does not decrease, and if $p_1 < a^* < 1$, $\Gamma(a^*)$ strictly increases, and

so does the fixed point. As the argument does not use $p_2 \leq p_1$, it applies to both agents.

A.10 Proof of Lemma 3

By Theorem 1, $\frac{\mathbb{P}(s_i=0)}{1-p} = \frac{1}{1+(\mathcal{E}_{\tilde{v}}(a)-1)(1-a)}$, where, by its proof, $a(p)$ is continuous and strictly increasing, so it maps $(p, p+\varepsilon)$ and $(p-\varepsilon, p)$ onto a right and a left neighbourhood of $a(p)$. Therefore agents reveal more for all $p_h \in (p, p+\varepsilon)$ and less for all $p_l \in (p-\varepsilon, p)$ if and only if $(\mathcal{E}_{\tilde{v}}(x)-1)(1-x)$ is strictly decreasing at $x = a$.

A.11 Proof of Proposition 4

By Theorem 1 and Proposition 1, the equilibrium is $(\{0, a\}; \{0, a\})$ with $a = a(p) > p$, where $a(p)$ is non-decreasing on $(0, 1)$, as $a(p) = 1$ for $p \geq \overline{p}$. When both agents play $\{0, a\}$ with prior $p < a$, the firm gets

$$U(p, a) := \frac{p^2}{a} + \frac{2p(a-p)}{a} v(a) = p + p\Big(1 - \frac{p}{a}\Big)\big(2v(a) - 1\big).$$

For a fixed p, U is non-decreasing in a, as $1 - \frac{p}{a}$ and $2v(a) - 1$ are non-negative and non-decreasing. For a fixed a, $\partial_p U = 1 + \big(1 - \frac{2p}{a}\big)\big(2v(a) - 1\big) > 2\big(1 - v(a)\big) \geq 0$, as $1 - \frac{2p}{a} > -1$ and $0 < 2v(a) - 1 \leq 1$. Hence, for $p_1 < p_2$,

$$u_F(p_1) = U\big(p_1, a(p_1)\big) \leq U\big(p_1, a(p_2)\big) < U\big(p_2, a(p_2)\big) = u_F(p_2).$$

A.12 Proof of Proposition 13

Since $v(-x) = 1 - v(x)$, $s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1) = \frac{s_1+s_2}{2} + H(s_1 - s_2)$, where $H(x) := x\hat{v}(x)$ is even, so the firm's expected utility is

$$u_F = \frac{p_1 + p_2}{2} + \mathbb{E}[H(s_1 - s_2)].$$

For $x \in (0,1]$, $0 < h(x) := H'(x) = \hat{v}(x) + xv'(x) \leq 2\hat{v}(x) \leq 2x$, since $v'(x)$ is decreasing on $[0,1]$ and $v'(0) \leq 1$, as $0.5 \geq \hat{v}(1) \geq \frac{v'(0)}{2}$ by the concavity of $v'(x)$. Also, $H(x) \leq \frac{x}{2}$.

Full revelation. $u_F = \frac{p_1+p_2}{2} + H(1)(p_1 + p_2 - 2p_1p_2)$, so $\partial_{p_1} u_F = 0.5 + H(1)(1 - 2p_2) > 0.5 - H(1) \geq 0$, and similarly for p_2.

Region (i). $u_F = \frac{p_1+p_2}{2} + (1-p_2)H(p_1) + p_2H(1-p_1)$, so $\partial_{p_2} u_F = 0.5 - H(p_1) + H(1-p_1) > 0$ and, as $p_2 \leq p_1$ and $h(x) \leq 2x$,

$$\partial_{p_1} u_F = 0.5 + (1 - p_2)h(p_1) - p_2h(1 - p_1) > 0.5 - 2p_1(1 - p_1) \geq 0.$$

Region (ii). Let $b = b(p_1)$ and $y := b - p_1$. By Step 1 of the proof of Proposition 2, $0 < y < p_1$ and $y' = -\frac{\partial_{p_1}\varphi}{\varphi'} > 0$. Then

$$u_F = \frac{p_1 + p_2}{2} + H(p_1) - \frac{p_2}{b}\big(H(p_1) - H(y)\big),$$

where $0 < H(p_1) - H(y) \leq \frac{p_1}{2} < \frac{b}{2}$. Hence $\partial_{p_2} u_F > 0$ and

$$\partial_{p_1} u_F = 0.5 + \Big(1 - \frac{p_2}{b}\Big)h(p_1) + \frac{p_2}{b}h(y)y' + \frac{p_2(1 + y')}{b^2}\big(H(p_1) - H(y)\big) > 0.$$

Region (iii). Let $a = a(p_2)$, which is non-decreasing by Step 2 of the proof of Proposition 2, with $p_2 \leq p_1 < a$ and $a \geq p^* > 0.5$. Then

$$u_F = \frac{p_1 + p_2}{2} + p_2H(1) + \frac{p_1}{a}K, \quad \text{where} \quad K := (1 - p_2)H(a) + p_2\big(H(1 - a) - H(1)\big),$$

so $\partial_{p_1} u_F = 0.5 + \frac{K}{a} \geq 0.5 - \frac{p_2H(1)}{a} > 0$ and, with $K_a := \partial_a K$,

$$\partial_{p_2} u_F = 0.5 + H(1) + \frac{p_1}{a}\big(H(1 - a) - H(1) - H(a)\big) + \frac{p_1a'}{a^2}(aK_a - K).$$

The first three terms sum to at least $0.5 - H(a) > 0$, and

$$aK_a - K = (1-p_2)a^2 v'(a) + p_2\big(\hat{v}(1) - \hat{v}(1-a) - a(1-a)v'(1-a)\big) \geq 0,$$

since, by the concavity of $v'(x)$ and $a \geq 0.5$,

$$\hat{v}(1) - \hat{v}(1-a) = \int_{1-a}^{1} v'(x)dx \geq \frac{a}{2}v'(1-a) \geq a(1-a)v'(1-a).$$

Example 10. Both signals solve the indifference conditions $N = 0$ from (3), which I solved numerically. For instance, for $p_2 = 0.84, 0.86, 0.873$, $(a, b) \approx (0.9375, 0.9996)$, $(0.9406, 0.9925)$ and $(0.9435, 0.9876)$, and $u_F \approx 0.96145$, 0.96092 and 0.96076.

A.13 Proof of Lemma 8

Write $u_i(p_i, p_j)$ for agent i's equilibrium payoff when his prior is p_i and agent j's prior is p_j. For $G_j \in \mathcal{G}(p_j)$, let $\overline{v}_{G_j}(x)$ be agent i's value function against G_j and $\overline{v}^{cv}_{G_j}$ its concavification on $[0,1]$. By Kamenica and Gentzkow (2011), $\overline{v}^{cv}_{G_j}(p_i) = \max_{H \in \mathcal{G}(p_i)} \mathbb{E}_H[\overline{v}_{G_j}]$ is agent i's best payoff against G_j, where the maximum is attained as $\overline{v}_{G_j}$ is continuous. The game is constant-sum, so agent j's equilibrium signal G_j^* minimizes i's payoff:

$$u_i(p_i, p_j) = \min_{G_j \in \mathcal{G}(p_j)} \overline{v}^{cv}_{G_j}(p_i) = \overline{v}^{cv}_{G_j^*}(p_i).$$

Step 1: each $\overline{v}^{cv}_{G_j}$ is non-decreasing on $[0,1]$, and strictly increasing if $v(x)$ is. As $v(x)$ is increasing, so is $\overline{v}_{G_j}$, hence $\overline{v}_{G_j} \leq \overline{v}_{G_j}(1)$ on $[0,1]$. The constant $\overline{v}_{G_j}(1)$ is then a concave majorant of $\overline{v}_{G_j}$, so $\overline{v}^{cv}_{G_j} \leq \overline{v}_{G_j}(1)$, while $\overline{v}^{cv}_{G_j}(1) = \overline{v}_{G_j}(1)$, since $\{1\}$ is the only signal with mean 1. So the concave function $\overline{v}^{cv}_{G_j}$ attains its maximum at $x = 1$. For $0 \leq x < y \leq 1$,

write $y = \lambda x + (1-\lambda)\cdot 1$ with $\lambda = \frac{1-y}{1-x} \in [0,1)$; by concavity,

$$\overline{v}^{cv}_{G_j}(y) \geq \lambda\, \overline{v}^{cv}_{G_j}(x) + (1-\lambda)\, \overline{v}^{cv}_{G_j}(1) \geq \overline{v}^{cv}_{G_j}(x).$$

If $v(x)$ is strictly increasing, so is $\overline{v}_{G_j}$. Then every $H \in \mathcal{G}(x)$ with $x < 1$ puts positive mass on $[0,1)$, where $\overline{v}_{G_j} < \overline{v}_{G_j}(1)$, so $\overline{v}^{cv}_{G_j}(x) = \max_{H\in\mathcal{G}(x)} \mathbb{E}_H[\overline{v}_{G_j}] < \overline{v}_{G_j}(1) = \overline{v}^{cv}_{G_j}(1)$, and the last inequality is strict.

Step 2: monotonicity of u_i. Let $p_i < p_i'$, and let G_j^* be agent j's equilibrium signal for priors (p_i', p_j). By Step 1,

$$u_i(p_i, p_j) \leq \overline{v}^{cv}_{G_j^*}(p_i) \leq (<)\ \overline{v}^{cv}_{G_j^*}(p_i') = u_i(p_i', p_j).$$

Finally, since $v(x) + v(-x) = 1$, $u_i(p_i, p_j) = 1 - u_j(p_j, p_i)$, so, by the above applied to agent j, u_i is non-increasing (strictly decreasing) in p_j.

B. Proofs, Technical Details and Additions for Section 4

B.1 Proof of Theorem 2

Lemma 9. *Let $v(x)$ be well S-shaped and $\hat{v}(x) := v(x) - 0.5$. Then $\hat{v}(x) \leq x(2-x)\hat{v}(1) \leq x - \frac{x^2}{2}$ for all $x \in [0,1]$, and if $\hat{v}(x) = x - \frac{x^2}{2}$ for some $x \in (0,1)$, then $v = v^*$.*

Proof. For $0 \leq t < s < 1$, s is a convex combination of t and 1 with weight $\frac{1-s}{1-t}$ on t, so the concavity of $v'(x)$ and $v'(1) \geq 0$ give $v'(s) \geq \frac{1-s}{1-t} v'(t)$. Hence $r(t) := \frac{v'(t)}{1-t}$ is non-decreasing on $[0,1)$: the density $v'(t)$ has a monotone likelihood ratio relative to the triangular density $1-t$. Let $T(x) := \int_0^x (1-t)dt = x - \frac{x^2}{2}$. For $x \in (0,1)$,

$$\frac{\hat{v}(x)}{T(x)} = \frac{\int_0^x r(t)(1-t)dt}{\int_0^x (1-t)dt} \leq r(x) \leq \frac{\int_x^1 r(t)(1-t)dt}{\int_x^1 (1-t)dt} = \frac{\hat{v}(1) - \hat{v}(x)}{T(1) - T(x)},$$

so $\hat{v}(x) \leq \frac{T(x)}{T(1)}\hat{v}(1) = x(2-x)\hat{v}(1) \leq x - \frac{x^2}{2}$, as $\hat{v}(1) \leq 0.5$. If $\hat{v}(x) = x - \frac{x^2}{2}$ for some $x \in (0,1)$, then $\hat{v}(1) = 0.5$ and both inequalities in the display are equalities, so r, which is continuous on $[0,1)$ by property 1) in Appendix A.1, is constant there and equal to $2\hat{v}(1) = 1$. Hence $v'(t) = 1 - t$ on $[0,1]$, and $v = v^*$, as $v'(x)$ is even. □

By Theorem 1 and Proposition 1, the equilibrium is $(\{0,a\};\{0,a\})$, where $a = 1$ if and only if $p \geq \overline{p}$, and $p < a \leq a^* := \min(2p, 1)$ by Lemma 6 and $\overline{p} \geq 0.5$ (Lemma 5). As in the proof of Proposition 4, $u_F(p) = p + 2p\left(1 - \frac{p}{a}\right)\hat{v}(a)$, so, by Lemma 9,

$$u_F(p) \leq p + 2p\left(1 - \frac{p}{a}\right)\left(a - \frac{a^2}{2}\right) \leq p + 2p\left(1 - \frac{p}{a^*}\right)\left(a^* - \frac{a^{*2}}{2}\right) = u_F^{ws}(p),$$

where the second inequality holds as both factors are positive and strictly increasing in a on $(p,1]$. This proves (i), and equality holds if and only if $a = a^*$ and $\hat{v}(a) = a - \frac{a^2}{2}$.

(ii) If $p \geq 0.5$, then $a^* = 1$, so equality means $a = 1$, i.e., $p \geq \overline{p}$, and $v(1) = 1$.

(iii) If $p < 0.5$, then $a^* = 2p \in (0,1)$, so equality requires $\hat{v}(2p) = 2p - 2p^2$, hence $v = v^*$ by Lemma 9. Conversely, for v^*, $\tilde{v}(x) = -x^2$ on $[0,1]$, so $a = 2p$ by Theorem 1, and both inequalities hold with equality.

B.2 Proof of Corollary 3

(i) Here $\tilde{v}(x) = -x^2$ on $[0,1]$, so $\overline{p} = 0.5$ and $v(1) = 1$; apply Theorem 2(ii) for $p \geq 0.5$ and (iii) for $p < 0.5$.

(ii) By the Fact in Appendix B.13, the utility of the firm under the noiseless rule is $\frac{4p}{3}$ for $p \leq 0.5$ and $1 - \frac{(1-p)^2(4p-1)}{3p^2}$ for $p > 0.5$. For $p \leq 0.5$, by (i),

$$p + 2p^2(1-p) - \frac{4p}{3} = -\frac{p}{3}\left(6p^2 - 6p + 1\right) > 0 \Leftrightarrow p \in \left(\frac{3-\sqrt{3}}{6}, \frac{3+\sqrt{3}}{6}\right).$$

For $p > 0.5$, by (i),

$$2p - p^2 - 1 + \frac{(1-p)^2(4p-1)}{3p^2} = \frac{(1-p)^2(4p-1-3p^2)}{3p^2} = \frac{(1-p)^3(3p-1)}{3p^2} > 0.$$

B.3 Proof of Proposition 5

Throughout, G_u is the equilibrium signal constructed in Proposition 14, and proper u-partitions are as defined in Appendix B.4.

(i) Let $p \leq 0.5$, $u < 2p$, $k = \lceil \frac{2p}{u} \rceil \geq 2$ and $r = 2p - (k-1)u \in (0, u]$. The proper u-partition of $[0, 2p]$ has atoms iu with masses $\frac{k-i}{k(k+1)}$ and atoms $iu + r = 2p - (k-1-i)u$ with masses $\frac{i+1}{k(k+1)}$, $i = 0, \ldots, k-1$. For fixed k, as u increases, iu moves at rate i and $iu + r$ at rate $-(k-1-i)$, so $\Psi_u(t) := \int_0^t G_u(x)dx$ satisfies $\partial_u \Psi_u(t) = -M(t)$, where $M(t)$ sums rate times mass over the atoms below t. The pair $\{iu, iu + r\}$ contributes $\frac{2i+1-k}{k(k+1)}$ and the atom iu alone $\frac{i(k-i)}{k(k+1)}$, hence $M(t) = 0$ for $t \in (ju, ju + r]$ and $M(t) = \frac{(j+1)(j+1-k)}{k(k+1)} < 0$ for $t \in (ju + r, (j+1)u]$, $j \leq k-2$. So $\Psi_u(t)$ is non-decreasing in u, and strictly increasing for t in a set of positive measure, i.e. G_u is a strict mean-preserving spread of $G_{u'}$ for $u' < u$ with the same k. At $u = \frac{2p}{k}$, $r = u$ and G_u is the uniform distribution on $\{0, \frac{2p}{k}, \ldots, 2p\}$, which is also its limit as $u \uparrow \frac{2p}{k}$, where $k+1$ is used and $r \to 0$; so G_u is continuous in u and the ranking extends to $(0, 2p]$. By Proposition 14, $G_u = \{0, \min(2p, 1)\}$ if $u \geq 2p$ and $p \leq 0.5$, or $u \geq 2 - 2p$ and $p > 0.5$.

For the example after Proposition 5, let $p \geq \frac{5}{9}$ and $u \in [1-p, 2-2p)$. Then $k = 1$ in case (iii) of the full statement in the proof of Proposition 14, and $a = 2u - \frac{u^2}{1-p} \leq 1 - u$, as $3u - \frac{u^2}{1-p}$ is maximized at $u = \frac{3(1-p)}{2}$, where it equals $\frac{9(1-p)}{4} \leq 1$. The proper u-partition of $[0, a]$ is $\{0, a\}$ with equal probabilities, so G_u puts probabilities $\frac{1-p}{2-a}$, $\frac{1-p}{2-a}$ and $\frac{2p-a}{2-a}$ on $\{0, a, 1\}$. As $\frac{da}{du} = 2 - \frac{2u}{1-p} < 0$ for $u > 1 - p$, a higher u lowers the probability of 0 and raises that of 1, so no two of these signals are ranked in the Blackwell order.

(ii) Since $v(-x) = 1 - v(x)$, with $d = s_1 - s_2$,

$$u_F = \mathbb{E}[s_1 v(d) + s_2 v(-d)] = \mathbb{E}[\max(s_1, s_2)] - \mathbb{E}\big[|d|\big(1 - v(|d|)\big)\big],$$

and $v(x) = 1 - \frac{(u-x)^2}{2u^2}$ for $x \in [0, u]$. Let $p \leq 0.5$. For $u \geq 2p$, $G_u = \{0, 2p\}$ and

$$u_F = \frac{p}{2} + p\, v(2p) = \frac{3p}{2} - \frac{p(u-2p)^2}{2u^2},$$

which equals $\frac{3p}{2}$ at $u = 2p$ and is strictly decreasing on $[2p, 1]$. For $u < 2p$, $u_F \leq \mathbb{E}[\max(s_1, s_2)] < \frac{3p}{2}$: for $s_2 \in [0, 2p]$, the convex function $\max(\cdot, s_2)$ lies below its chord on $[0, 2p]$, strictly on $(0, 2p)$ if $s_2 \in (0, 2p)$, so $\mathbb{E}_{s_1}[\max(s_1, s_2)] \leq p + \frac{s_2}{2}$, strictly if $s_2 \in (0, 2p)$, as G_u has mass in $(0, 2p)$; averaging over s_2 gives the bound $\frac{3p}{2}$, which is strict for the same reason.

Let $p > 0.5$. For $u \in [2 - 2p, 1]$, $G_u = \{0, 1\}$ and $v(1) = 1$, so $u_F = u_F^*(p)$. For $u < 2 - 2p$, $v(x)$ does not implement a fully revealing equilibrium by the proof of Proposition 14 (ii), so $u_F < u_F^*(p)$ by Lemma 4.

(iii) The proper u-partition of $[0, a]$ with $k = \lceil \frac{a}{u} \rceil$ puts mass $\frac{1}{k}$ on each pair $\{iu, iu + r\} \subset [iu, (i+1)u]$, $i = 0, \ldots, k-1$, so it converges weakly to $U[0, a]$ as $u \to 0$. If $p \leq 0.5$, G_u is the proper u-partition of $[0, 2p]$ and converges to $U[0, 2p]$. If $p > 0.5$ and $u < \min(2p - 1, 2 - 2p)$, then with k from Proposition 14 (iii), we have $a \in ((k-1)u, ku]$ and $ku < 2 - 2p \leq (k+1)u$, so $a \in (2 - 2p - 2u, 2 - 2p)$ and $a < 1 - u$. Hence case (iii) applies, $a \to 2 - 2p$, the weight $\frac{2(1-p)}{2-a} \to \frac{1-p}{p}$, and G_u converges to the $(\frac{1-p}{p}, \frac{2p-1}{p})$ mix of $U[0, 2-2p]$ and $\delta_{\{1\}}$ — exactly the distributions from Appendix B.13.

B.4 Triangular selection rules

This subsection constructs the symmetric equilibrium used in Subsection 4.2 for the selection rules generated by i.i.d. $\varepsilon_i \sim U[0, u]$ with $u < 1$. Since $v(x)$ is piecewise

quadratic, $v'(x)$ is piecewise linear, so the functional equation from Appendix D.2 reduces to a difference equation, and one can find $G_i(s_i)$ s.t. $\overline{v}(s_j)$ is linear on $[a, b]$.

Definition 8. Say χ is a proper u-partition of the interval $[a, b]$ if its CDF is given by:

$$G(x) = \begin{cases} 0, x < a \\ \frac{i+1}{k+1}, x \in [a + i \cdot u, b - (k-1-i) \cdot u), i \in \{0, \dots, k-1\} \\ \frac{i+1}{k}, x \in [b - (k-1-i) \cdot u, a + (i+1) \cdot u), i \in \{0, \dots, k-2\} \\ 1, x \geq b \end{cases}, \text{with } k = \lceil \frac{b-a}{u} \rceil$$

Remark 7. This G solves the functional equation $G(x-u) + G(x+u) = 2G(x)$, i.e. $\mathbb{P}(x < \chi \leq x+u) = \mathbb{P}(x-u < \chi \leq x)$, for all $x \in [a, b)$, subject to $\text{supp}(\chi) \subseteq [a, b]$.

The proper u-partition of $[a, b]$ is the unique $G_i(s_i)$ supported in $[a, b]$ that makes $\overline{v}(s_j) = \int_0^1 v(s_j - s_i) dG_i(s_i)$ linear on $[a, b]$. Both agents playing it is often not an equilibrium, since $\overline{v}(s_j)$ beyond $[a, b]$ also matters, but it is the key ingredient: the next proposition constructs a symmetric equilibrium for all (u, p) from mixtures of proper u-partitions and, possibly, $\delta_{\{1\}}$.

Proposition 14. *Let $v(x)$ be the CDF of a triangular distribution on $[-u, u]$. Then a symmetric equilibrium signal strategy is:*

(i) If $p \leq 0.5$, the signal is a proper u-partition of $[0, 2p]$.

(ii) If $p > 0.5$ and $u \geq 2 - 2p$, the signal is $\{0, 1\}$.

(iii) Otherwise, the signal mixes a proper u-partition of $[0, a]$ for some a with a mass point at 1 and (possibly) with a proper u-partition of $[0, 1-u]$.

So binary support survives a slight violation of well S-shapedness for extreme priors, $p \leq 0.5u$ or $p \geq 1 - 0.5u$, but fails for all $p \in (0.5u, 1 - 0.5u)$. The last case, whose exact equations are in the proof below, shows how complex equilibria can be

for general $v(x)$: as the signal 1 must be treated separately, an equilibrium may require a three-way mixture of two proper u-partitions and a mass point.

B.5 Proof of Proposition 14

Full statement: let $v(x)$ be the CDF of a triangular distribution on $[-u, u]$; then there exists a symmetric equilibrium in which each agent's signal G is:

(i) Proper u-partition of $[0, 2p]$ if $p \leq 0.5$

(ii) $\{0, 1\}$, if $p > 0.5, u \geq 2 - 2p$.

(iii) $(\frac{2(1-p)}{2-a}, \frac{2p-a}{2-a})$ mix of the proper u-partition of $[0, a]$ with $\delta_{\{1\}}$, if (i) $p > 0.5, u < 2 - 2p$, and (ii) $a \leq 1 - u$, where $a = 2ku - \frac{k(k+1)u^2}{2(1-p)}$, $k = \lceil \frac{2-2p}{u} \rceil - 1$.

(iv) If, in the solution of the equations in (iii), $a > 1 - u$, the true equilibrium G is constructed as follows: let $k = \lceil \frac{1}{u} \rceil$, and let $\tilde{G}$ be a $(\frac{k-1}{k}, \frac{1}{k})$ mix of the proper u-partition of $[0, 1 - u]$ and $\delta_{\{1\}}$. Then let G be the

$$\left(\frac{(2p - a)k}{(k-1)(1-u) + 2 - ka}, \frac{(k-1)(1-u) + 2 - 2pk}{(k-1)(1-u) + 2 - ka} \right)$$

mix of $\tilde{G}$ and the proper u-partition of $[0, a]$, where $a \in (1 - u, (k-1)u)$ is the unique solution of $(1 - a)\overline{v}(0) + a\overline{v}(1) = \overline{v}(a)$.

If $u \geq 1$, then $v(x)$ is quadratic and $\tilde{v}(x) = -x^2$ on $[0, 1]$; hence I refer the reader to Proposition 1 and Theorem 1. Consider now $u < 1$, where $v(x)$ is the CDF of the

triangular distribution on $[-u, u]$:

$$v(x) = \begin{cases} 0, & x < -u, \\ \frac{(u+x)^2}{2u^2}, & x \in [-u, 0), \\ 1 - \frac{(u-x)^2}{2u^2}, & x \in [0, u), \\ 1, & x \geq u, \end{cases} \qquad v'(x) = \begin{cases} 0, & x < -u, \\ \frac{u+x}{u^2}, & x \in [-u, 0), \\ \frac{u-x}{u^2}, & x \in [0, u), \\ 0, & x \geq u, \end{cases}$$

i.e. $v'(x) = \frac{\max(u-|x|,0)}{u^2}$.

Here I use the continuity of $v'(x)$, which makes $\overline{v}'(x)$ continuous.

(i) Suppose that $p \leq 0.5, u < 1$, and let $G(x)$ be the CDF of the proper u-partition χ on $[0, 2p]$ with $k = \lceil \frac{2p}{u} \rceil$. Let's analyze $\overline{v}'(x) = \int_0^1 v'(x-s)dG(s) = \int_0^{2p} v'(x-s)dG(s)$:

$$\begin{aligned} \int_0^{2p} v'(x-s)dG(s) &= \sum_{i=0}^{k-1} v'(x-iu)\mathbb{P}(\chi = iu) + \sum_{i=0}^{k-1} v'(x-2p+iu)\mathbb{P}(\chi = 2p-iu) \\ &= \sum_{i=0}^{k-1} v'(x-iu)\frac{k-i}{k(k+1)} + \sum_{i=0}^{k-1} v'(x-2p+iu)\frac{k-i}{k(k+1)} \\ &= \sum_{i=0}^{k-1} \frac{\max(u-|x-iu|,0)}{u^2}\frac{k-i}{k(k+1)} \\ &\quad + \sum_{i=0}^{k-1} \frac{\max(u-|x-2p+iu|,0)}{u^2}\frac{k-i}{k(k+1)} \\ &= \underbrace{\sum_{i=0}^{k} \frac{\max(u-|x-iu|,0)}{u^2}\frac{k-i}{k(k+1)}}_{S_L} \\ &\quad + \underbrace{\sum_{i=0}^{k} \frac{\max(u-|x-2p+iu|,0)}{u^2}\frac{k-i}{k(k+1)}}_{S_R}. \end{aligned}$$

Consider S_L and say that $a_i = \max(u - |x - iu|, 0)$ is *active* if $a_i > 0$. Then, as long as

$x \in (iu, (i+1)u)$, only a_i and a_{i+1} are active, so

$$S_L = \frac{(u-(x-iu))(k-i)+(u-((i+1)u-x))(k-i-1)}{u^2k(k+1)} = \frac{ku-x}{u^2k(k+1)}.$$

By the continuity of S_L at $x = iu$, $S_L = \frac{ku-x}{u^2k(k+1)}$ on $[0, ku]$, and hence on $[0, 2p]$, since $ku \geq 2p$. Analogously, $S_R = \frac{ku+x-2p}{u^2k(k+1)}$ on $[0, 2p]$, hence $\overline{v}'(x) = \frac{2ku-2p}{u^2k(k+1)}$ on $[0, 2p]$.

For $x > 2p$, $S_L = \frac{(ku-x)^+}{u^2k(k+1)}$ and $S_R = \frac{k(u-x+2p)^+}{u^2k(k+1)}$.

$$\overline{v}'(x) = \begin{cases} \frac{2ku-2p}{u^2k(k+1)}, & x \in [0, 2p], \\ \frac{2ku-x(1+k)+2kp}{u^2k(k+1)}, & x \in (2p, ku], \\ \frac{k(u-x+2p)}{u^2k(k+1)}, & x \in (ku, 2p+u], \\ 0, & x \geq 2p+u. \end{cases}$$

$\overline{v}'(x)$ is continuous, decreasing on $[2p, 1]$ and constant on $[0, 2p]$, hence $\overline{v}(x)$ is concave on $[0, 1]$, and linear on $[0, 2p]$, so indeed a pair of proper u-partitions of $[0, 2p]$ is an equilibrium. Also, $\overline{v}'(x)$ is convex on $[2p, 1]$.

(ii) Suppose that $p > 0.5$, $2 - 2p \leq u < 1$; then $\overline{v}(x) = (1-p)v(x) - pv(1-x)$. Then

$$\overline{v}'(x) = \frac{(1-p)\max(u-x, 0) + p\max(u+x-1, 0)}{u^2},$$

where the numerator is a piecewise linear convex function, hence $\overline{v}(x)$ is concave and then convex, and we need to check that $\overline{v}'(0) \leq \overline{v}(1) - \overline{v}(0)$.

$\overline{v}'(0) = \frac{1-p}{u}$, $\overline{v}(1) - \overline{v}(0) = v(1) - v(0) = 0.5$, so $\{0, 1\}$ is a best reply iff $2 - 2p \leq u$.

(iii) Suppose $p > 0.5, u < 2 - 2p$, and $a \leq 1 - u$, where $a = 2ku - \frac{k(k+1)u^2}{2(1-p)}$ for $k = \lceil \frac{2-2p}{u} \rceil - 1$. Then consider the $(\frac{2(1-p)}{2-a}, 1 - \frac{2(1-p)}{2-a})$ mix of the proper u-partition of $[0, a]$ with $\delta_{\{1\}}$; notice that this is a proper G, since the proper u-partition of $[0, a]$ has a mean $0.5a$, and $\frac{2(1-p)}{2-a} \cdot 0.5a + 1 - \frac{2(1-p)}{2-a} = p$.

First, $\lceil \frac{2-2p}{u} \rceil - 1 = \lceil \frac{a}{u} \rceil$, so k is the same in the proper u-partition of $[0, a]$:

$$\left\lceil \frac{a}{u} \right\rceil = \left\lceil 2k - \frac{k(k+1)u}{2(1-p)} \right\rceil = k,$$

since $k < \frac{2(1-p)}{u} \leq k+1$, hence $k \leq \frac{k(k+1)u}{2(1-p)} < k+1$.

Since $\overline{v}(x)$ is linear in the strategy of the other agent,

$$\overline{v}'(x) = \frac{2(1-p)}{2-a} S(x) + \frac{2p-a}{2-a} \cdot \frac{\max(u+x-1, 0)}{u^2},$$

where $S(x)$ is $\overline{v}'(x)$ from (i) against the proper u-partition χ of $[0, a]$. Hence $\overline{v}(x)$ is linear on $[0, a]$ with slope $\overline{v}'(0) = \frac{2(1-p)}{2-a} \cdot \frac{2ku-a}{u^2 k(k+1)}$. Since $v(1-s) = 1$ for $s \leq a \leq 1-u$,

$$\overline{v}(1) - \overline{v}(0) = \frac{2(1-p)}{2-a} \mathbb{E}[v(\chi)] + \frac{2p-a}{2(2-a)}, \qquad \mathbb{E}[v(\chi)] = 1 - \frac{1}{2(k+1)} - \frac{(ku-a)^2}{2u^2 k(k+1)},$$

and the equation $\overline{v}(1) - \overline{v}(0) = \overline{v}'(0)$ reduces to a linear equation in a with the root $a = 2ku - \frac{k(k+1)u^2}{2(1-p)}$. Hence $\int_a^1 (\overline{v}'(x) - \overline{v}'(a)) dx = 0$.

On $[a, 1]$ both summands of $\overline{v}'(x)$ are convex: the first by (i), the second as a maximum of linear functions. Hence $\overline{v}'(x) - \overline{v}'(a)$ is convex on $[a, 1]$, vanishes at a and integrates to zero, so $\exists c : \overline{v}'(x) \leq \overline{v}'(a)\ \forall x \in [a, c]$ and $\overline{v}'(x) \geq \overline{v}'(a)\ \forall x \in [c, 1]$. Therefore $\overline{v}(x) \leq \overline{v}(a) + (x-a)\overline{v}'(a)$ on $[a, 1]$, with equality at $x = 1$, so $[0, a] \cup \{1\}$ is the concavification support, and (G, G) is an equilibrium.

(iv) Notice that G is a feasible distribution, since the mean of $\tilde{G}$ is $\frac{k-1}{k} \frac{1-u}{2} + \frac{1}{k} = \frac{Q}{2k}$ with $Q := (k-1)(1-u) + 2$, the mean of the proper u-partition of $[0, a]$ is $\frac{a}{2}$, and $\beta := \frac{(2p-a)k}{Q-ka}$ solves $\beta \frac{Q}{2k} + (1-\beta)\frac{a}{2} = p$.

Suppose that a from (iii) satisfies $a > 1-u$. I first show that if the other agent plays the $(\frac{k-1}{k}, \frac{1}{k})$ mix of the proper u-partition of $[0, 1-u]$ and $\delta_{\{1\}}$, where $k = \lceil \frac{1}{u} \rceil$, then

$\overline{v}(x)$ is linear on $[0,(k-1)u]$. Again, consider $\overline{v}'(x)$, noting that $\lceil \frac{1-u}{u} \rceil = k-1$:

$$\overline{v}'(x) = \frac{k-1}{k}\Bigg[\underbrace{\sum_{i=0}^{k-1} \frac{\max(u-|x-iu|,0)}{u^2}\frac{k-1-i}{k(k-1)}}_{S_L} + \underbrace{\sum_{i=0}^{k-1} \frac{\max(u-|x-1+(i+1)u|,0)}{u^2}\frac{k-1-i}{k(k-1)}}_{S_R}\Bigg] + \frac{1}{k}\frac{\max(u-|x-1|,0)}{u^2}.$$

By the proof of (i), $\overline{v}'(x)$ is constant on $[0,1-u]$ (since the last term is zero on that interval) and equal to $\frac{k-1}{k}\frac{2(k-1)u-(1-u)}{u^2k(k-1)} = \frac{(2k-1)u-1}{u^2k^2}$. On $[1-u,(k-1)u]$, the only active terms are a_{k-2} in S_L, a_0 in S_R and the last term.

$$\begin{aligned}\overline{v}'(x) &= \frac{k-1}{k}\left[\frac{((k-1)u-x)}{u^2}\frac{1}{k(k-1)} + \frac{1-x}{u^2}\frac{k-1}{k(k-1)}\right] + \frac{1}{k}\frac{x+u-1}{u^2} \\ &= \frac{((k-1)u-x)+(k-1)(1-x)+k(x+u-1)}{k^2u^2} = \frac{(2k-1)u-1}{k^2u^2}.\end{aligned}$$

Finally, on $[(k-1)u,1]$, only a_0 in S_R and the last term are active.

$$\overline{v}'(x) = \frac{k-1}{k}\frac{1-x}{u^2}\frac{k-1}{k(k-1)} + \frac{1}{k}\frac{x+u-1}{u^2} = \frac{(k-1)(1-x)+k(x+u-1)}{k^2u^2} = \frac{x+ku-1}{k^2u^2}.$$

So against $\tilde{G}$, $\overline{v}'(x)$ is constant on $[0,(k-1)u]$ and increasing linear on $[(k-1)u,1]$, while against the proper u-partition of $[0,a]$ it is constant on $[0,a]$ and convex on $[a,1]$ by (i). Hence, if the other agent plays G, $\overline{v}(x)$ is linear on $[0,a]$ and $\overline{v}'(x)$ is convex on $[a,1]$, and as in (iii), if a solves $(1-a)\overline{v}(0)+a\overline{v}(1) = \overline{v}(a)$, then $[0,a]\cup\{1\}$ is the concavification support, and (G,G) is an equilibrium.

It remains to show that such an $a \in (1-u,(k-1)u)$ exists and is unique, and that $\beta \in (0,1)$. Let a_3 and $k_3 = \lceil \frac{2-2p}{u} \rceil - 1$ be a and k from (iii). Since $a_3 \leq k_3 u$ and

$k_3 \le k-1$, $a_3 > 1-u > (k-2)u$ requires $k_3 = k-1$, and then $a_3 > 1-u$ is equivalent to $p < p_u := 1 - \frac{k(k-1)u^2}{2((2k-1)u-1)}$. Let χ be the proper u-partition of $[0,a]$ and, for a signal H of the other agent, let s_H be the slope of $\overline{v}(x)$ on $[0,a]$ and $C_H := \overline{v}(1) - \overline{v}(0)$, so that the condition on a is $s_G = C_G$. A direct computation gives

$$s_{\tilde{G}} - C_{\tilde{G}} = -\frac{((k-1)u-1)^2}{2k^2u^2}, \qquad \frac{d}{da}(s_\chi - C_\chi) = -\frac{1-a}{(k-1)u^2},$$

and $s_\chi = \frac{2(k-1)u-a}{u^2k(k-1)}$ by (i). Hence

$$\Lambda(a) := (Q-ka)(s_G - C_G) = -(2p-a)\frac{((k-1)u-1)^2}{2ku^2} + (Q-2kp)(s_\chi - C_\chi)$$

has the sign of $s_G - C_G$, as $Q - ka \ge (k+1)(1-(k-1)u) > 0$, and $\Lambda(a)$ is quadratic in a with the leading coefficient $\frac{Q-2kp}{2(k-1)u^2}$. Since $p < p_u$ and $\frac{Q}{2k} - p_u = \frac{(k-1)((k-1)u-1)^2}{2k((2k-1)u-1)} > 0$, $2kp < Q$ and Λ is strictly convex. Moreover,

$$\Lambda(1-u) = \frac{(1+u)((2k-1)u-1)}{k(k-1)u^2}(p_u - p) > 0,$$
$$\Lambda((k-1)u) = -\frac{(k+1)(2p-1)((k-1)u-1)^2}{2ku^2} < 0,$$

so Λ has a unique root in $(1-u,(k-1)u)$. And $a < 1 < 2p$ and $2kp < Q$ give $\beta \in (0,1)$.

B.6 Proof of Lemma 4

Since $v(-x) = 1 - v(x)$, for $s_1 \ge s_2$,

$$\begin{aligned} s_1 v(s_1-s_2) + s_2 v(s_2-s_1) &= s_2 + (s_1-s_2)v(s_1-s_2) \le s_2 + (s_1-s_2)v(1) \\ &= v(1)\max(s_1,s_2) + (1-v(1))\min(s_1,s_2) \le \max(s_1,s_2), \end{aligned}$$

where the first inequality holds as $v(x)$ is increasing and $s_1 - s_2 \leq 1$, and the second as $v(1) \leq 1$, strictly if $s_1 \neq s_2$ and $v(1) < 1$. Hence $u_F^*(p) \leq \max_{G_1, G_2 \in \mathcal{G}(p)} \mathbb{E}[\max(s_1, s_2)]$. For $s_2 \in [0,1]$, $\max(s, s_2)$ lies below its secant $s_2 + s(1 - s_2)$ on $[0,1]$, strictly on $(0,1)$ if $s_2 \in (0,1)$. Hence $\mathbb{E}_{G_1}[\max(s_1, s_2)] \leq s_2 + p(1 - s_2)$, with equality for all s_2 if G_1 is full revelation, and for $s_2 \in (0,1)$ only then. Averaging over s_2, $\mathbb{E}[\max(s_1, s_2)] \leq 2p - p^2$, with equality if and only if at least one agent plays full revelation. If both agents play full revelation and $v(1) = 1$, then $s_1 - s_2 \in \{-1, 0, 1\}$ and both inequalities above are equalities, so $u_F^*(p) = 2p - p^2$.

By definition, $\overline{u}_F(p) \leq u_F^*(p)$. Let (G_1^*, G_2^*) be an equilibrium with $\mathbb{E}[s_1 v(s_1 - s_2) + s_2 v(s_2 - s_1)] = u_F^*(p)$. Then all inequalities above are equalities, so some agent plays full revelation, and since then $s_1 \neq s_2$ with positive probability, $v(1) = 1$. As all equilibria form a symmetric product set, $(\{0,1\}; \{0,1\})$ is then an equilibrium too. Conversely, if $(\{0,1\}; \{0,1\})$ is an equilibrium and $v(1) = 1$, then $s_1 - s_2 \in \{-1, 0, 1\}$ and both inequalities are equalities, so $\overline{u}_F(p) = u_F^*(p)$.

B.7 Robustness across unequal priors

Lemma 10. *Let $S \subseteq (0,1)$. A selection rule $v(x)$ implements a fully revealing equilibrium for all $(p_1, p_2) \in S^2$ if and only if it does so for all (p, p) with $p \in S$.*

The *only if* direction is straightforward; as for *if*, suppose that $v(x)$ implements a fully revealing equilibrium for some (p, p) with $p \in S$, and let agent j with prior $p_j = p$ play full revelation. Agent i's value function does not depend on his own prior, and since full revelation is a best reply for $p_i = p$, its concavification is the secant from 0 to 1, so full revelation is a best reply for every $p_i \in (0,1)$. Hence, if $v(x)$ implements a fully revealing equilibrium for all (p, p) with $p \in S$, then it does so for all $(p_1, p_2) \in S^2$.

B.8 Proof of Lemma 10

Let agent j with prior p_j play full revelation. Then $\overline{v}(s_i) - 0.5 = (1-p_j)\hat{v}(s_i) - p_j\hat{v}(1-s_i)$ does not depend on p_i, and full revelation is a best reply of agent i for any $p_i \in (0,1)$ if and only if $\overline{v}$ lies below its secant from 0 to 1:

$$(1-p_j)\hat{v}(x) - p_j\hat{v}(1-x) \leq \hat{v}(1)(x-p_j) \quad \forall x \in (0,1).$$

Hence $(\{0,1\};\{0,1\})$ is an equilibrium for (p_1,p_2) if and only if this condition holds for $p_j \in \{p_1,p_2\}$, so it is an equilibrium for all $(p_1,p_2) \in S^2$ if and only if the condition holds for all $p_j \in S$, i.e., if and only if it is an equilibrium for all (p,p), $p \in S$.

B.9 Proof of Proposition 6

Let $a := \hat{v}(1)$; since $v(x) \leq 1$ is increasing, $a \in [0, 0.5]$. By the proof of Lemma 10, $v(x)$ implements a fully revealing equilibrium for (p,p) if and only if $(1-p)\hat{v}(x) - p\hat{v}(1-x) \leq a(x-p)$ for all $x \in (0,1)$. Applying it also at $1-x$ and summing the two inequalities with weights $1-p$ and p gives $(1-2p)\big(\hat{v}(x) - ax\big) \leq 0$. Taking $p = p_l$ and $p = p_h$ yields $\hat{v}(x) = ax$, which proves (ii); then the condition holds with equality for all $p \in (0,1)$, and Lemma 10 extends full revelation to all pairs of priors, which proves (i).

B.10 Proof of Proposition 7

(i) Let a and $g(x)$ be as in Definition 6. Then the condition $(1-p)\hat{v}(x) - p\hat{v}(1-x) \leq a(x-p)$ $\forall x \in (0,1)$ from the proof of Lemma 10 simplifies to

$$(1-p)g(x) - pg(1-x) \leq 0 \quad \forall x \in (0,1).$$

Since this condition is linear in p, it holds $\forall p \in [0.5, 1)$ if and only if it holds for

$p = 0.5$ and $p = 1$; the latter means $g(1-x) \geq 0 \; \forall x \in (0,1)$.

$p = 0.5$ implies $0.5g(x) - 0.5g(1-x) \leq 0 \; \forall x \in (0,1)$. Replacing x with $1-x$ gives $0.5g(x) - 0.5g(1-x) \geq 0 \; \forall x \in (0,1)$, so $v(x)$ is balanced.

Finally, since $\hat{v}(x)$ should be increasing, on the one hand, for $x < y$, $g(x) + ax \leq g(y) + ay$; on the other hand, for $1 - y < 1 - x$,

$$g(1-y) + a(1-y) \leq g(1-x) + a(1-x) \;\Leftrightarrow\; g(y) - ay \leq g(x) - ax,$$

hence $|g(x) - g(y)| \leq a(y-x) = a|x-y|$, and the converse is immediate.

The proof for $p \in (0, 0.5]$ is done analogously.

(ii) Since all equilibria form a product set, if full revelation is each agent's unique best reply to full revelation, it is the unique equilibrium.

The uniqueness of the best reply is equivalent to $(1-p)\hat{v}(x) - p\hat{v}(1-x) < a(x-p) \; \forall x \in (0,1)$, and it can be rewritten as $(1-2p)g(x) < 0 \; \forall x \in (0,1)$, if $v(x)$ is balanced. Hence, if $g(x) < 0 \; \forall x \in (0,1)$, this condition holds $\forall p < 0.5$, and vice versa.

B.11 Proof of Corollary 4

Suppose that agent j plays $\{0,1\}$. Then $\overline{v}(s_i) = (1-p_j)v(s_i) + p_j v(s_i - 1)$, and $\{0,1\}$ is a best reply of agent i only if the secant from 0 to 1 lies weakly above $\overline{v}(s_i)$ on $[0,1]$, which requires $\overline{v}'(0) \leq \overline{v}(1) - \overline{v}(0) \leq \overline{v}'(1)$. Since $v'(x)$ is even, $\overline{v}'(0) = (1-p_j)v'(0) + p_j v'(1)$ and $\overline{v}'(1) = (1-p_j)v'(1) + p_j v'(0)$, so this requires $(2p_j - 1)\big(v'(0) - v'(1)\big) \geq 0$, i.e., $p_j \geq 0.5$. Hence in a fully revealing equilibrium both priors are at least 0.5.

If $v(x)$ is generated by i.i.d. shocks with a square-integrable density f, then $v'(x) = \int f(t)f(t+x)dt$ is continuous, and by the Cauchy–Schwarz inequality,

$$v'(1) = \int f(t)f(t+1)dt \leq \int f^2(t)dt = v'(0),$$

with equality only if $f(t+1) = f(t)$ almost everywhere, which is impossible for a probability density. Hence $v'(0) > v'(1)$.

B.12 Costly signals

Suppose that signals are costly: agent i pays $\kappa\, c(G_i)$, where $\kappa > 0$ and $c(G_i) = \int_0^1 \phi\, dG_i - \phi(p_i)$ for a strictly convex ϕ, continuous on $[0,1]$, so that his payoff is $\mathbb{E}[v(s_i - s_j)] - \kappa\, c(G_i)$. The firm does not bear the cost, and her utility is unchanged.

Lemma 11. *For any priors and selection rule, equilibria exist and form a product set.*

Proof. Since agent i's cost does not depend on G_j, both agents have the same best replies as in the zero-sum game in which agent 1 receives $W(G_1, G_2) := \mathbb{E}[v(s_1 - s_2)] - \kappa c(G_1) + \kappa c(G_2)$ and agent 2 receives $-W(G_1, G_2)$. W is affine in each G_i and continuous in the weak* topology, as v and ϕ are continuous, and each $\mathcal{G}(p_i)$ is convex and weak* compact. By Sion's minimax theorem, this game has a value and optimal strategies, and its equilibria are exactly the pairs of optimal strategies. □

Unlike the value of this zero-sum game, the agents' own payoffs need not be the same in all equilibria.

B.12.1 The variance cost

Let $\phi(x) = x^2$, so that $c(G_i) = \mathrm{Var}_{G_i}[s_i]$ is the expected reduction in the residual variance of θ_i. As the linear part of $v(x)$ adds only a linear function of s_i to agent i's value function, best replies depend on $v(x)$ only through $\tilde{v}(x)$ and

$$\lambda := \frac{\kappa}{|v(1) - 0.5 - v'(0)|},$$

and the normalized value function against G_j is $f(s_i) := \overline{v}(s_i) - \lambda s_i^2$.

Proposition 15. *Let $v(x)$ be well S-shaped, the priors (p, p), and $c(G_i) = Var_{G_i}[s_i]$.*

(i) If $p \geq \overline{p}_\lambda := 1 - \frac{1-\lambda}{\mathcal{E}_{\tilde{v}}(1)}$, the unique equilibrium is $(\{0,1\};\{0,1\})$.

(ii) If $\lambda \geq \frac{|\tilde{v}(p)|}{p^2}$, the unique equilibrium is $(\{p\};\{p\})$.

(iii) Otherwise, the unique equilibrium is $(\{0,a\};\{0,a\})$, where $a \in (p,1)$ solves $p = a - \frac{|\tilde{v}(a)|-\lambda a^2}{|\tilde{v}'(a)|}$, and $\mathbb{P}(s_i = 0) = \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\left(1 - \frac{\lambda a^2}{|\tilde{v}(a)|}\right)$.

The conditions in (i) and (ii) are mutually exclusive.

For $\lambda = 0$, (ii) never applies, and Proposition 15 reduces to Theorem 1. Since λ is the cost relative to the curvature of $v(x)$, noise that flattens $v(x)$ makes the same cost weigh more. For instance, for $\varepsilon_i \sim U[0,u]$ with $u \geq 1$, $\tilde{v}(x) = -x|x|$ and $\lambda = 2\kappa u^2$: agents reveal fully if $p \geq \frac{1+\lambda}{2}$, pool if $\lambda \geq 1$, and otherwise play $\{0, \frac{2p}{1+\lambda}\}$ with $\mathbb{P}(s_i = 0) = \frac{1-\lambda}{2}$. Without the cost, informativeness does not depend on $u \geq 1$; with it, it decreases in u.

Proof. Exclusivity. By the concavity of $\tilde{v}(x)$ on $[0,1]$ (property 2 in Appendix A.1), $\tilde{v}(p) \leq \tilde{v}(1) + (p-1)\tilde{v}'(1) = -1 + (1-p)\mathcal{E}_{\tilde{v}}(1)$. So if (i) holds, $\lambda \leq 1 - (1-p)\mathcal{E}_{\tilde{v}}(1) \leq |\tilde{v}(p)| < \frac{|\tilde{v}(p)|}{p^2}$.

Uniqueness. By Lemma 11 and symmetry, all equilibria form a product set $E \times E$. So if $G^* \in E$ is the unique best reply to itself, then $E = \{G^*\}$, as every $G \in E$ is a best reply to G^*.

Best reply to $\{0,a\}$. Let agent j play $\{0,a\}$ with $a \in (p,1]$, and let $\gamma := \frac{a-p}{a} \in (0,1)$. Then

$$f(s) = \gamma\tilde{v}(s) + (1-\gamma)\tilde{v}(s-a) - \lambda s^2.$$

Let $l(s)$ be the line through $(0, f(0))$ and $(a, f(a))$, whose slope is $\frac{\tilde{v}(a)-\lambda a^2}{a}$, and $d(s) := l(s) - f(s)$ on $[0,a]$, so that $d(0) = d(a) = 0$. As $\tilde{v}'(x)$ is even,

$$d'(s) = \frac{\tilde{v}(a) - \lambda a^2}{a} - \gamma\tilde{v}'(s) - (1-\gamma)\tilde{v}'(a-s) + 2\lambda s,$$

which is convex on $[0, a]$, since $\tilde{v}'(x)$ is concave. If $d'(a) \leq 0$, then $d \geq 0$ on $[0, a]$: with $z := \inf\{s \in [0, a] : d'(s) \leq 0\}$, convexity gives $d' \leq 0$ on $[z, a]$, while $d' > 0$ on $[0, z)$. If $a < 1$ and $d'(a) = 0$, then for $s \in (a, 1]$,

$$f'(s) = \gamma\tilde{v}'(s) + (1-\gamma)\tilde{v}'(s-a) - 2\lambda s < \gamma\tilde{v}'(a) + (1-\gamma)\tilde{v}'(0) - 2\lambda a = f'(a),$$

as $\tilde{v}'(x)$ is strictly decreasing on $[0, 1]$, so f lies strictly below its tangent l on $(a, 1]$. Hence, if $d'(a) \leq 0$ for $a = 1$, or $d'(a) = 0$ for $a < 1$, then $\{0, a\}$ is a best reply to itself. Since $f'(a) = \gamma\tilde{v}'(a) - 2\lambda a$,

$$d'(a) = 0 \;\Leftrightarrow\; \gamma\tilde{v}'(a) = \frac{\tilde{v}(a)}{a} + \lambda a \;\Leftrightarrow\; \gamma = \frac{1}{\mathcal{E}_{\tilde{v}}(a)}\Big(1 - \frac{\lambda a^2}{|\tilde{v}(a)|}\Big) \;\Leftrightarrow\; p = a - \frac{|\tilde{v}(a)| - \lambda a^2}{|\tilde{v}'(a)|},$$

where the last step uses $\gamma a = a - p$.

Step 1: other best replies. Suppose that $d'(a) = 0$ and $d(c) = 0$ for some $c \in (0, a)$. Then c minimizes d on $[0, a]$, so $d'(c) = 0$, and as in the proof of Theorem 1, convexity and $d(0) = d(c) = d(a) = 0$ give $d' \equiv 0$ on $[0, a]$. So $\gamma\tilde{v}'(s) + (1-\gamma)\tilde{v}'(a-s)$ is affine on $[0, a]$; as a sum of two concave functions, it is affine only if both are, so $\tilde{v}'(s) = -Ks$ on $[0, a]$ for some $K > 0$, and $d' \equiv 0$ then requires $K(1-2\gamma) = 2\lambda$, i.e., $\frac{p}{a} = 1 - \gamma = \frac{1}{2} + \frac{\lambda}{K}$. Hence, either $d > 0$ on $(0, a)$, and $\{0, a\}$ is the unique best reply to itself, or $\tilde{v}(x) = -\frac{K}{2}x|x|$ on $[-a, a]$ and $p = a\big(\frac{1}{2} + \frac{\lambda}{K}\big)$, and every $G \in \mathcal{G}(p)$ supported on $[0, a]$ is a best reply to $\{0, a\}$.

Step 2: the quadratic case. Take such a $G \neq \{0, a\}$ and let $w(s) := \int_0^a \tilde{v}(s-t)dG(t) - \lambda s^2$ be the normalized value function against it. For $s \in [0, a]$, $w'(s) = -K\int_0^a |s - t|dG(t) - 2\lambda s$, hence $w'(a) = -K(a-p) - 2\lambda a$, while $w(0) = \frac{K}{2}\mathbb{E}_G[t^2]$ and $w(a) = -\frac{K}{2}\mathbb{E}_G[(a-t)^2] - \lambda a^2$. So the slope of the secant from $(0, w(0))$ to $(a, w(a))$ exceeds $w'(a)$ by

$$\frac{K}{a}\Big(\frac{a^2}{2} - \mathbb{E}_G[t^2]\Big) + \lambda a > \frac{K}{a}\Big(\frac{a^2}{2} - ap\Big) + \lambda a = 0,$$

since $t^2 \leq at$ on $[0, a]$, with equality only at $t \in \{0, a\}$, and $p = a\left(\frac{1}{2} + \frac{\lambda}{K}\right)$. Hence $w(s)$ lies strictly above the secant for $s < a$ close enough to a, so, for such $s > p$, $\{0, s\}$ yields a strictly higher payoff against G than $\{0, a\}$. Thus $\{0, a\}$ is not a best reply to G, $G \notin E$, and the equilibrium is unique.

(i) For $a = 1$, $\gamma = 1 - p$ and $d'(1) = -1 + \lambda + (1 - p)\mathcal{E}_{\tilde{v}}(1)$, so $d'(1) \leq 0$ if and only if $p \geq \overline{p}_\lambda$, and then $\{0, 1\}$ is a best reply to itself. If $p > \overline{p}_\lambda$, then $d > 0$ on $(0, 1)$: otherwise, $d(c) = 0$ for some $c \in (0, 1)$ gives $d'(c) = 0$, so $d' \leq 0$ on $[c, 1]$ by convexity, and $\int_c^1 d'(s)ds = 0$ gives $d' \equiv 0$ on $[c, 1]$, contradicting $d'(1) < 0$. So $\{0, 1\}$ is the unique best reply to itself, and the equilibrium is unique. If $p = \overline{p}_\lambda$, then $d'(1) = 0$, and Steps 1 and 2 with $a = 1$ give uniqueness.

(ii) Against $\{p\}$, $f(s) = \tilde{v}(s - p) - \lambda s^2$, and $f'(p) = -2\lambda p$. So $\{p\}$ is a best reply if and only if f lies below its tangent at p, i.e., if and only if $\tilde{v}(s - p) \leq \lambda(s - p)^2$ on $[0, 1]$. For $s > p$, this holds strictly, as $\tilde{v} < 0$ on $(0, 1]$. For $s < p$, with $t = p - s$, it reads $\frac{|\tilde{v}(t)|}{t^2} \leq \lambda$, and as this ratio is non-decreasing on $(0, 1]$ by Lemma 5, it holds for all $t \in (0, p]$ if and only if $\lambda \geq \frac{|\tilde{v}(p)|}{p^2}$. Then every best reply to $\{p\}$ is supported on $[0, p]$ and has mean p, so it is $\{p\}$, and the equilibrium is unique.

(iii) Let $p < \overline{p}_\lambda$ and $\lambda < r(p)$, where $r(t) := \frac{|\tilde{v}(t)|}{t^2}$ is continuous and non-decreasing on $(0, 1]$. Let $h(a) := a - \frac{|\tilde{v}(a)| - \lambda a^2}{|\tilde{v}'(a)|} = a\left(1 - \frac{1 - \lambda/r(a)}{\mathcal{E}_{\tilde{v}}(a)}\right)$, continuous on $(0, 1]$, with $h(1) = \overline{p}_\lambda > p$. Let $a_0 := \sup\{t \in (0, 1] : r(t) \leq \lambda\}$, and $a_0 := 0$ if this set is empty; by continuity and monotonicity of r, $a_0 < p$. If $a_0 > 0$, then $r(a_0) = \lambda$ and $h(a_0) = a_0 < p$. If $a_0 = 0$, then $\lambda < r(a)$ for all $a \in (0, 1]$, so $h(a) \in [\frac{a}{2}, a)$ by Lemma 5, and $h(a) \to 0$ as $a \to 0$. In both cases, by continuity, $h(a) = p$ for some $a \in (a_0, 1)$. As $r(a) > \lambda$ there, $p = h(a) < a$, so $\gamma \in (0, 1)$, and $d'(a) = 0$ by the equivalence above. Hence $\{0, a\}$ is a best reply to itself, with $\mathbb{P}(s_i = 0) = \gamma$ as stated, and Steps 1 and 2 give uniqueness. □

B.12.2 Robust implementation

For the rules of Section 4, let ϕ be continuously differentiable on $[0,1]$, including the endpoints, and let $\psi(s) := (1-s)\phi(0) + s\phi(1) - \phi(s)$, which is positive on $(0,1)$ by strict convexity, with $\lim_{s\to 0} \frac{\psi(s)}{s} = \phi(1) - \phi(0) - \phi'(0) > 0$ and $\lim_{s\to 1} \frac{\psi(s)}{1-s} = \phi'(1) - \phi(1) + \phi(0) > 0$.

Proposition 16. *Let ϕ be continuously differentiable on $[0,1]$.*

(i) If $v(x) = 0.5 + ax$, the unique equilibrium is $(\{p_1\};\{p_2\})$ for all (p_1, p_2).

(ii) Let $v(x)$ be strictly positively [negatively] balanced with $\liminf_{s\to 0_+} \frac{|g(s)|}{s} > 0$. Then $m := \inf_{s\in(0,1)} \frac{|g(s)|}{\psi(s)} > 0$, and full revelation is the unique equilibrium for all (p_1, p_2) with $p_1, p_2 > 0.5 + \frac{\kappa}{2m}$ $\left[p_1, p_2 < 0.5 - \frac{\kappa}{2m}\right]$.

(iii) Let $v(x)$ be positively or negatively balanced with $\liminf_{s\to 0_+} \frac{|g(s)|}{s} = 0$. Then $(\{0,1\};\{0,1\})$ is not an equilibrium for any (p_1, p_2).

For the variance cost, $\psi(s) = s(1-s)$; for $v^*(x)$, $g(s) = \frac{s(1-s)}{2}$, so $m = \frac{1}{2}$ and (ii) gives full revelation for $p > 0.5 + \kappa$, in line with Proposition 15(i), as $\lambda = 2\kappa$. If instead $\phi'(0) = -\infty$, as for entropy, then $\frac{\psi(s)}{s} \to \infty$ as $s \to 0$, while $\frac{|g(s)|}{s} \le a$ for every balanced rule, so by $(*)$ below, no balanced rule implements full revelation for any $\kappa > 0$.

Proof. (i) Agent i's payoff is $0.5 + a(p_i - p_j) - \kappa c(G_i)$, and by strict convexity, $c(G_i) > 0$ unless $G_i = \{p_i\}$.

(ii)–(iii) Let agent j play $\{0,1\}$. With $\hat{v}(x) = ax + g(x)$ on $[0,1]$ and $g(x) = g(1-x)$, agent i's value function is, up to a constant,

$$(1-p_j)\hat{v}(s) - p_j\hat{v}(1-s) = a(s-p_j) + (1-2p_j)g(s),$$

and he maximizes the expectation of this function minus $\kappa\phi(s)$. By Kamenica and Gentzkow (2011), $\{0,1\}$ is a best reply if and only if this function lies below its chord

on $[0,1]$. As $g(0) = g(1) = 0$, this reads

$$(2p_j - 1)g(s) \geq \kappa\psi(s) \quad \forall s \in (0,1), \tag{$*$}$$

which does not depend on p_i; if $(*)$ is strict on $(0,1)$, $\{0,1\}$ is the unique best reply.

(ii) Consider the positive case. The ratio $\frac{g}{\psi}$ is continuous and positive on $(0,1)$, and $\liminf_{s\to 0} \frac{g(s)}{\psi(s)} > 0$ by the limit of $\frac{\psi(s)}{s}$ above. As $g(s) = g(1-s)$, $\liminf_{s\to 1} \frac{g(s)}{1-s} > 0$ too, so $\liminf_{s\to 1} \frac{g(s)}{\psi(s)} > 0$, and hence $m > 0$. If $p_j > 0.5 + \frac{\kappa}{2m}$, then $(2p_j - 1)g(s) \geq (2p_j - 1)m\psi(s) > \kappa\psi(s)$ on $(0,1)$, so $(*)$ is strict. Hence, if both priors exceed $0.5 + \frac{\kappa}{2m}$, full revelation is each agent's unique best reply to itself, and by Lemma 11, it is the unique equilibrium. The negative case is symmetric.

(iii) Consider the positive case. If $p_j \leq 0.5$, the left side of $(*)$ is non-positive, and the right side is positive. If $p_j > 0.5$, take $s_n \to 0$ with $\frac{g(s_n)}{s_n} \to 0$; then $(2p_j - 1)\frac{g(s_n)}{s_n} \to 0$, while $\kappa\frac{\psi(s_n)}{s_n} \to \kappa\big(\phi(1) - \phi(0) - \phi'(0)\big) > 0$, so $(*)$ fails. The negative case is symmetric. □

B.13 Equations for the noiseless rule

Fact (Boleslavsky and Cotton (2015) Lemma 3)**.** *The unique symmetric equilibrium for* $v(x) = 0.5 + 0.5 \cdot \text{sign}(x)$ *has the form* (G, G)*, where for* $p \geq 0.5$

$$G(x) = \begin{cases} \frac{x}{2p}, & x \leq 2 - 2p, \\ \frac{1}{p} - 1, & 2 - 2p \leq x < 1, \\ 1, & x = 1, \end{cases}$$

while for $p \leq 0.5$

$$G(x) = \begin{cases} \frac{x}{2p}, & x \leq 2p, \\ 1, & x \in [2p, 1]. \end{cases}$$

Hence the firm's equilibrium utility is

$$u_F^0(p) = \mathbb{E}_G[\max(s_1, s_2)] = \begin{cases} \frac{4p}{3}, & 0 < p \leq 0.5, \\ 1 - \frac{(1-p)^2(4p-1)}{3p^2}, & 0.5 < p < 1. \end{cases}$$

C. Proofs, Technical Details and Additions for Subsection 4.4 and Section 5

C.1 Censorship under a general prior

For an arbitrary prior F, let $F^{-1}(u) := \inf\{x \in [0,1] : F(x) \geq u\}$ for $u \in (0,1]$, $F^{-1}(0) := 0$, and let $F^{-1}(u^+)$ be its right limit, with $F^{-1}(1^+) := F^{-1}(1)$. For $q \in [0,1)$ let $m(q) := \frac{1}{1-q}\int_q^1 F^{-1}(u)du$, and $m(1) := \overline{\theta}$. With $U \sim U[0,1]$, the quantile censorship G_q reveals the type $F^{-1}(U)$ if $U < q$ and sends $m(q)$ otherwise:

$$G_q(x) = \begin{cases} \min\{F(x), q\}, & x < m(q), \\ 1, & x \geq m(q). \end{cases}$$

Proposition 9 follows from the next lemma: with a positive density on $[\underline{\theta}, \overline{\theta}]$, F^{-1} is continuous and strictly increasing, the second alternative in (i) cannot occur, $\theta^* = F^{-1}(q^*)$, and the change of variables $u = F(s)$ turns (4) into (1).

Lemma 12. *Let $v(x)$ be well S-shaped and F any prior.*

(i) There are $q^* \in (0,1]$ and $\theta^* \in [F^{-1}(q^*), F^{-1}(q^{*+})]$ such that (G_{q^*}, G_{q^*}) is an equilibrium

and either $\theta^* < m^* := m(q^*)$ *and*

$$\int_0^{q^*} \big[\tilde{v}(m^* - F^{-1}(u)) - \tilde{v}(\theta^* - F^{-1}(u)) - (m^* - \theta^*)\tilde{v}'(m^* - F^{-1}(u))\big]\,du + \tilde{v}(m^* - \theta^*)(1 - q^*) = 0, \tag{4}$$

or $q^* = 1$*, so that both agents reveal fully, and* $\int_0^1 v'(x - t)\,dF(t)$ *is nondecreasing on* $[\underline{\theta}, \overline{\theta}]$*.*

(ii) Every q *and* $\theta \in [F^{-1}(q), F^{-1}(q^+)]$ *with* $\theta < m(q)$ *that solve* (4) *give an equilibrium* (G_q, G_q)*.*

(iii) If $v'(x)$ *is not linear near* 0*, the equilibrium is unique; all solutions in (ii) then have the same* q *and induce the same signal.*

Lemma 13. *Let* $v(x)$ *be well S-shaped and* F *any prior on* $[0, 1]$*. Against every rival signal* K*, some quantile censorship* G_q*,* $q \in [0, 1]$*, is a best reply.*

Proof. Let $\underline{\theta} < \overline{\theta}$, $w := w_K$, let L_x be the tangent to w at x, and let $e(q, \theta) := L_{m(q)}(\theta) - w(\theta) = \int_\theta^{m(q)} (w'(z) - w'(m(q)))dz$, which is continuous on Γ.

(a) If $e(0, \underline{\theta}) \geq 0$, then pooling $G_0 = \{\mu\}$ is a best reply. Let $D := L_\mu - w$, so $D(\underline{\theta}) \geq 0 = D(\mu) = D'(\mu)$. If $w'(x) > w'(\mu)$ for some $x > \mu$, then $w' < w'(\mu)$ on $[0, \mu)$, so $D(\underline{\theta}) < D(\mu) = 0$, a contradiction. So $D \geq 0$ on $[\mu, 1]$. On $[\underline{\theta}, \mu]$, D first increases and then decreases, so $D \geq \min(D(\underline{\theta}), D(\mu)) \geq 0$. Hence $L_\mu \geq w$ on $[\underline{\theta}, \overline{\theta}]$, and $\int w\,dH \leq L_\mu(\mu) = w(\mu)$ for every mean-preserving contraction H of F.

(b) If w' is non-decreasing on $[\underline{\theta}, \overline{\theta}]$, then $F = G_1$ is a best reply by Step 2 of the proof of Lemma 12.

(c) Otherwise, $e(0, \underline{\theta}) < 0$, and as in Step 3 there are $x_1 < x_2 < \overline{\theta}$ with $w'(x_1) > w'(x_2)$. So w' is strictly decreasing on $[x_2, 1]$, and $e(q, \theta) > 0$ for $(q, \theta) \in \Gamma$ with $x_2 < \theta < m(q)$, which holds near $(\overline{q}, \overline{\theta})$. Since the path from $(0, \underline{\theta})$ to $(\overline{q}, \overline{\theta})$ is continuous, e has a zero

(q, θ) with $\theta < m(q)$, and G_q is a best reply by Step 1. □

C.2 Proof of Lemma 12

Preliminaries. By property 1) in Appendix A.1, v is C^1 on $[-1, 1]$ with a concave derivative, but in general it is not C^2 (e.g. $v(x) = 0.5 + x - 0.5x|x|$). For a distribution K on $[0, 1]$, let $w_K(x) := \int_0^1 \tilde{v}(x-y)dK(y)$. Since $v(x) = 0.5 + v'(0)x + |v(1) - 0.5 - v'(0)|\tilde{v}(x)$ and every feasible signal has mean $\mathbb{E}[\theta] =: \mu$, w_K has the same best replies as $\overline{v}(\cdot, K)$. It is C^1 on $[0, 1]$, and $w'_K(x) = \int_0^1 \tilde{v}'(x-y)dK(y)$ is concave. Only these two properties of w_K are used below; in particular, no second derivative is needed.

Quantile facts: F^{-1} is non-decreasing and left-continuous, and $F^{-1}(u) \leq x \Leftrightarrow u \leq F(x)$. Let $\overline{q} := F(\overline{\theta}-)$, so that $F^{-1} < \overline{\theta}$ on $(0, \overline{q})$ and $F^{-1} = \overline{\theta}$ on $(\overline{q}, 1]$. As F^{-1} is integrable, m is continuous on $[0, 1)$, and $F^{-1}(q^+) \leq m(q) \leq \overline{\theta}$ gives $m(q) \to \overline{\theta} = m(1)$ as $q \to 1$. For $q < \overline{q}$, F^{-1} is not a.e. constant on $(q, 1)$, so $m(q) > F^{-1}(q^+)$; for $q \geq \overline{q}$, $m(q) = \overline{\theta}$ and $G_q = F$. G_q is the distribution of $\mathbb{E}[F^{-1}(U)|\, U\mathbf{1}\{U < q\}]$, so it is feasible.

The curve. Let $\Gamma := \{(q, \theta) : q \in [0, \overline{q}],\ F^{-1}(q) \leq \theta \leq F^{-1}(q^+)\}$, with $F^{-1}(\overline{q}^+) := \overline{\theta}$, i.e. the graph of F^{-1} on $[0, \overline{q}]$ with its jumps filled in. As $\phi(q) := q + F^{-1}(q)$ is strictly increasing, for each $r \in [0, \overline{q} + \overline{\theta}]$ there is a unique $q(r)$ with $\phi(q(r)) \leq r \leq \phi(q(r)^+)$, and $q(r)$ is 1-Lipschitz, since $q(r) > q(r')$ gives $r - r' \geq q(r) - q(r') + F^{-1}(q(r)) - F^{-1}(q(r')^+) \geq q(r) - q(r')$. So $r \mapsto (q(r), r - q(r))$ is a continuous path that traverses Γ from $(0, 0)$ to $(\overline{q}, \overline{\theta})$. By the quantile facts, $\theta < m(q)$ on $\Gamma \setminus \{(\overline{q}, \overline{\theta})\}$.

For $(q, \theta) \in \Gamma$, let $w_q := w_{G_q}$, let L_q be the tangent to w_q at $m(q)$, and let $E(q, \theta) := L_q(\theta) - w_q(\theta)$. Since $\tilde{v}$ is odd with $\tilde{v}'(0) = 0$, $E(q, \theta)$ is the left-hand side of (4), and

$$E(q, \theta) = \int_\theta^{m(q)} \big(w'_q(z) - w'_q(m(q))\big)dz.$$

E is continuous on Γ, as m is continuous and $\tilde{v}, \tilde{v}'$ are uniformly continuous, and

$E(\overline{q}, \overline{\theta}) = 0$.

Step 1 (verification). Let K be any distribution on $[0,1]$, $w := w_K$, $(q,\theta) \in \Gamma$ with $\theta < m := m(q)$, and L the tangent to w at m with $L(\theta) = w(\theta)$. Then G_q is a best reply to K. Indeed, let $D := L - w$, so that $D(\theta) = D(m) = D'(m) = 0$ and $D' = w'(m) - w'$. If $w'(x) > w'(m)$ for some $x > m$, then $w' < w'(m)$ on $[0,m)$ by concavity, so D is strictly increasing on $[0,m]$, contradicting $D(\theta) = D(m)$. Hence D is non-decreasing, and non-negative, on $[m,1]$. The set of $x \leq m$ with $w'(x) \geq w'(m)$ is an interval $[\alpha, m]$, so D is non-decreasing on $[\theta,\alpha]$ and non-increasing on $[\max(\alpha,\theta), m]$, and $D(\theta) = D(m) = 0$ gives $D \geq 0$ on $[\theta, m]$. If $\alpha > \theta$, then $w'(\theta) < w'(\alpha)$, so w' is non-decreasing on $[0,\theta]$ and $w'(\theta) < w'(m)$. If $\alpha \leq \theta$, then $D = 0$ on $[\theta, m]$, so w' is constant on $[\theta, m]$, hence maximal there and non-decreasing on $[0,\theta]$. In both cases

$$p(x) := \begin{cases} w(x), & x \leq \theta, \\ L(x), & x \geq \theta, \end{cases}$$

is convex, $p \geq w$, and $p = w$ on $[0,\theta] \cup \{m\}$. For $u < q$, $F^{-1}(u) \leq F^{-1}(q) \leq \theta$, and for $u > q$, $F^{-1}(u) \geq F^{-1}(q^+) \geq \theta$, where p is affine. Hence $\mathrm{supp}(G_q) \subseteq [0,\theta] \cup \{m\}$, and

$$\int p\, dF = \int_0^q p(F^{-1}(u))du + (1-q)p(m) = \int p\, dG_q = \int w\, dG_q.$$

So every mean-preserving contraction H of F satisfies $\int w\, dH \leq \int p\, dH \leq \int p\, dF = \int w\, dG_q$.

Step 2 (full revelation). If w_K' is non-decreasing on $[\underline{\theta}, \overline{\theta}]$, then F is a best reply to K: w_K is convex on $[\underline{\theta}, \overline{\theta}]$, which contains the support of every mean-preserving contraction H of F, so $\int w_K\, dH \leq \int w_K\, dF$.

Step 3 (existence). Let $\underline{\theta} < \overline{\theta}$. At the start of the path, $E(0,\theta) = \tilde{v}(\mu - \theta) < 0$ for $\theta \in [0, \underline{\theta}]$, as $\theta \leq \underline{\theta} < \mu$. Let $g := w_F'$, which is concave.

If g is non-decreasing on $[\underline{\theta}, \overline{\theta}]$, then $F = G_1$ is a best reply to F by Step 2, so (G_1, G_1) is an equilibrium, which is the second alternative in (i).

Otherwise, there are $\underline{\theta} \leq x_1 < x_2 < \overline{\theta}$ with $g(x_1) - g(x_2) =: 2\gamma > 0$ (if $x_2 = \overline{\theta}$, move x_2 slightly to the left, by continuity). Let $Y_q := F^{-1}(U)\mathbf{1}\{U < q\} + m(q)\mathbf{1}\{U \geq q\} \sim G_q$. As $q \to \overline{q}$, $Y_q \to Y_{\overline{q}}$ in probability, since they differ only on $\{q \leq U < \overline{q}\}$ and by $|m(q) - m(\overline{q})|$. Since $G_{\overline{q}} = F$ and $\tilde{v}'$ is uniformly continuous, $w'_q \to g$ uniformly. Hence there is a neighborhood N of $(\overline{q}, \overline{\theta})$ in Γ on which $w'_q(x_1) > w'_q(x_2)$ and $\theta > x_2$. For $(q, \theta) \in N$, the concave w'_q is then strictly decreasing on $[x_2, 1] \supseteq [\theta, m(q)]$, so $E(q, \theta) > 0$ whenever $\theta < m(q)$. By the intermediate value theorem along the path, E has a zero $(q^*, \theta^*) \in \Gamma \setminus \{(\overline{q}, \overline{\theta})\}$, where $\theta^* < m(q^*)$, and $q^* > 0$ since $E < 0$ for $q = 0$. By Step 1 with $K = G_{q^*}$, (G_{q^*}, G_{q^*}) is an equilibrium. This proves (i), and Step 1 with $K = G_q$ proves (ii).

The second alternative needs an atom at $\overline{\theta}$: the left derivative of g at $\overline{\theta}$ is $\int \tilde{v}''_-(\overline{\theta} - t)dF(t)$, where $\tilde{v}''_-(z) < 0$ for $z \in (0, 1]$ by property 2) and $\tilde{v}''_-(0) = |\tilde{v}''_+(0)|$. So it requires $(F(\overline{\theta}) - F(\overline{\theta}-))|\tilde{v}''_+(0)| \geq \int_{[0,\overline{\theta})} |\tilde{v}''_-(\overline{\theta} - t)|dF(t)$, which fails if F has no atom at $\overline{\theta}$ or if $\tilde{v}''_+(0) = 0$.

Step 4 (uniqueness). Now suppose that $v'(x)$ is not linear near 0. First, let (q, θ) be as in (ii), so $q < 1$, and let $w := w_q$, $m := m(q)$, with D, α, p as in Step 1 for $K = G_q$. By property 4) in Appendix A.1, w' is affine on no interval that G_q charges, and $G_q(\{m\}) \geq 1 - q > 0$. Hence $\alpha > \theta$, as otherwise w' is constant on $[\theta, m]$. Also $D > 0$ on $(\theta, 1] \setminus \{m\}$: a zero of D in $(m, 1]$, or in $[\alpha, m)$, makes w' constant on an interval with endpoint m, and a zero in (θ, α) would give $D = 0$ on $[\theta, x]$, i.e. $w' = w'(m)$ there, while $w' < w'(m)$ on $[0, \alpha)$. Finally, w' is strictly increasing on $[0, \theta]$: it is non-decreasing there and concave, so if it were constant on a subinterval of $[0, \theta]$, it would attain its maximum there, but $w'(\theta) < w'(m)$.

Let H be a best reply to G_q, so that both inequalities in Step 1 are equalities. The

first one gives $\text{supp}(H) \subseteq [0,\theta] \cup \{m\}$. Writing $p(x) = p(0) + p'(0)x + \int_0^1 (x-y)^+ dp'(y)$ and $\Psi_K(y) := \int_0^y K(x)dx$, the second one gives $\int_0^1 \big(\Psi_F(y) - \Psi_H(y)\big)dp'(y) = 0$, where $\Psi_H \leq \Psi_F$. As dp' charges every subinterval of $[0,\theta]$, $\Psi_H = \Psi_F$ on $[0,\theta]$, so $H = F$ on $[0,\theta)$. The rest of H, of mass $1 - F(\theta-)$, lies on $\{\theta, m\}$, and its first moment is $\int_{[\theta,1]} x\, dF$. As G_q also equals F on $[0,\theta)$ and puts $q - F(\theta-)$ on θ and $1-q$ on m, these two moment conditions, with $\theta \neq m$, give $H = G_q$. So G_q is the unique best reply to itself.

Second, if $g = w_F'$ is non-decreasing on $[\underline{\theta}, \overline{\theta}]$, it is strictly increasing there: a concave non-decreasing g that is constant on a subinterval is constant up to $\overline{\theta}$, i.e. affine on an interval that F charges, contradicting property 4). So w_F is strictly convex on $[\underline{\theta}, \overline{\theta}]$, and the same Ψ argument gives $H = F$ for every best reply H to F.

So in both cases of (i), the equilibrium signal is the unique best reply to itself. As all equilibria form a product set, every equilibrium strategy is a best reply to it, so the equilibrium is unique, and every solution in (ii) induces the same signal. A signal G_q with $\theta < m(q)$ determines q, as its top atom is at $m(q)$ with mass $1-q$, and the revealed types lie below $m(q)$. So q is unique, and θ can vary only within $[F^{-1}(q), F^{-1}(q^+)]$, i.e. within a gap in $\text{supp}(F)$.

C.3 Median censorship

This subsection shows that, for $v(x) = 0.5 + x - 0.5x|x|$ and every prior, $(G_{1/2}, G_{1/2})$ is an equilibrium, and $q = \frac{1}{2}$ is the only solution of (4) with $\theta < m(q)$. Here $\tilde{v}(x) = -x|x|$. For $(q,\theta) \in \Gamma$ with $\theta < m$ and $u < q$, $F^{-1}(u) \leq \theta < m$, so the integrand in (4) is

$$-(m - F^{-1}(u))^2 + (\theta - F^{-1}(u))^2 + 2(m-\theta)(m - F^{-1}(u)) = (m-\theta)^2,$$

and the left-hand side of (4) equals $(m-\theta)^2(2q-1)$. So $q=\frac{1}{2}$ is the only solution with $\theta < m(q)$. If $\overline{q} > \frac{1}{2}$, then $(\frac{1}{2}, F^{-1}(\frac{1}{2})) \in \Gamma$ with $F^{-1}(\frac{1}{2}) < m(\frac{1}{2})$ solves it, and $(G_{1/2}, G_{1/2})$ is an equilibrium by Lemma 12 (ii). If $\overline{q} \leq \frac{1}{2}$, then $G_{1/2} = F$, and $w'_F(x) = -2\int |x-t| dF(t)$ has right derivative $2(1-2F(x)) \geq 0$ on $[\underline{\theta}, \overline{\theta})$, as $F(x) \leq F(\overline{\theta}-) = \overline{q} \leq \frac{1}{2}$ there. So (F, F) is an equilibrium by Step 2 of the proof of Lemma 12. If $\underline{\theta} = \overline{\theta}$, the claim is trivial.

C.4 A convexity lemma

The following lemma extends the converse of the Hermite–Hadamard inequality (see Niculescu and Persson (2003)), which is the case of a uniform K.

Lemma 14. *Let a CDF K and a function h be continuous on an interval J, with K strictly increasing. If $h\big(\mathbb{E}_K[\theta|\theta \in J']\big) \leq \mathbb{E}_K[h(\theta)|\theta \in J']$ for every interval $J' \subseteq J$, then h is convex.*

Proof. Let $x < z$ in J, let ℓ be the secant of h through x and z, and suppose that $g := h - \ell$ has a positive maximum on $[x, z]$, first attained at $c \in (x, z)$. For $\delta > 0$ small enough that $\mathbb{E}_K[\theta|\theta \in [c-\delta, z]] > c$, there is $\delta' \in (0, z-c)$ such that $J' := [c-\delta, c+\delta']$ satisfies $\mathbb{E}_K[\theta|\theta \in J'] = c$, as K is continuous and strictly increasing. Since $g < g(c)$ on $[c-\delta, c)$, which has positive K-mass, and ℓ is affine, $\mathbb{E}_K[h(\theta)|\theta \in J'] = \ell(c) + \mathbb{E}_K[g(\theta)|\theta \in J'] < h(c)$, a contradiction. Hence $h \leq \ell$ on $[x, z]$. □

C.5 Proof of Corollary 5

Let $c = 2p - \underline{\theta}$ and $G = U[\underline{\theta}, c]$. As F is concave, it lies above the uniform CDF on $[\underline{\theta}, \overline{\theta}]$, so $p \leq \frac{\theta+\overline{\theta}}{2}$, i.e. $c \leq \overline{\theta}$. Moreover, $F - G$ is concave on $[\underline{\theta}, c]$ and non-negative at $\underline{\theta}$, while $F - G = F - 1 \leq 0$ on $[c, \overline{\theta}]$, so $F - G$ changes sign at most once, from positive to negative. Since F and G have the same mean, G is a mean-preserving contraction of F by the single-crossing condition, i.e. it is feasible. Against G, $\overline{v}(s) = G(s)$, which is concave on $[\underline{\theta}, \overline{\theta}]$, so by Jensen's inequality every feasible H gives $\mathbb{E}_H[G] \leq G(p) = 0.5 = \mathbb{E}_G[G]$,

and (G, G) is an equilibrium. Finally,

$$u_F = \mathbb{E}[\max(s_1, s_2)] = \underline{\theta} + \frac{2}{3}(c - \underline{\theta}) = \frac{4p - \underline{\theta}}{3}.$$

For uniqueness, let (G, H) be an equilibrium, so that (H, H) and (H, G) are equilibria too. As in Subsection C.6.4, H has no atom in $(\underline{\theta}, \overline{\theta}]$, and an atom $a > 0$ at $\underline{\theta}$ would not be a best reply to H either: pooling it with a small mass ε of H above $\underline{\theta}$ raises the payoff per unit of the atom from $\frac{a}{2}$ to at least a, at a total cost of at most $\varepsilon < \frac{a^2}{2}$. So H is atomless. As H is a best reply to G and $G(s) = \min\{\frac{s-\underline{\theta}}{c-\underline{\theta}}, 1\}$ is concave, $\mathbb{E}_H[G] \leq G(p) = 0.5$ with equality only if $\operatorname{supp}(H) \subseteq [\underline{\theta}, c]$. As G is a best reply to H and every mean-preserving contraction of G is feasible, pooling the types of G in any interval $J \subseteq [\underline{\theta}, c]$ does not pay, so H is convex on $[\underline{\theta}, c]$ by Lemma 14 with $K = G$. Hence $H \leq G$ on $[\underline{\theta}, c]$, as G is the secant of H there, and $\int_{\underline{\theta}}^{c} (G - H)dx = \mathbb{E}_H[s] - \mathbb{E}_G[s] = 0$, so $H = G$.

C.6 Proofs of the benchmark results

Throughout, a rule is a general selection rule, and $V_G(x) = \int H_1(x, y)\, dG(y)$ is the agent's value function against a rival who plays G. For a CDF K, let $I_K(x) := \int_{-\infty}^{x} K(t)\, dt$; for distributions with the same mean, $G \leq_{cx} F$ if and only if $I_G \leq I_F$.

C.6.1 Two tools

Lemma 15. *In every symmetric profile (G, G), the firm's utility is $u_F = p + 2\operatorname{Cov}_G\big(s, V_G(s)\big)$.*

Proof. With s_1, s_2 i.i.d. G, $u_F = \mathbb{E}[s_2] + \mathbb{E}[(s_1 - s_2)H_1(s_1, s_2)]$. Since $\kappa := H_1 - \frac{1}{2}$ is anti-symmetric, $\mathbb{E}[s_2\kappa(s_1, s_2)] = -\mathbb{E}[s_1\kappa(s_1, s_2)]$, so $\mathbb{E}[(s_1 - s_2)H_1(s_1, s_2)] = 2\mathbb{E}[s_1\kappa(s_1, s_2)] = 2\mathbb{E}\big[s_1\big(V_G(s_1) - \frac{1}{2}\big)\big]$, and $\mathbb{E}[V_G(s)] = \frac{1}{2}$. □

Lemma 16. *Let F be any prior and $V : [\underline{\theta}, \overline{\theta}] \to [0, 1]$. There is a rule with $V_F = V$ on*

$\mathrm{supp}(F)$ *if and only if* $V(\theta) \leq_{cx} U[0,1]$ *for* $\theta \sim F$*, i.e.,*

$$\mathbb{E}_F[V(\theta)] = \tfrac{1}{2} \qquad \textit{and} \qquad \mathbb{E}_F\left[(V(\theta) - t)^+\right] \leq \tfrac{(1-t)^2}{2} \ \textit{for all } t \in [0,1].$$

Then one such rule is $H_1(x,y) = \mathbb{P}(S_x > S_y) + \frac{1}{2}\mathbb{P}(S_x = S_y)$*, where the agents' scores are independent, the score* S_x *of signal* x *has mean* $V(x)$*, and* $S_\theta \sim U[0,1]$ *for* $\theta \sim F$*.*

Proof. Necessity: $\mathbb{E}_F[V] = \frac{1}{2}$ by symmetry. For every set A, the winner's type lies in A with probability $2\int_A V\, dF$, at most the probability $1 - (1 - F(A))^2$ that some type lies in A. Applied to the upper level sets of V, this bounds the sum of the top m quantiles of $V(\theta)$ by $m - \frac{m^2}{2}$, the same sum for $U[0,1]$, at every mass m of such a set, and hence at every $m \in [0,1]$: between two such masses, $V(\theta)$ has an atom, so the left side is linear in m, while the right side is concave. With equal means, this is the stated condition. Sufficiency: by Strassen's theorem, there is a kernel $\kappa(\cdot|z)$ with mean z that spreads the distribution of $V(\theta)$ into $U[0,1]$; let $S_x \sim \kappa(\cdot|V(x))$, and for x with $V(x)$ outside the support of the law of $V(\theta)$, let S_x be any law on $[0,1]$ with mean $V(x)$. Against a fully revealing rival, whose score is $U[0,1]$, $\mathbb{P}(S_x > S_y) = \mathbb{E}[S_x] = V(x)$, and the rule is symmetric and constant-sum by construction. □

The second lemma is the feasibility claim in Subsection 5.2: noise can only make winning probabilities less dispersed than exact ranking does, and the noiseless rule is the boundary case, with $V_F = F$ and $F(\theta) \sim U[0,1]$.

C.6.2 Proof of Proposition 8

(i) For any signals, $u_F = \mathbb{E}[s_2 + (s_1 - s_2)H_1(s_1,s_2)] \leq \mathbb{E}[\max(s_1,s_2)] \leq \mathbb{E}[s_1 + s_2 - s_1 s_2] = 2p - p^2$, where the second inequality holds as $\mathbb{E}[\max(\theta_1,\theta_2)|s_1,s_2] = s_1 + s_2 - s_1 s_2$, and it is an equality if and only if $\min(s_1,s_2)\big(1 - \max(s_1,s_2)\big) = 0$ almost surely. If both agents put mass on $(0,1)$, this fails with positive probability, so in an equilibrium

that attains the first best, some agent, say agent 1, reveals fully. As the game is symmetric and constant-sum, its equilibrium strategies form the same set for both agents, so $(\{0,1\};\{0,1\})$ is an equilibrium too. The first inequality is an equality only if $H_1(1,s) = 1$ for almost every signal $s < 1$ of agent 2, and agent 2 sends some $s < 1$, as $p < 1$. As H is nondecreasing in the own signal and constant-sum, $H_1(1,0) = 1 - H_1(0,1) \geq 1 - H_1(s,1) = H_1(1,s) = 1$. Against a fully revealing rival, the agent's value function is $V(x) = (1-p)H_1(x,0) + p\,H_1(x,1)$, with $V(0) = \frac{1-p}{2}$ and $V(1) = 1 - \frac{p}{2}$ when $H_1(1,0) = 1$, so full revelation is a best reply if and only if V lies below its chord from 0 to 1, which is $\frac{1-p+x}{2}$. Conversely, if $H_1(1,0) = 1$ and this holds, $(\{0,1\};\{0,1\})$ is an equilibrium, and the firm selects a good agent over a bad one for sure, which attains $2p - p^2$.

(ii) The condition in (i) is linear in p, so it holds for all $p \in (0,1)$ if and only if it holds in the limits $p \to 0$ and $p \to 1$, which give $H_1(x,0) \leq \frac{1+x}{2}$ and $H_1(x,1) \leq \frac{x}{2}$. If both are strict on $(0,1)$, then against a fully revealing rival with any prior p_j, $V(x) < \frac{1-p_j+x}{2}$ on $(0,1)$ while V equals the chord at 0 and 1, so full revelation is the unique best reply for every own prior, as every other signal puts mass on $(0,1)$. So $(\{0,1\};\{0,1\})$ is an equilibrium for all $(p_1,p_2) \in (0,1)^2$; if (G_1,G_2) is another, then by interchangeability so is $(G_1,\{0,1\})$, so G_1 is a best reply to full revelation, and $G_1 = \{0,1\}$, and likewise $G_2 = \{0,1\}$.

(iii) The pass/fail rule is nondecreasing in the own signal, $H_1(1,0) = 1$, $H_1(x,0) = \frac{1}{2} < \frac{1+x}{2}$ and $H_1(x,1) = 0 < \frac{x}{2}$ for $x \in (0,1)$, so (ii) applies.

C.6.3 Proof of Proposition 10 and Remark 4

Let $S := \operatorname{supp}(F)$ and $\bar{F}(x) := F(x^-) + \frac{1}{2}F(\{x\})$, the winning probability of type x against a fully revealing rival when the higher type is selected and ties are split, as symmetry forces. In terms of F, the condition of Remark 4 says that F is convex on every interval

contained in S, every atom of F below $\overline{\theta}$ is an isolated point of S, and the chords of $\bar{F}$ across the gaps of S have nondecreasing slopes.

(i) *Sufficiency, for any support.* Let ψ be convex with $\psi = \bar{F}$ on S, and let ϕ equal $\bar{F}$ on S and be affine on each gap (a, b) of S. Then ϕ is convex: on each interval contained in S it equals ψ, and at the ends of a gap its slopes are ordered, as the chord of the convex ψ over $[a, b]$ has slope between $\psi'(a^+)$ and $\psi'(b^-)$, and consecutive chords of ψ have nondecreasing slopes. As $\bar{F}$ is nondecreasing on S, $\phi \geq \bar{F}(a)$ on a gap (a, b). Let $r(x) := \max\left(S \cap [\underline{\theta}, x]\right)$ and let the rule select the higher rounded signal, tossing a coin at ties; it is nondecreasing in the own signal. Against a fully revealing rival, $V_F = \bar{F}$ on S and $V_F(x) = \bar{F}(r(x)) \leq \phi(x)$ off S. So for every $G \leq_{cx} F$, $\int V_F \, dG \leq \int \phi \, dG \leq \int \phi \, dF = \int V_F \, dF$, and full revelation is an equilibrium. Distinct types are never rounded to the same point, so the higher type is always selected.

(ii) *Necessity, for any support and a rule nondecreasing in the own signal.* Under full revelation, the first best requires $H_1(x, y) = 1$ for $F \otimes F$-almost all $x > y$, and $H_1(x, x) = \frac{1}{2}$ by symmetry, so $V_F = \bar{F}$ F-almost everywhere and at every atom; as V_F is nondecreasing, $V_F(x) \in [F(x^-), F(x)]$ for every $x \in S$. Three deviations, each a mean-preserving contraction of F, rule out the three ways in which $\bar{F}|_S$ can fail to extend convexly. (a) An atom $a < \overline{\theta}$ of mass m that is a limit of S from the right: pooling a fraction α of the atom with all types in $(a, a + \delta)$, of mass $\mu_\delta > 0$, gives a signal in $(a, a + \delta)$ with winning probability at least $F(a)$, so the atom gains at least $\alpha \frac{m}{2}$, while the other types lose at most $\mu_\delta(F(a + \delta) - F(a))$, which is smaller for small δ and $\alpha = \mu_\delta$. (b) An atom $a < \overline{\theta}$ that is a limit of S from the left only: then a is the left end of a gap (a, b), and pooling the types in $(a - \delta, a)$, of mass μ_δ, with a mass $w = O(\mu_\delta \delta)$ of the types at b into the signal a raises the former from at most $F(a^-)$ to $\bar{F}(a) = F(a^-) + \frac{m}{2}$ and costs the latter at most w. (c) Points $x < y < z$ of S with $\bar{F}(y)$ above the chord of $\bar{F}$ through x and z: pooling types near x and near z (at x and z themselves if they are

atoms) into the signal y in the right proportions pays, as V_F is continuous at non-atoms of S by (a) and (b). If none of (a)–(c) applies, $\bar{F}|_S$ is convex on S and continuous at every limit point of S below $\overline{\theta}$, and its extension by chords is convex. This proves the converse in Remark 4.

(iii) *Interval support.* Let $S = [\underline{\theta}, \overline{\theta}]$ and let (G_1, G_2) attain the first best. The selected type must be the higher type almost surely, and selection depends only on the signals, so for almost all pairs of signals the conditional supports of θ_1 and θ_2 are ordered. If a signal of agent 1 pooled two types $a < b$ of S, the conditional supports of agent 2 would avoid (a, b), whose F-mass is positive. So both agents reveal fully, and (ii) applies: by (a), F has no atom below $\overline{\theta}$, and by (c), F is convex on $[\underline{\theta}, \overline{\theta})$. Conversely, then $\bar{F} = F$ on $[\underline{\theta}, \overline{\theta})$, and an atom at $\overline{\theta}$ only raises $\bar{F}(\overline{\theta})$ above $F(\overline{\theta}^-)$, which keeps $\bar{F}$ convex on $[\underline{\theta}, \overline{\theta}]$; with no gaps, the rounding rule of (i) is the noiseless rule. If F is moreover atomless, full revelation is the unique equilibrium of the noiseless rule by Appendix C.6.4.

C.6.4 The noiseless rule with an atomless convex prior

Let $I = [\underline{\theta}, \overline{\theta}]$. For a CDF K let $\Psi_K(y) := \int_{\underline{\theta}}^{y} K(x)dx$, so that a signal G is feasible if and only if $\Psi_G \leq \Psi_F$ on I with equality at $\overline{\theta}$, and for convex ϕ and feasible G,

$$\int \phi \, dF - \int \phi \, dG = \int \big(\Psi_F(y) - \Psi_G(y)\big) d\phi'(y) \geq 0.$$

Against an atomless distribution H, the payoff of a signal s is $H(s)$. In particular, against full revelation, a deviation to G yields $\mathbb{E}_G[F]$, and full revelation yields $\mathbb{E}_F[F] = 0.5$.

Sufficiency. If F is convex on I, the display with $\phi = F$ gives $\mathbb{E}_G[F] \leq \mathbb{E}_F[F] = 0.5$ for every feasible G, so (F, F) is an equilibrium.

Necessity. Pooling the types in an interval $J \subseteq I$ into their mean m_J is feasible and changes the deviator's payoff by $\mathbb{P}(\theta \in J)\big(F(m_J) - \mathbb{E}_F[F(\theta)|\theta \in J]\big)$, so in a fully revealing equilibrium $F(m_J) \leq \mathbb{E}_F[F(\theta)|\theta \in J]$ for every J. First, F is strictly increasing

on I: otherwise $F = \gamma \in (0,1)$ on a maximal interval $[L,R] \subset I$ with $L < R$, and for $J = \{\theta : \gamma - r \leq F(\theta) \leq \gamma + \frac{r}{2}\}$, $\mathbb{E}_F[F(\theta)|\theta \in J] = \gamma - \frac{r}{4}$, while $m_J \to \frac{2L+R}{3} > L$ as $r \to 0$, so $F(m_J) \geq \gamma$ for small r. Then F is convex on I by Lemma 14 with $K = h = F$.

Uniqueness. Let F be convex, so that it is strictly increasing on I, and let (F,H) be an equilibrium. As all equilibria form a symmetric product set, (H,H) and (H,F) are equilibria too, and each agent gets 0.5 in all of them. Against H, the payoff of a signal s is $H(s^-) + \frac{1}{2}\big(H(s) - H(s^-)\big)$. First, H is atomless. It has no atom at $\underline{\theta}$ or $\overline{\theta}$, as $\Psi_H \leq \Psi_F$ with equality at $\overline{\theta}$ and F is atomless. If H had an atom $a > 0$ at $m \in (\underline{\theta}, \overline{\theta})$, then $\Psi_H(m) < \Psi_F(m)$, since otherwise $\Psi_F - \Psi_H \geq 0$ would have a minimum at m with left derivative $F(m) - H(m^-)$ above its right derivative $F(m) - H(m)$. So replacing the atom by masses $\frac{2a}{3}$ at $m + \varepsilon$ and $\frac{a}{3}$ at $m - 2\varepsilon$ is feasible for small ε, as $\Psi_F - \Psi_H > 0$ near m, and against H it raises the payoff by $a^2(\frac{2}{3} - \frac{1}{2}) + o(1) > 0$, contradicting that H is a best reply to H. Second, H is convex on I: as F is a best reply to H, pooling the types in an interval $J \subseteq I$ does not pay, i.e., $H(m_J) \leq \mathbb{E}_F[H(\theta)|\theta \in J]$, and Lemma 14 applies with $K = F$ and $h = H$. Finally, as F and H are convex, $\int F\,dH \leq \int F\,dF$ and $\int H\,dH \leq \int H\,dF$, and all four integrals equal 0.5. By the identity above, $\Psi_F = \Psi_H$ on the supports of dF' and dH', and there $F = H$, since each such point in $(\underline{\theta}, \overline{\theta})$ is a minimum of the differentiable function $\Psi_F - \Psi_H \geq 0$, while $F = H$ at $\underline{\theta}$ and $\overline{\theta}$. On each open interval off these supports, F and H are affine and agree at the endpoints. Hence $H = F$, and (F,F) is the unique equilibrium.

C.6.5 Rules of the signal gap

This proves the claim after Proposition 10: with $\text{supp}(F) = [\underline{\theta}, \overline{\theta}]$, a rule of the signal gap attains the first best only if it is the noiseless rule. Let $d = s_1 - s_2$. For any rule of

the signal gap and feasible G_1, G_2, since $v(-x) = 1 - v(x)$,

$$u_F = \mathbb{E}[\max(s_1, s_2)] - \mathbb{E}\big[|d|\big(1 - v(|d|)\big)\big] \leq \mathbb{E}[\max(s_1, s_2)] \leq \mathbb{E}[\max(\theta_1, \theta_2)].$$

By Proposition 10, the first best requires full revelation by both agents and an F that is convex on $[\underline{\theta}, \overline{\theta})$ with no atom below $\overline{\theta}$, so F has a density on $[\underline{\theta}, \overline{\theta})$; then also $|d|\big(1-v(|d|)\big) = 0$ almost surely. As $0 < |d| < \varepsilon$ with positive probability for every $\varepsilon > 0$, $v(x) = 1$ for arbitrarily small $x > 0$, i.e., $v(x)$ is the noiseless rule, as it is increasing and $v(x) \leq 1$.

C.6.6 Proof of Theorem 3

As full revelation is a best reply and V_F is upper semicontinuous, there is a convex $\phi \geq V_F$ on $[\underline{\theta}, \overline{\theta}]$ with $\phi = V_F$ on $\operatorname{supp}(F)$ (Dworczak and Martini, 2019). Let $g := \phi - L$. It is convex, $g(\overline{\theta}) \leq 0$ as $\phi(\overline{\theta}) = V_F(\overline{\theta}) \leq 1 = L(\overline{\theta})$, and $\int g\, dF = \frac{1}{2} - \frac{1}{2} = 0$. If $g(x_1) \leq 0$, then $g \leq \max(g(x_1), g(\overline{\theta})) \leq 0$ on $[x_1, \overline{\theta}]$, so $g > 0$ exactly on some $[\underline{\theta}, x^*)$ and $g \leq 0$ on $[x^*, \overline{\theta}]$. Hence $\int x\, g\, dF = \int (x - x^*) g\, dF \leq 0$, with equality only if $g = 0$ F-almost everywhere, and by Lemma 15, $u_F = p + 2\operatorname{Cov}_F(\theta, \phi(\theta)) \leq p + 2\operatorname{Cov}_F(\theta, L(\theta)) = p + \frac{\operatorname{Var}_F(\theta)}{\overline{\theta} - p}$. By Lemma 16, L is implementable if and only if $L(\theta) \leq_{cx} U[0,1]$, i.e., $F \leq_{cx} L^{-1}(U[0,1]) = \bar{U}$. The implementing kernel of Lemma 16, written in signal units, is a mean-preserving spread of each signal whose truthful scores are distributed as $\bar{U}$; Lemma 17 gives a monotone implementation. Under L, the agent is indifferent among all feasible signals; mixing the rule with H^* of Proposition 11, whose V_F is strictly convex below the top type (see Appendix C.6.7), makes full revelation the unique equilibrium at a payoff arbitrarily close to the bound.

Every concave prior satisfies $F \leq_{cx} \bar{U}$: it lies above the uniform CDF on $[\underline{\theta}, \overline{\theta}]$, so $p \leq \frac{\underline{\theta}+\overline{\theta}}{2}$, and $F - \bar{U}$ is concave on $[\underline{\theta}, \overline{\theta}]$, vanishes at $\overline{\theta}$ and is non-positive below $\underline{\theta}$, so it crosses zero once, from below. A convex prior other than the uniform has $p > \frac{\underline{\theta}+\overline{\theta}}{2}$,

so $2p - \overline{\theta} > \underline{\theta}$, and F puts mass below the support of $\bar{U}$.

Lemma 17. *Let F be any prior; in (iii), let F be atomless.*

(i) Every nondecreasing V that satisfies the condition of Lemma 16 equals V_F on $\mathrm{supp}(F)$ *for a rule nondecreasing in the own signal. One such rule is a lottery over coarse gradings: the firm draws a partition of $[\underline{\theta}, \overline{\theta}]$ into intervals, selects the agent whose signal lies in the higher interval, and tosses a coin if both lie in the same one; a signal at an atom of F draws its quantile uniformly from the atom's quantile interval, so the partition may split an atom.*

(ii) In particular, whenever $F \leq_{cx} \bar{U}$, the bound of Theorem 3 is attained by such a rule.

(iii) L is generated by a single random pass mark C, drawn independently of the signals, under which the higher signal is selected if it is at least C and a coin is tossed otherwise, if and only if $\sqrt{I_F}$ is concave on $[\underline{\theta}, \overline{\theta}]$; the pass mark then has the CDF $\mathbb{P}(C \leq z) = \frac{I_F(z)}{(\overline{\theta}-p)\,F(z)^2}$. Every concave prior satisfies this condition.

Proof. (i) In quantiles $q = F(x)$, a nondecreasing V that satisfies Lemma 16 is a nondecreasing $v(q) = V(F^{-1}(q))$ majorized by the identity, the interim rule of the noiseless comparison. By Kleiner et al. (2021, Theorems 1 and 3), the extreme points of this set equal the identity outside a collection of disjoint intervals and are constant inside each, and each is implemented by selecting the higher quantile across intervals and randomizing uniformly within an interval; this rule is nondecreasing in the own signal, and so is every mixture of such rules. If F has an atom at x, v is constant on its quantile interval $[F(x^-), F(x)]$, and drawing the quantile of signal x uniformly from that interval, independently across agents, gives signal x the average of v over the interval, which is $V(x)$; the drawn quantile is stochastically increasing in the signal, so the rule stays nondecreasing. (ii) follows from (i), as L is increasing. (iii) Under a pass mark with CDF K, $H_1(x, y) = \frac{1}{2} + K(\max(x, y))\big(\mathbb{1}\{x > y\} - \frac{1}{2}\big)$, which is nondecreasing

in x, and $V_F(x) = \frac{1}{2} + \frac{1}{2}F(x)K(x) - \frac{1}{2}\int_x^{\overline{\theta}} K\,dF$. So $V_F' = Kf + \frac{1}{2}FK'$, i.e., $(F^2K)' = 2F\,V_F'$, and $V_F = L$ if and only if $F^2K = I_F/(\overline{\theta} - p)$, using $I_F(\overline{\theta}) = \overline{\theta} - p$ for $K(\overline{\theta}) = 1$. This K is a CDF if and only if $I_F/F^2 = I_F/(I_F')^2$ is nondecreasing, i.e., $(I_F')^2 \geq 2I_F I_F''$, i.e., $\sqrt{I_F}$ is concave. If F is concave, it lies below its tangent at z, which vanishes at or below $\underline{\theta}$, so $I_F(z) \leq \frac{F(z)^2}{2f(z)}$, which is the same inequality. □

So for concave priors the best fully revealing rule is a lottery among agents who *fail* a random pass mark. A lottery among agents who *clear* a bar instead makes V_F flat at the top, so V_F is not convex and agents gain by pooling their best types.

C.6.7 Proof of Proposition 11

Sufficiency: convexity of each $H_1(\cdot, y)$ makes V_F convex for every F, so full revelation is a best reply to itself by Jensen's inequality. Necessity: for atomless full-support priors $F_n \to \delta_y$, full revelation is a best reply to itself, so pooling the types in any interval does not pay, and V_{F_n} is convex by Lemma 14; as H is continuous, V_{F_n} converges pointwise to $H_1(\cdot, y)$, which is therefore convex. Optimality: for $x > y$, $H_1(\cdot, y)$ is convex on $[y, 1]$ with $H_1(y, y) = \frac{1}{2}$ and $H_1(1, y) \leq 1$, so it lies below the chord, $H_1(x, y) \leq \frac{1}{2} + \frac{x-y}{2(1-y)} = H_1^*(x, y)$. H^* is admissible: for $x < y$, $H_1^*(x, y) = \frac{1-y}{2(1-x)}$, which is convex in x and has the same slope $\frac{1}{2(1-y)}$ at $x = y$ as the linear branch, and $H_1^*(x, y) + H_1^*(y, x) = 1$. Since $u_F = \mathbb{E}[m + (M - m)H_1(M, m)]$ with $M = \max(\theta_1, \theta_2)$ and $m = \min(\theta_1, \theta_2)$, pointwise dominance gives the ranking. Uniqueness: H_1^* is continuous except at $(1, 1)$, so V_F is continuous on $[\underline{\theta}, 1)$, and $V_F''(x) = \int_{(x,1]} \frac{1-y}{(1-x)^3} dF(y) > 0$ on $[\underline{\theta}, t)$, where $t := \sup(\mathrm{supp}(F) \cap [0, 1))$; on $[t, 1)$, which contains no type, V_F is affine, and if F has an atom at 1, V_F jumps up there, as $H_1^*(x, 1) = 0$ for $x < 1$ and $H_1^*(1, 1) = \frac{1}{2}$. A mean-preserving contraction $G \neq F$ pools some types $a < b$; if $a < t$, strict convexity on $[a, \min(b, t))$ makes the pooling strictly costly, and otherwise $a = t$ and $b = 1$, and the jump at 1 does. So $\int V_F\,dG < \int V_F\,dF$, F

is the unique best reply to F, and by interchangeability every equilibrium (G_1, G_2) has (G_1, F) as an equilibrium, so $G_1 = F$, and likewise $G_2 = F$.

C.6.8 Proof of Proposition 12 and Corollary 6

The rule. The greatest common mean-preserving contraction $G^*_{\bar{s}}$ of F and $\bar{U}_{\bar{s}}$ is defined by $I_{G^*_{\bar{s}}}$ being the convex minorant of $\min(I_F, I_{\bar{U}_{\bar{s}}})$. For $\bar{s} \geq 2p - \underline{\theta}$, the uniformizing rule of width $\bar{s}$ scores an agent with signal x by an independent draw S_x with mean x: a draw from a martingale kernel that spreads $G^*_{\bar{s}}$ into $\bar{U}_{\bar{s}}$ if $x \in \text{supp}(G^*_{\bar{s}})$, and $S_x = x$ otherwise; the higher score is selected. For concave F, Lemma 18 gives a monotone implementation.

Equilibrium and payoff. Against $G^*_{\bar{s}}$, the rival's score is distributed as $\bar{U}_{\bar{s}}$, whose CDF equals $L_{\bar{s}}$ on $[2p - \bar{s}, \bar{s}]$. For $x \leq \bar{s}$, S_x lies in this interval, as $\underline{\theta} \geq 2p - \bar{s}$, so the agent wins with probability $\mathbb{E}[L_{\bar{s}}(S_x)] = L_{\bar{s}}(x)$; for $x > \bar{s}$, he wins with probability $1 < L_{\bar{s}}(x)$. Hence every feasible G' yields at most $\int L_{\bar{s}}\, dG' = \frac{1}{2}$, which $G^*_{\bar{s}}$ attains, and $u_F(\bar{s})$ follows from Lemma 15. If the value function equals $L_{\bar{s}}$ on $\text{supp}(G)$, Lemma 16, applied to G in place of F, requires $G \leq_{cx} \bar{U}_{\bar{s}}$, and $u_F = p + \text{Var}(G)/(\bar{s} - p)$ is largest at the greatest common contraction.

Shape. For concave F and $\bar{s} \in (2p - \underline{\theta}, \overline{\theta})$, $F - \bar{U}_{\bar{s}}$ is non-positive below $\underline{\theta}$, concave on $[\underline{\theta}, \bar{s}]$ and negative on $[\bar{s}, \overline{\theta})$, so $I_F - I_{\bar{U}_{\bar{s}}}$ changes sign once, from negative to positive, and the convex minorant of the minimum bridges the crossing by a common supporting line, touching I_F at t_1 (which may be the kink of I_F at $\underline{\theta}$ if F has an atom there) and $I_{\bar{U}_{\bar{s}}}$ at t_2. At the ends, $\bar{U}_{2p-\underline{\theta}} \leq_{cx} F$ (proof of Corollary 5) and $F \leq_{cx} \bar{U}_{\overline{\theta}}$ (Appendix C.6.6), so $G^*_{2p-\underline{\theta}} = \bar{U}_{2p-\underline{\theta}}$ and $G^*_{\overline{\theta}} = F$.

Corollary 6. For concave F, the best fully revealing rule attains $p + \frac{\text{Var}_F(\theta)}{\overline{\theta} - p}$ by Theorem 3, and the noiseless rule gives $\frac{4p - \underline{\theta}}{3} = p + \frac{p - \underline{\theta}}{3}$ by Corollary 5. The former is larger if and only if $\text{Var}_F(\theta) > \frac{(\overline{\theta} - p)(p - \underline{\theta})}{3}$.

C.6.9 Proof of Theorem 4

Let (G, G) be a symmetric equilibrium, $s \sim G$ and $V := V_G$. By Lemma 15, $u_F = p + 2\,\mathrm{Cov}(s, V(s))$. The maximum in the statement exists, as $\mathrm{Var}(G^*_{\bar{s}})$ is continuous in $\bar{s}$.

Step 1 (convexity on the support). By Lebesgue's differentiation theorem for G, at G-almost every x the value function is approximately continuous: for every $\varepsilon > 0$, the share of the G-mass within δ of x at which V differs from $V(x)$ by more than ε vanishes as $\delta \to 0$. Let A be the set of such points in $\mathrm{supp}(G)$, so that $G(A) = 1$, and take $s_1 < s_2 < s_3$ in A. For small δ, there are sets of positive mass within δ of s_1 and of s_3 on which V is within ε of $V(s_1)$ and of $V(s_3)$, and a point x within δ of s_2 with $|V(x) - V(s_2)| \leq \varepsilon$. Removing small masses from the two sets and merging them, in the proportion that puts their mean at x, is a contraction of G, hence feasible, and optimality of G bounds $V(x)$ by the weighted average of the removed values. Letting $\varepsilon, \delta \to 0$ gives $V(s_2) \leq \frac{s_3 - s_2}{s_3 - s_1} V(s_1) + \frac{s_2 - s_1}{s_3 - s_1} V(s_3)$. So V is the restriction to A of a convex function φ on the convex hull of A, and $V(s) = \varphi(s)$ almost surely.

Step 2 (implementability). As in the proof of Lemma 16, with G in place of F, $V(s) \leq_{cx} U[0, 1]$.

Step 3 (monotone pooling). Let Q be the quantile function of G and $r(t) := \varphi(Q(t))$, which is quasi-convex on $[0, 1]$, as φ is convex and Q nondecreasing. Let q^* be the largest minimizer of the running average $c(q) := \frac{1}{q} \int_0^q r$ on $[0, 1]$, with $c(0) := r(0^+)$, and $c^* := c(q^*)$. Minimality of c^* and quasi-convexity of r, whose sublevel sets are intervals, give three facts: $r \geq c^*$ almost everywhere on $(q^*, 1]$, as otherwise c would fall below c^* to the right of q^*; r is nondecreasing there; and on $[0, q^*]$, $r - c^*$ is positive below some j and nonpositive above it, as otherwise c would fall below c^* just left of q^*. Now pool the lowest q^* quantiles into one signal, their mean μ, with winning probability c^*. The new signal distribution is a contraction of G, so it is still $\leq_{cx} F$; the new winning probability is a conditional expectation of $V(s)$, so it is still $\leq_{cx} U[0, 1]$ by Step 2; and

$\mathbb{E}[(s-p)V(s)]$ rises by $-\int_0^{q^*}(Q(t)-\mu)(r(t)-c^*)\,dt = -\int_0^{q^*}(Q(t)-Q(j))(r(t)-c^*)\,dt \geq 0$, so u_F weakly rises. The new value function is the restriction of $\max(\varphi, c^*)$ to $[\mu,\infty)$, which is convex and nondecreasing there, as $\varphi(\mu) \leq c^*$ by Jensen's inequality. As Steps 4–6 use only these properties, assume from now on that φ is nondecreasing and that $\mathrm{Cov}(s, V(s)) > 0$, as otherwise $u_F \leq p$, below every $u_F(\bar{s})$. This also proves the claim in Subsection 4.4 that monotonicity in the own signal is without loss for the firm.

Step 4 (the width). Let $\beta := \frac{\mathrm{Cov}(s,V(s))}{\mathrm{Var}(V(s))}$, $\bar{s} := p + \frac{\beta}{2}$, $\psi := p + \beta(\varphi - \frac{1}{2})$, which is convex and nondecreasing, and $Y := \psi(s)$. By Step 2, $Y \leq_{cx} U[p - \frac{\beta}{2}, p + \frac{\beta}{2}] = \bar{U}_{\bar{s}}$, so $\mathbb{E}Y = p$; by the choice of β, $\mathrm{Cov}(Y, s-Y) = \beta\,\mathrm{Cov}(V,s) - \beta^2\mathrm{Var}(V) = 0$; and $\mathbb{E}[(Y-p)(s-p)] = 2(\bar{s}-p)\,\mathrm{Cov}(s, V(s))$. So it suffices to show that $\mathbb{E}[(Y-p)(s-p)] \leq \mathrm{Var}(G^*_{\bar{s}})$.

Step 5 (a dual bound). Suppose that w_1, w_2 are convex, $\int w_1\,dF + \int w_2\,d\bar{U}_{\bar{s}} = \mathrm{Var}(G^*_{\bar{s}})$, a is concave, and

$$R(s,y) := w_1(s) + w_2(y) - (y-p)(s-p) - a(y)(s-y) \geq 0 \qquad \text{for all } s, y.$$

As $G \leq_{cx} F$ and $Y \leq_{cx} \bar{U}_{\bar{s}}$,

$$\mathbb{E}[(Y-p)(s-p)] = \mathbb{E}w_1(s) + \mathbb{E}w_2(Y) - \mathbb{E}[a(Y)(s-Y)] - \mathbb{E}R(s,Y) \leq \mathrm{Var}(G^*_{\bar{s}}) - \mathbb{E}[a(Y)(s-Y)].$$

Such functions exist. For $\bar{s} \in (2p - \underline{\theta}, \overline{\theta})$, let t_1, t_2 be as in Proposition 12, let C be the chord of $(x-p)^2$ over $[t_1, t_2]$, and let

$$w_1(x) := \begin{cases} (x-p)^2, & x \leq t_1, \\ C(x), & x \geq t_1, \end{cases} \qquad w_2(x) := \begin{cases} 0, & x \leq t_2, \\ (x-p)^2 - C(x), & x \geq t_2, \end{cases}$$

$$a(y) := \min\{y - p,\ t_1 - p,\ t_1 + t_2 - p - y\}.$$

Here w_1 is affine where $I_{G^*} < I_F$ (above t_1), so $\int w_1\,dF = \int w_1\,dG^*$; w_2 is affine

where $I_{G^*} < I_{\bar{U}_{\bar{s}}}$ (below t_2), so $\int w_2\, d\bar{U}_{\bar{s}} = \int w_2\, dG^*$; and $w_1 + w_2 = (x-p)^2$ on $\mathrm{supp}(G^*) \subseteq [\underline{\theta}, t_1] \cup [t_2, \bar{s}]$, so the two integrals sum to $\mathrm{Var}(G^*_{\bar{s}})$. For $y \le t_1$, $R(s,y)$ is the gap between w_1 and its tangent at y; for $y \ge t_2$, $R(s,y) = w_1(s) - C(s) \ge 0$; and for $t_1 < y < t_2$, $R(s,y) = (s-t_1)(t_2-y)$ if $s \ge t_1$ and $R(s,y) = (s-t_1)(s-y)$ if $s \le t_1$, both nonnegative. For $\bar{s} \le 2p - \underline{\theta}$, take $w_1 = 0$, $w_2 = (x-p)^2$ and $a(y) = p - y$, so that $R = 0$ and the value is $\mathrm{Var}(\bar{U}_{\bar{s}}) = \mathrm{Var}(G^*_{\bar{s}})$, as $\bar{U}_{\bar{s}} \le_{cx} F$; for $\bar{s} \ge \overline{\theta}$, take $w_1 = (x-p)^2$, $w_2 = 0$ and $a(y) = y - p$, so that $R = (s-y)^2$ and the value is $\mathrm{Var}(F) = \mathrm{Var}(G^*_{\bar{s}})$, as $F \le_{cx} \bar{U}_{\bar{s}}$. In these two cases, $p + \mathrm{Var}(G^*_{\bar{s}})/(\bar{s} - p)$ equals $p + \frac{\bar{s}-p}{3}$, which is increasing, and $p + \frac{\mathrm{Var}(F)}{\bar{s}-p}$, which is decreasing, so a width outside $[2p - \underline{\theta}, \overline{\theta}]$ is weakly beaten by the nearest end of the family.

Step 6 (the sign). Let $\nu(B) := \mathbb{E}[(s-Y)\mathbb{1}\{Y \in B\}]$, so that $\nu(\mathbb{R}) = 0$ and $\int y\, d\nu = \mathrm{Cov}(Y, s-Y) = 0$. As $x - \psi(x)$ is concave, the signals with $s > \psi(s)$ lie in an interval J that contains every other signal between two of them. If J contains no signal, $\nu = 0$. Otherwise, let (y_a, y_b) be the interior of the convex hull of the values $\psi(s)$ of signals $s \in J$. If $\psi(s) \in (y_a, y_b)$, then $\psi(s') < \psi(s) < \psi(s'')$ for some signals $s', s'' \in J$, so $s' < s < s''$ as ψ is nondecreasing, and $s \in J$. Hence $\nu \ge 0$ on (y_a, y_b) and $\nu \le 0$ outside $[y_a, y_b]$. With ℓ the chord of a over $[y_a, y_b]$ (a supporting line if $y_a = y_b$), $a - \ell$ is nonnegative on $[y_a, y_b]$ and nonpositive outside it, so $\mathbb{E}[a(Y)(s-Y)] = \int (a-\ell)\, d\nu \ge 0$. With Step 5, $\mathbb{E}[(Y-p)(s-p)] \le \mathrm{Var}(G^*_{\bar{s}})$, which proves the bound.

Step 7 (interior widths). Let F be concave and not uniform. Two facts are used. For distributions with mean p, $\mathrm{Var}(K_1) - \mathrm{Var}(K_2) = 2\int (I_{K_1} - I_{K_2})\, dx$. For a uniform distribution of half-width r around p, $\frac{\partial I}{\partial r} = \frac{(x-p+r)(r-x+p)}{4r^2}$ on $[p-r, p+r]$ and 0 elsewhere. As F is concave and not uniform, $p < \frac{\underline{\theta}+\overline{\theta}}{2}$, $F \le_{cx} \bar{U}_{\overline{\theta}}$ and $\bar{U}_{2p-\underline{\theta}} \le_{cx} F$, with $I_{\bar{U}_{\overline{\theta}}} - I_F > 0$ and $I_F - I_{\bar{U}_{2p-\underline{\theta}}} > 0$ on $(\underline{\theta}, \overline{\theta})$, since in both cases the CDFs cross once.

Near the full revelation end, let $\eta := \overline{\theta} - \bar{s}$ and $h := \max_x (I_F - I_{\bar{U}_{\bar{s}}})^+$. Then $\max(I_F - h,\ x-p,\ 0)$ is a convex minorant of $\min(I_F, I_{\bar{U}_{\bar{s}}})$, so $\mathrm{Var}(F) - \mathrm{Var}(G^*_{\bar{s}}) \le 2(\overline{\theta} - \underline{\theta})h$.

Write $I_F - I_{\bar{U}_{\bar{s}}} = -(I_{\bar{U}_{\overline{\theta}}} - I_F) + (I_{\bar{U}_{\overline{\theta}}} - I_{\bar{U}_{\bar{s}}})$. The derivative formula, with $r - x + p \le \overline{\theta} - x$ and $x - p + r \le 2r$, bounds the second term by $\frac{\eta(\overline{\theta}-x)^+}{2(\bar{s}-p)}$. Fix $\delta > 0$: on $[\underline{\theta}, \overline{\theta} - \delta]$ the first term is bounded away from zero, so there is no violation for small η; on $[\overline{\theta} - \delta, \overline{\theta}]$ the violation is at most $\frac{\eta\delta}{2(\bar{s}-p)}$. Hence $h = o(\eta)$, and

$$u_F(\bar{s}) - u_F(\overline{\theta}) \ge \frac{\mathrm{Var}(F)\,\eta}{(\bar{s}-p)(\overline{\theta}-p)} - \frac{2(\overline{\theta}-\underline{\theta})\,o(\eta)}{\bar{s}-p} > 0 \quad \text{for small } \eta.$$

Near the noiseless end, let $\bar{s}_0 := 2p - \underline{\theta}$, $\eta := \bar{s} - \bar{s}_0$, and $h' := \max_x (I_{\bar{U}_{\bar{s}}} - I_F)^+$. Symmetrically, $\max(I_{\bar{U}_{\bar{s}}} - h', x - p, 0)$ is a convex minorant of the minimum, so $\mathrm{Var}(\bar{U}_{\bar{s}}) - \mathrm{Var}(G^*_{\bar{s}}) \le 4(\bar{s}-p)h'$. Write $I_{\bar{U}_{\bar{s}}} - I_F = (I_{\bar{U}_{\bar{s}}} - I_{\bar{U}_{\bar{s}_0}}) - (I_F - I_{\bar{U}_{\bar{s}_0}})$. The derivative formula, with $r - x + p \le 2r$ and $x - p + r \le x - \underline{\theta} + \eta$, bounds the first term by $\frac{\eta(x-\underline{\theta}+\eta)^+}{2(p-\underline{\theta})}$; below $\underline{\theta}$, the violation is at most $I_{\bar{U}_{\bar{s}}}(\underline{\theta}) = O(\eta^2)$; and near $\overline{\theta}$ the first term vanishes for $x \ge \bar{s}$, while on $[\underline{\theta} + \delta, \bar{s}]$, a compact subset of $(\underline{\theta}, \overline{\theta})$ as $\bar{s}_0 < \overline{\theta}$, the second is bounded away from zero and the first is at most $\frac{\eta(\overline{\theta}-\underline{\theta}+\eta)}{2(p-\underline{\theta})} \to 0$. So $h' = o(\eta)$. As $\mathrm{Var}(\bar{U}_{\bar{s}}) = \frac{(\bar{s}-p)^2}{3}$ and $G^*_{\bar{s}_0} = \bar{U}_{\bar{s}_0}$,

$$u_F(\bar{s}) - u_F(\bar{s}_0) \ge \frac{\bar{s}-p}{3} - \frac{\bar{s}_0-p}{3} - 4h' = \frac{\eta}{3} - o(\eta) > 0 \quad \text{for small } \eta.$$

So neither end maximizes $u_F(\bar{s})$, and the maximum is attained only at interior widths.

C.6.10 The random pass mark

For concave priors, a random pass mark implements the whole uniformizing family, and its equilibrium is unique.

Lemma 18. *Let F be concave, $\bar{s} \in [2p - \underline{\theta}, \overline{\theta}]$, and $G^* := G^*_{\bar{s}}$. Draw a pass mark C*

independently of the signals with

$$\mathbb{P}(C \le z) = K_{\bar{s}}(z) := \min\left\{1, \frac{I_{G^*}(z)}{(\bar{s}-p)\,G^*(z)^2}\right\},$$

and select the higher signal if it is at least C, tossing a coin otherwise. Then $K_{\bar{s}}$ is a CDF with $K_{\bar{s}} = 1$ on $[t_2, \infty)$, and (G^*, G^*) is an equilibrium in which the agent's value function is $L_{\bar{s}}$ on $[\underline{\theta}, \bar{s}]$, so the rule attains $u_F(\bar{s})$. If F is not uniform and $\bar{s} < \overline{\theta}$, it is the unique equilibrium.

Proof. Equilibrium. Against a rival G that is atomless on $(\underline{\theta}, \infty)$, integrating by parts gives $V_G(x) = K(x)G(x) + \frac{1}{2}\int_x^{\overline{\theta}} G\,dK$ for $x > \underline{\theta}$, so $V_G' = \frac{1}{2}K'G + Kg$ there. With $G = G^*$ and $K = K_{\bar{s}}$, $(G^{*2}K)' = 2dG^*$ with $d = \frac{1}{2(\bar{s}-p)}$, i.e., $V_{G^*}' = d$, and $V_{G^*}(\bar{s}) = K(\bar{s}) = 1$, so $V_{G^*} = L_{\bar{s}}$ on $(\underline{\theta}, \bar{s}]$. If G^* has an atom at $\underline{\theta}$, signals there tie with each other, and $V_{G^*}(\underline{\theta}) = V_{G^*}(\underline{\theta}^+)$, as the gap $\frac{1}{2}K(\underline{\theta}^+)G^*(\{\underline{\theta}\})$ vanishes: $I_{G^*}(x) \le x - \underline{\theta}$, while $G^*(x)$ stays above the mass of the atom, so $K(\underline{\theta}^+) = 0$. So $V_{G^*} = L_{\bar{s}}$ on $[\underline{\theta}, \bar{s}]$, while $V_{G^*} = 1 < L_{\bar{s}}$ above $\bar{s}$; hence every feasible signal yields at most $\int L_{\bar{s}}\,dG' = \frac{1}{2}$. On $[t_2, \bar{s}]$, $I_{G^*} = \frac{d}{2}(z - 2p + \bar{s})^2$ and $G^* = d(z - 2p + \bar{s})$, so $K = 1$. K is nondecreasing if and only if $\sqrt{I_{G^*}}$ is concave, as in Lemma 17(iii), which holds: $\sqrt{I_F}$ is concave on $[\underline{\theta}, t_1]$, as the argument of Lemma 17(iii) uses only concavity of F, I_{G^*} is affine on $[t_1, t_2]$, $\sqrt{I_{G^*}}$ is affine on $[t_2, \bar{s}]$, and I_{G^*} is continuously differentiable on $(\underline{\theta}, \infty)$.

Uniqueness. Write $K := K_{\bar{s}}$ and $m_R := G^*(t_1)$, which is $F(t_1)$ unless G^* keeps only part of an atom at $t_1 = \underline{\theta}$. Against a rival G, a signal x wins with probability $V_G(x) = \frac{1}{2} + \frac{1}{2}K(x)G(x^-) - \frac{1}{2}\int_{(x,\infty)} K\,dG$. If (G, G') is an equilibrium, then by interchangeability so is (G, G^*), so G is a best reply to G^* and G^* is a best reply to G. The first gives $\mathrm{supp}(G) \subseteq [\underline{\theta}, \bar{s}]$. The second is used through deviations from G^* that remain feasible: any mean-preserving contraction of G^*, and any small mean-preserving perturbation of G^* compactly supported in $(t_1, \overline{\theta})$, where $I_{G^*} < I_F$.

Step 1. Two-sided perturbations inside $(t_2, \bar{s})$ make V_G affine there. As $K = 1$ there,

V_G is the mid-CDF of G, so G has a constant density d' on $(t_2, \bar{s})$ and no atoms there. Spreading mass from near $\bar{s}$ to just above it, where $V_G = 1$, is unprofitable only if G has no atom at $\bar{s}$. So $V_G = \ell$ on $(t_2, \bar{s})$ with $\ell(x) := 1 - d'(\bar{s} - x)$.

Step 2. G has no atom in $(\underline{\theta}, t_1)$: at an atom a, V_G jumps up by $K(a)\,G(\{a\})$, so pooling a small interval of types around a with mean just above a would pay. Hence V_G is continuous on $(\underline{\theta}, t_1)$. Merging a small mass of G^* at a signal $u < t_2$ with pooled mass at $x \in (t_2, \bar{s})$ into a point $y \in (t_2, x)$ is a contraction of G^* and changes the payoff by a positive multiple of $\ell(u) - V_G(u)$. Hence $V_G \geq \ell$ G^*-almost everywhere below t_2.

Step 3. If $d' > d$, then $G < G^*$ on $(t_2, \bar{s})$, and $G \leq 1 - d'(\bar{s} - t_2) < m_R = G^*$ on (t_1, t_2), while $\int_{\underline{\theta}}^{t_1} (G - G^*)\,dx = I_G(t_1) - I_F(t_1) \leq 0$. So $\int (G - G^*)\,dx < 0$, contradicting equal means. Hence $d' \leq d$.

Step 4. The equilibrium value gives $\frac{1}{2} = \int V_G\,dG^* \geq \int \ell\,dG^* = \ell(p) = 1 - \frac{d'}{2d}$, by Steps 1 and 2. Hence $d' = d$, $\ell = L_{\bar{s}}$, $G = G^*$ on $(t_2, \bar{s}]$, and, as the inequality is now an equality, $V_G = L_{\bar{s}}$ G^*-almost everywhere below t_2.

Step 5. K is nondecreasing and positive on $(\underline{\theta}, \infty)$. If $\bar{s} = 2p - \underline{\theta}$, then $G^* = \bar{U}_{\bar{s}}$ and Step 4 already gives $G = G^*$. If $t_1 = \underline{\theta}$ otherwise, then G^* has an atom at $\underline{\theta}$, and $V_G(\underline{\theta}) = L_{\bar{s}}(\underline{\theta}) = V_{G^*}(\underline{\theta})$ with $G = G^*$ above t_2 gives $\int_{(\underline{\theta}, t_2]} K\,dG = 0$, so G has no mass on $(\underline{\theta}, t_2]$ and $G = G^*$. Otherwise, letting $x \uparrow t_1$ in $V_G(x) = L_{\bar{s}}(x) = V_{G^*}(x)$, with $G = G^*$ above t_2 and G^* putting no mass on $[t_1, t_2]$, gives $K(t_1)\big(G(t_1^-) - m_R\big) = \int_{[t_1, t_2]} K\,dG \geq K(t_1)\,G([t_1, t_2]) = K(t_1)\big(m_R - G(t_1^-)\big)$, so $G(t_1^-) \geq m_R$; as $G(t_1^-) \leq G(t_2) = m_R$, G has no mass on $[t_1, t_2]$. On $[\underline{\theta}, t_1)$, with $E := G - F$, the same comparison and an integration by parts give $2K(x)E(x) = -\int_x^{t_1} E\,dK =: -\Phi(x)$. For every $x_0 \in (\underline{\theta}, t_1)$, $K \geq K(x_0) > 0$ on $[x_0, t_1]$, so $|\Phi(x)| \leq \int_x^{t_1} \frac{|\Phi|}{2K(x_0)}\,dK$ there, and $\Phi \equiv 0$ on $[x_0, t_1]$ by Grönwall's inequality. So $E \equiv 0$ on $(\underline{\theta}, t_1)$, and also at $\underline{\theta}$ by right-continuity. Hence $G = G^*$, and every equilibrium is (G^*, G^*). □

C.6.11 Continuous rules

Lemma 19. *Let F be concave and not uniform, let $\bar{s}^*$ be the best width, and let K be the pass-mark CDF of Lemma 18 at $\bar{s}^*$, restricted to signals in $[\underline{\theta}, \overline{\theta}]$. For $\varepsilon > 0$, let*

$$H_1^\varepsilon(x, y) := \tfrac{1}{2} + K(\max(x, y))\, \rho\Big(\frac{x - y}{\varepsilon}\Big), \qquad \rho(t) := \max\{-\tfrac{1}{2}, \min\{\tfrac{t}{2}, \tfrac{1}{2}\}\}.$$

H^ε is continuous and equals the pass-mark rule when $|x - y| \geq \varepsilon$. It has an equilibrium, and as $\varepsilon \to 0$, every equilibrium strategy of H^ε converges weakly to $G^*_{\bar{s}^*}$, and the firm's payoff in every equilibrium converges to $u_F(\bar{s}^*)$. Likewise, if F is convex and atomless with $\operatorname{supp}(F) = [\underline{\theta}, \overline{\theta}]$ *and $K \equiv 1$, every equilibrium strategy converges weakly to F, and the firm's payoff converges to the first best.*

Proof. Write H for the pass-mark rule and $G^* := G^*_{\bar{s}^*}$. K, with $K(\underline{\theta}) := K(\underline{\theta}^+)$, is continuous on $[\underline{\theta}, \overline{\theta}]$, so H^ε is continuous, and an equilibrium exists by Sion's theorem; each equilibrium strategy guarantees $\frac{1}{2}$. H is continuous off the diagonal, $H^\varepsilon \to H$ uniformly off every neighborhood of the diagonal, and at a diagonal point (y, y) all limits of H^ε lie between $\frac{1}{2} - \frac{1}{2}K(y)$ and $\frac{1}{2} + \frac{1}{2}K(y)$. Hence, if $\mu_\varepsilon \to \mu$ weakly and μ puts no mass on the diagonal, $\int H^\varepsilon \, d\mu_\varepsilon \to \int H \, d\mu$; the same holds if μ puts mass on the diagonal only at $(\underline{\theta}, \underline{\theta})$ and $K(\underline{\theta}^+) = 0$.

Let G_ε be equilibrium strategies; the feasible set is weakly compact, so take a convergent subsequence, $G_\varepsilon \to G$. (i) G is a best reply to G^* under H: $\iint H^\varepsilon \, dG_\varepsilon \, dG^* \geq \frac{1}{2}$, and $G \otimes G^*$ puts mass on the diagonal only where G^* has an atom, which is at most at $\underline{\theta}$, where $K(\underline{\theta}^+) = 0$; so $\int V_{G^*} \, dG \geq \frac{1}{2}$. (ii) G^* is a best reply to G under H against every deviation G' that has no atom in common with G except possibly at $\underline{\theta}$: $\iint H^\varepsilon \, dG' \, dG_\varepsilon \leq \frac{1}{2}$, and $G' \otimes G$ puts no mass on the diagonal, so $\int V_G \, dG' \leq \frac{1}{2}$, while $\int V_G \, dG^* = \frac{1}{2}$. The uniqueness part of the proof of Lemma 18 uses only (i) and (ii): every deviation in its Steps 1 and 2 can avoid the countably many atoms of G

other than $\underline{\theta}$. Hence $G = G^*$, and the whole family converges to G^*. The firm's payoff is $\iint c_\varepsilon \, dG_\varepsilon \, dG'_\varepsilon$ with $c_\varepsilon(x, y) := y + (x - y)H_1^\varepsilon(x, y)$, which converges uniformly, as $|c_\varepsilon - c| \leq \varepsilon$, to $c(x, y) := y + (x - y)H_1(x, y)$, a continuous function; so the payoff converges to $u_F(\bar{s}^*)$.

For convex F, (i) and (ii) hold with F in place of G^*, as full revelation is an equilibrium of the noiseless rule and F is atomless; $V_G(x) = G(x^-) + \frac{1}{2}G(\{x\})$ is the mid-CDF. G has no atom: it has none at $\underline{\theta}$ or $\overline{\theta}$, as $G \leq_{cx} F$ and F is atomless, and at an interior atom x_0, V_G jumps up, and pooling the types near x_0 into a point just above x_0 that is not an atom of G would pay. So $V_G = G$ is continuous, and by (ii) and Dworczak and Martini (2019), G is convex on $[\underline{\theta}, \overline{\theta}]$. By (i), (ii) and constant-sum, $\int G \, dF = \int F \, dG = \frac{1}{2} = \int F \, dF = \int G \, dG$, so $\int F \, d(F - G) = \int G \, d(F - G) = 0$. As F and G are convex and $G \leq_{cx} F$, the measures df and dg put no mass where $I_G < I_F$. On a component (c, d) of that set, F and G are then affine; at c and d, $I_F - I_G$ attains its minimum 0, which forces $G(c) = F(c)$ and $G(d) = F(d)$; so $F = G$ on (c, d), a contradiction. Hence $G = F$, and the payoff converges to $\mathbb{E}\max(\theta_1, \theta_2)$. □

D. Extensions

This appendix relaxes the remaining assumptions. Appendices D.1 and D.2 leave the well S-shaped class: the first studies its mirror image, and the second describes equilibria around its boundary. Appendix D.3 removes the rival.

D.1 Reverse S-shaped selection rules

The well S-shaped class has a mirror image, in which the density of the selection rule is convex rather than concave. A concave density puts an agent's marginal gain from a higher signal near a tie, whereas a convex density puts it at large margins.

Definition 9. A function $v : [-1, 1] \to [0, 1]$ is called *well reverse S-shaped* if $v(x) - 0.5$ is

odd and increasing, and $v'(x)$ is convex on $[-1,1]$ and uniquely minimized at $x = 0$.

The following lemma transfers the equilibrium analysis to this class.

Lemma 20. *(i) For every well reverse S-shaped $v_2(x)$, there is a well S-shaped $v_1(x)$ with $\tilde{v}_2(x) = -\tilde{v}_1(x)$.*

(ii) Let $\tilde{v}_2(x) = -\tilde{v}_1(x)$. If $(G_1(s_1), G_2(s_2))$ is an equilibrium for $v_1(x)$ and priors (p_1, p_2), then $(\check{G}_1, \check{G}_2)$ is an equilibrium for $v_2(x)$ and priors $(1-p_1, 1-p_2)$, where $\check{G}_i$ is the distribution of $1 - s_i$ for $s_i \sim G_i$.

Hence, for equal priors, the unique equilibrium of a well reverse S-shaped $v(x)$ is $(\{b,1\};\{b,1\})$, where b increases in p analogously to Theorem 1: each agent reveals good news rather than conceding bad news, and full revelation is reached for $p \leq \frac{1}{\mathcal{E}_{\tilde{v}}(1)}$, mirroring Proposition 1. In particular, such rules can deliver the first best for $p < 0.5$: on $[0,1]$, $v(x) = 0.5 + (0.5-\eta)x + \eta x|x|$ with $\eta \in (0, 0.5]$ equals $0.5 + 0.5x - \eta x(1-x)$, so it is revelation-accurate and strictly negatively balanced (Proposition 7).

The guarantee of Proposition 4, however, fails for many such rules.

Example 11. For $v(x) = 0.5 + 0.5x^3$, $u_F(p)$ is non-monotone in p.

For these rules, informativeness often decreases in p unless the elasticity increases sharply, mirroring Lemma 3 and Corollary 1, and since $v(x)$ is convex on $[0,1]$, this effect can outweigh the increase in the prior. Equilibria for unequal priors are flipped too: under a well reverse S-shaped rule with $v'(x)$ not linear near 0, the agent with the high prior reveals fully and the other pools at his prior, the opposite of the pattern in Subsection 3.2.4.

D.2 Around the boundary of the well S-shaped class

This appendix describes what changes as the selection rule leaves the well S-shaped class: under a weaker condition, the equilibrium keeps the forms of Subsection 3.2.4,

whereas further away equilibrium signals may have non-binary support.

D.2.1 Binary support equilibrium

Although $\overline{v}(s_i)$ need not stay S-shaped outside the class, the equilibrium forms of Lemma 1 and Proposition 2 survive outside it. Their proofs use the shape of $v(x)$ only through the shape of agent i's value function against agent j's equilibrium signals, $\{p_j\}$ and $\{0, c\}$. Against $\{0, c\}$, the sign of $\overline{v}''(s_i)$ on $(0, c)$ depends on $w(s_i)$ and $w(c - s_i)$, where $w(x) := -v''(x)$, so only pairs of points with sum at most 1 matter.

Proposition 17. *Let $v(x)$ be a selection rule with $v'(x)$ differentiable on $(0, 1]$, $w(x) > 0$ on $(0, 1]$, and $\frac{w(c-x)}{w(x)}$ strictly decreasing in x on $(0, c)$ for every $c \in (0, 1]$. Then for all priors, the equilibrium is unique, and each agent i plays $\{p_i\}$ or $\{0, a_i\}$ with $a_i \in (p_i, 1]$.*

These conditions hold for every well S-shaped $v(x)$ with strictly concave and differentiable $v'(x)$, but they do not require $v'(x)$ to be concave. Unlike well S-shapedness, however, they are not straightforward to verify, since the monotonicity of $\frac{w(c-x)}{w(x)}$ must hold for every $c \in (0, 1]$. For $\xi \sim N(0, \sigma^2)$, they hold if and only if $\sigma \geq \frac{1}{2}$, a quarter of the variance that well S-shapedness requires ($\sigma \geq 1$). This bound is nearly tight: in simulations, for $\sigma = 0.46$ and $(p_1, p_2) = (0.62, 0.48)$, agent 1 sends three signals in every equilibrium.

D.2.2 Non-binary support equilibrium

Outside the well S-shaped class, equilibrium signals need not have binary support, and their structure varies considerably with the selection rule.

Lemma 21. *Let $v(x)$ be real-analytic on $[-1, 1]$ with $v''(x) > 0$ for all $x \in [-1, 0)$. Then for all $p_i, p_j \in (0, 1)$ and $G_j \in \mathcal{G}(p_j)$, every optimal signal for $(\overline{v}(s_i, G_j), p_i)$ has finite support.*

These conditions hold for the selection rules generated by i.i.d. Gumbel or normal shocks of any variance, so for these natural examples, equilibrium signals are discrete. By the identity theorem, an analytic $\overline{v}(s_i)$ coincides with a line either on a finite set or on all of $[0,1]$, and the latter is ruled out by the convexity of $v(x)$ on $[-1,0]$. Still, simulations[6] show that the number of support points grows as the noise decreases, mirroring the classical result for games on the unit square with bell-shaped kernels (see Karlin (1959)), in which optimal strategies are finitely supported and the number of support points grows without bound as the kernel sharpens.

In contrast, a selection rule that is not analytic can induce an equilibrium signal with a continuous part, already for exponential shocks.

Example 12. Let $v(x)$ be generated by i.i.d. $\varepsilon_i \sim Exp(\lambda)$, i.e. the Laplace CDF $v(x) = 0.5 + 0.5 \cdot \text{sign}(x)\left(1 - e^{-\lambda|x|}\right)$. Then for all $p \leq 0.5$, both agents playing

$$G = \frac{1}{2(1+\lambda p)}\left(\delta_{\{0\}} + \delta_{\{2p\}}\right) + \frac{\lambda p}{1+\lambda p}\,U[0,2p]$$

is a symmetric equilibrium.

In general, such equilibria are hard to find explicitly. If an equilibrium signal of agent j has an interval $[a,b]$ in its support, $\overline{v}(s_j)$ must be linear on $[a,b]$, which is an integral equation for G_i^*:

$$\overline{v}'(s_j) = \int_0^1 v'(s_j - s_i)\,dG_i^*(s_i) = const \quad \forall s_j \in [a,b].$$

Closed-form solutions are known only for special kernels,[7] such as those with a rational Fourier transform, as in Example 12, or piecewise polynomial ones, which include the triangular selection rules of Appendix B.4.

[6] For $p = 0.3$ and $\xi \sim N(0,\sigma^2)$, the equilibrium G_i^* has 5 support points for $\sigma = 0.1$ and 9 for $\sigma = 0.05$.

[7] This is a Fredholm equation of the first kind with a difference kernel on a finite interval; see Sakhnovich (2015) for the general theory and Polyanin and Manzhirov (2008) for solvable kernels.

D.3 A single agent

Without a rival, the firm compares the agent with an outside option of known value $c \in (p, 1)$: she hires him if his score exceeds c, i.e., with probability $a(s) := \mathbb{P}(s + \varepsilon \geq c)$, where ε has a continuous CDF F with $F(0) = 0.5$. Her gain over not hiring is $W := \mathbb{E}_G[a(s)(s - c)]$, and her first best is $p(1 - c)$. The agent maximizes $\mathbb{E}_G[a(s)]$ over $G \in \mathcal{G}(p)$.

Proposition 18. *Let $c \in (p, 1)$.*

(i) Without noise, $W = 0$. With noise, $W \leq p \min(0.5, 1 - c)$.

(ii) If $\varepsilon \sim U[-u, u]$ with $u < c$ and $c + u \leq 1$, the agent's unique best reply is $\{0, c + u\}$, and $W = \frac{pu}{c+u}$. In particular, if $c > 0.5$ and $u = 1 - c$, he reveals fully, and the firm attains her first best.

(iii) The firm can attain her first best only if $c \geq 0.5$.

Without noise, the agent splits the prior into $\{0, c\}$, and the firm hires him exactly when she is indifferent, so she gains nothing from the evidence. Noise devalues a narrow lead over c, as it devalues a narrow lead over a rival in the main model, so the agent must prove more. The bound in (i) holds because the agent can always split into $\{0, c\}$, which wins with probability at least $\frac{p}{c}a(c) = \frac{p}{2c}$: at the outside option, the noise makes hiring a coin flip. The firm thus hires with probability at least $\frac{p}{2c}$, each hire costs her c, and she can hire at most the mass p of good types. By (ii), uniform noise of half-width $\min(c, 1 - c)$ attains this bound, strictly for $c > 0.5$ and in the limit for $c \leq 0.5$: it is the widest noise that never hires a revealed bad agent and never rejects a revealed good one. As after Proposition 1, a tie is a coin flip, and this is why full revelation requires $c \geq 0.5$.[8] A privately known threshold can likewise force a single

[8]The bound uses only $a(c) \geq 0.5$, so it holds for any noise with median 0. With mean-zero noise

sender to reveal fully (Kolotilin et al., 2017); here, the threshold is random to both sides, and the noise is unbiased.

More generally, if the density of ε is unimodal with its peak at 0 and is not constant on $[0, c]$, then $a(s)$ is convex on $[0, c]$ and concave on $[c, 1]$, and the agent splits the prior into $\{0, y\}$ with $y > c$. The firm then gains if and only if p exceeds

$$\hat{p} := \frac{y\,c\,(1 - F(c))}{a(y)(y - c) + c\,(1 - F(c))}$$

Here $1 - F(c)$ is the probability that the noise hires a revealed bad agent, so $\hat{p} = 0$ if $\varepsilon \leq c$, and for normal noise, $\hat{p}$ is close to 0 for small variances and rises with the variance. Two differences from the main model stand out. Without a rival, the noiseless firm learns nothing useful, whereas with two agents, she already learns much (Boleslavsky and Cotton, 2015). And a single agent's threshold is fixed, so a unimodal density suffices for bad news signal, whereas a rival's randomization makes the threshold random, and the well S-shaped class requires a concave density. The comparison also shows the role of commitment: a firm facing a single agent that could commit to any rule would hire only on conclusive evidence and attain her first best without noise. With binary types, a firm comparing two agents can do the same with a pass/fail rule (Proposition 8); with a continuum of types, she cannot, and her best rule blurs the signals (Theorem 4).

D.4 Proofs

D.4.1 Proof of Lemma 20

(i) Consider $v_1(x) = 0.5 + 0.5x - a\tilde{v}_2(x)$. The convexity of $\tilde{v}_2'(x)$ and the uniqueness of its minimum at 0 are simply inverted into the concavity of $v_1'(x)$ and the uniqueness of

instead, rare shocks far beyond the posterior scale can move the mean without affecting decisions, and the firm can approach her first best for every c.

its maximum at 0, and $v_1(x) - 0.5$ is trivially odd and increasing for sufficiently small a, since $\tilde{v}_2'(x)$ is bounded. So $v_1(x)$ is well S-shaped for such a, while

$$\tilde{v}_1(x) = \frac{0.5x - a\tilde{v}_2(x) - x(0.5 - a\tilde{v}_2'(0))}{|0.5 - a - (0.5 - a\tilde{v}_2'(0))|} = -\tilde{v}_2(x).$$

(ii) Since a normalization changes payoffs by a positive factor and terms independent of the signals, $v_k(x)$ and $\tilde{v}_k(x)$ have the same equilibria, $k \in \{1, 2\}$. The map $G \mapsto \check{G}$, where $\check{G}$ is the distribution of $1 - s$ for $s \sim G$, is a bijection between $\mathcal{G}(p)$ and $\mathcal{G}(1 - p)$. Since $\tilde{v}_1(x)$ is odd, $\tilde{v}_2\big((1 - s_i) - (1 - s_j)\big) = -\tilde{v}_1(s_j - s_i) = \tilde{v}_1(s_i - s_j)$, so each agent's payoff from $(\check{H}_1, \check{H}_2)$ under $\tilde{v}_2$ equals that from (H_1, H_2) under $\tilde{v}_1$ for all H_1, H_2. Hence $(\check{G}_1, \check{G}_2)$ is an equilibrium for $\tilde{v}_2$ and priors $(1 - p_1, 1 - p_2)$.

D.4.2 Proof of Proposition 17

Let $G_j = \{0, c\}$ with $c \in (p_j, 1]$, so that $\mathbb{P}(s_j = c) = q := \frac{p_j}{c} \in (0, 1)$. Since $v''(x)$ is odd, $\overline{v}''(s) = (1 - q)v''(s) + q\, v''(s - c) = -(1 - q)w(s) - q\, w(s - c) < 0$ on $(c, 1]$, and on $(0, c)$,

$$\overline{v}''(s) = -(1 - q)w(s) + q\, w(c - s) = w(s)\left(q\, \frac{w(c - s)}{w(s)} - (1 - q)\right),$$

which, as the ratio is strictly decreasing, is first positive and then negative, with at most one zero. If $G_j = \{p_j\}$, then $\overline{v}''(s) = v''(s - p_j)$ is positive on $[0, p_j)$ and negative on $(p_j, 1]$. In both cases, there is $c^* \in [0, 1]$ such that $\overline{v}$ is strictly convex on $[0, c^*]$ and strictly concave on $[c^*, 1]$, so, as in the proof of Lemma 1, agent i's best reply is unique and is either $\{p_i\}$ or $\{0, a_i\}$ with $a_i \in (p_i, 1]$.

Now restrict each agent i to these signals, indexed by $a_i \in [p_i, 1]$, where $a_i = p_i$ stands for $\{p_i\}$. Payoffs are continuous in (a_1, a_2), and, by the above, the best reply within this family is unique and is also a best reply among all signals. By the maximum theorem, each agent's best reply is continuous in the other agent's index, so Brouwer's

theorem gives a pair of mutual best replies, which is an equilibrium. Since all equilibria form a product set and each agent's best reply to the other's equilibrium signal is unique, this equilibrium is unique.

D.4.3 Proof of Lemma 21

Since $v(x)$ is real-analytic on $[-1,1]$, it extends analytically to $\{z \in \mathbb{C} : dist(z,[-1,1]) < \delta\}$ for some $\delta > 0$, hence $\overline{v}(s_i) = \int_0^1 v(s_i - s_j)dG_j(s_j)$ extends analytically to $\{z \in \mathbb{C} : dist(z,[0,1]) < \delta\}$. Let $l(s_i)$ be a line supporting $\overline{v}^{cv}(s_i)$ at p_i. Then $l(s_i) \geq \overline{v}(s_i)$ on $[0,1]$, and every optimal signal is supported on $\{s_i \in [0,1] : \overline{v}(s_i) = l(s_i)\}$. If this set were infinite, it would have an accumulation point in $[0,1]$, and by the identity theorem $\overline{v}(s_i) = l(s_i)$ on $[0,1]$, so $\overline{v}''(0) = 0$. However, $\overline{v}''(0) = \int_0^1 v''(-s_j)dG_j(s_j) > 0$, since $v''(-s_j) > 0$ for $s_j \in (0,1]$ and G_j puts positive mass on $(0,1]$ as $p_j > 0$.

D.4.4 Proof of Proposition 18

(i) Without noise, $a(s) = \mathbb{1}_{s \geq c}$ lies below $\frac{s}{c}$ on $[0,1]$, touching it only at 0 and c, so the agent's unique best reply is $\{0,c\}$, and $W = \frac{p}{c}(c-c) = 0$. With noise, the split $\{0,c\}$ is feasible since $p < c$, and it gives the agent $(1-\frac{p}{c})a(0)+\frac{p}{c}a(c) \geq \frac{p}{2c}$, as $a(c) = 1-F(0) = 0.5$ by continuity. Hence $\mathbb{E}_G[a(s)] \geq \frac{p}{2c}$ in equilibrium, and since $a \leq 1$ and $s \geq 0$,

$$W = \mathbb{E}_G[a(s)s] - c\,\mathbb{E}_G[a(s)] \leq p - c \cdot \frac{p}{2c} = \frac{p}{2}.$$

Also $W \leq \mathbb{E}_G[(s-c)^+] \leq p(1-c)$, since $(s-c)^+ \leq s(1-c)$ on $[0,1]$.

(ii) Here $a(s) = 0$ on $[0, c-u]$, $a(s) = \frac{s-c+u}{2u}$ on $[c-u, c+u]$, and $a(s) = 1$ on $[c+u, 1]$. Let $l(s) := \frac{s}{c+u}$. Then $l > a$ on $(0, c-u]$, $l - a$ is affine on $[c-u, c+u]$, positive at $c-u$ and zero at $c+u$, and $l > a$ on $(c+u, 1]$. So l supports the concavification of a and touches a only at 0 and $c+u$, and by Kamenica and Gentzkow (2011), the agent's unique best reply is $\{0, c+u\}$, with $\mathbb{P}(s = c+u) = \frac{p}{c+u}$. As $a(0) = 0$ and

$a(c+u)=1$, $W=\frac{p}{c+u}(c+u-c)=\frac{pu}{c+u}$. If $c>0.5$ and $u=1-c$, then $u<c$, $c+u=1$, and $W=p(1-c)$.

(iii) Since $a\leq 1$ and $(s-c)^{+}\leq s(1-c)$ with equality only at $s\in\{0,1\}$, $W=p(1-c)$ requires $G=\{0,1\}$, $a(0)=0$ and $a(1)=1$. Full revelation is a best reply only if a lies below its chord from $(0,0)$ to $(1,1)$, i.e., $a(s)\leq s$ on $[0,1]$, and at $s=c$ this gives $0.5\leq c$.